

Table 4: Notation used throughout the framework.

	Symbol	Meaning
Task discovery	$\mathcal{D}$	Input tabular dataset.
	N	Number of observations in $\mathcal{D}$.
	$\mathcal{X}$	Observation (feature) space.
	x_i	The i -th observation (sample) in $\mathcal{D}$, $i \in \{1, \dots, N\}$.
	y_i	Label of x_i ; 0 for normal, 1 for anomalous.
	X, Y	Random variables for an observation and its label, with realizations x_i, y_i .
	K	Number of candidate tasks discovered from $\mathcal{D}$.
	$\mathcal{T}$	Set of candidate tasks.
	τ_k	The k -th candidate task.
	$\mathcal{D}_k^{\text{norm}}$	Normal samples assigned to task k .
	$\mathcal{D}_k^{\text{anom}}$	Anomalous samples assigned to task k .
	$\mathcal{Z}$	Set of natural boundaries in $\mathcal{D}$ (e.g., days, users).
	g	Boundary mapping $\mathcal{X} \rightarrow \mathcal{Z}$ used for natural-boundary task discovery.
	h	Mapping $\mathcal{X} \rightarrow \{1, \dots, K\}$ used for clustering-based task discovery.
	t	A boundary, $t \in \mathcal{Z}$.
Task evaluation	$P_t(X, Y)$	Joint distribution over observations and labels under boundary t .
	$P_t(X Y=0)$	Distribution of normal observations under boundary t .
	F	Pool of single-task expert (STE) models used in task evaluation.
	f	A single model, $f \in F$.
	$ F $	Number of models in F .
	M^f	Cross-task STE performance matrix for model f , of size $K \times K$.
	$M_{i,j}^f$	Performance of f trained on task i and evaluated on task j .
Task selection	b_k	Random-baseline performance on task k (e.g., 0.5 for ROC-AUC).
	S_k^*	Best self-performance on task k across models, $\max_{f \in F} M_{k,k}^f$.
	γ_l	Minimum gain over b_k required for self-learnability.
	γ_t	Retained-performance fraction of S_k^* used to count covering tasks.
	P_c	Maximum allowed number of other tasks that cover task k (FC2).
	$\mathcal{C}_k^f$	Set of tasks whose STE under f covers task k above $\gamma_t S_k^*$.
	γ_d	Relative dominance threshold over S_k^* .
Task ordering	P_d	Maximum allowed number of tasks dominated by task k .
	$\mathcal{O}_k^f$	Set of tasks dominated by task k under model f .
	δ_r	Maximum allowed average difference between two tasks profiles.
	$\mathcal{S}$	Set of scenario candidates produced by task selection.
	s_i	A scenario candidate, $s_i \in \mathcal{S}$.
	m	Number of scenario candidates in $\mathcal{S}$.
	Π	Family of ordering strategies considered by the framework.
Final scenario and ordering selection	Π^f	Orderings produced for model f on a given scenario.
	π	A single ordering of the K tasks in a scenario.
	π_{SD}^f	Smooth-drift ordering for model f .
	π_{AD}^f	Abrupt-drift ordering for model f .
	π_C^f	Curriculum (easy-to-hard) ordering for model f .
	π_{RC}^f	Reverse-curriculum (hard-to-easy) ordering for model f .
	π_{GI}^f	Increasing-generalization ordering for model f .
	π_{GD}^f	Decreasing-generalization ordering for model f .
	$D_{i,j}$	Cross-task dissimilarity used by drift orderings.
	d_i	Difficulty of task i used by curriculum orderings, $d_i = 1 - M_{i,i}^f$.
	g_i	Generalization score of task i used by generalization orderings.
	s^*	Selected scenario after cross-model concordance, $s^* \in \mathcal{S}$.
	$\Pi^{f,s}$	Orderings produced for model f on scenario s .
	l	Index over the six ordering families (SD, AD, C, RC, GI, GD).
	W_π	Kendall's coefficient of concordance for ordering family π across the models in F , $W_\pi \in [0, 1]$.
	Π^*	Set of final consensus orderings, one per family.
	π_l^*	Final consensus ordering for family l , obtained by Borda count on s^* .
	$r_{l,f}(t)$	Rank of task t in the ordering π_l produced by model f .
	$B_l(t)$	Borda score of task t in family l , $B_l(t) = \sum_{f \in F} r_{l,f}(t)$.
	P	Task-by-task performance matrix; $P_{t,k}$ is the performance on task k after training up to task t .

B Additional descriptions

B.1 Detailed framework overview

Figure 3 provides the full version of the framework overview shown compactly in the main paper.

B.2 Task discovery details

Task discovery aims to identify candidate tasks from the full dataset. Each candidate task should correspond to a coherent regime of normal behavior together with the anomalies used to evaluate detection performance within that regime.

More formally, let the fully preprocessed dataset be denoted by $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathcal{X}$ is an observation and $y_i \in \{0, 1\}$ indicates whether the observation is normal ($y_i = 0$) or anomalous ($y_i = 1$). The goal of task discovery is to partition or group $\mathcal{D}$ into a collection of candidate tasks $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_K\}$, where each task τ_k corresponds to a coherent regime of normality together with the anomalies used to evaluate detection under that regime. More precisely, each task can be written as $\tau_k = (\mathcal{D}_k^{\text{norm}}, \mathcal{D}_k^{\text{anom}})$, with $\mathcal{D}_k^{\text{norm}} \subseteq \{(x_i, y_i) \in \mathcal{D} : y_i = 0\}$ and $\mathcal{D}_k^{\text{anom}} \subseteq \{(x_i, y_i) \in \mathcal{D} : y_i = 1\}$.

The framework supports multiple task discovery mechanisms because the relevant source of task structure is inherently dataset-dependent. In some cases, tasks may be induced by natural boundaries, such as days, users, or operating conditions. If such metadata are available, one may define a mapping $g : \mathcal{X} \rightarrow \mathcal{Z}$, where $\mathcal{Z}$ is a set of boundary labels, and construct tasks from the level sets of g . In other datasets, however, no such explicit structure exists, and tasks must instead be inferred from the data through a grouping rule $h : \mathcal{X} \rightarrow \{1, \dots, K\}$, for example via clustering, change-point detection, or other similarity-based procedures.

Importantly, the mere existence of natural boundaries does not imply that they define meaningful continual learning tasks. A boundary-induced partition is only justified if it corresponds to a substantive change in the underlying data-generating distribution. Formally, if $P_t(X, Y)$ denotes the distribution associated with boundary t , then distinct boundaries $t \neq t'$ are not automatically informative when $P_t(X, Y) \approx P_{t'}(X, Y)$, or, more specifically in the anomaly detection setting, when $P_t(X \mid Y = 0) \approx P_{t'}(X \mid Y = 0)$. For example, data collected on different days may still represent essentially the same normal regime.

Accordingly, task discovery seeks to identify a sequence of tasks $\tau_1, \tau_2, \dots, \tau_K$ such that each task is meaningful from a continual learning standpoint, regardless of whether it coincides with the original dataset boundaries. In particular, the objective is to construct tasks that induce sufficiently distinct and non-trivial learning conditions. A newly introduced task τ_k should therefore not be reducible to a trivial recombination of previous tasks $\{\tau_1, \dots, \tau_{k-1}\}$, but should require genuine adaptation because its associated normal regime and evaluation anomalies differ in a material way from those previously encountered.

We note that, in some circumstances, task discovery may be optionally skipped in our framework. For instance, a practitioner may want to extract a multi-dataset scenario with one concept per dataset. In that context, task discovery would not be appropriate since it represents an unnecessary filtering stage.

B.3 Preprocessing

For tabular data, this includes removing unusable columns, handling missing values, encoding categorical variables, scaling numerical attributes when required by the downstream model, and preserving any metadata that may define a natural task boundary. Before attempting to create a continual scenario, we also perform a dataset-level feasibility check. The purpose is to avoid constructing elaborate scenarios from a dataset that cannot be solved by the candidate anomaly detectors even as a single task. If no reasonable detector can separate normal and anomalous examples under a non-continual protocol, then failures in the continual setting would not be informative about forgetting or adaptation.

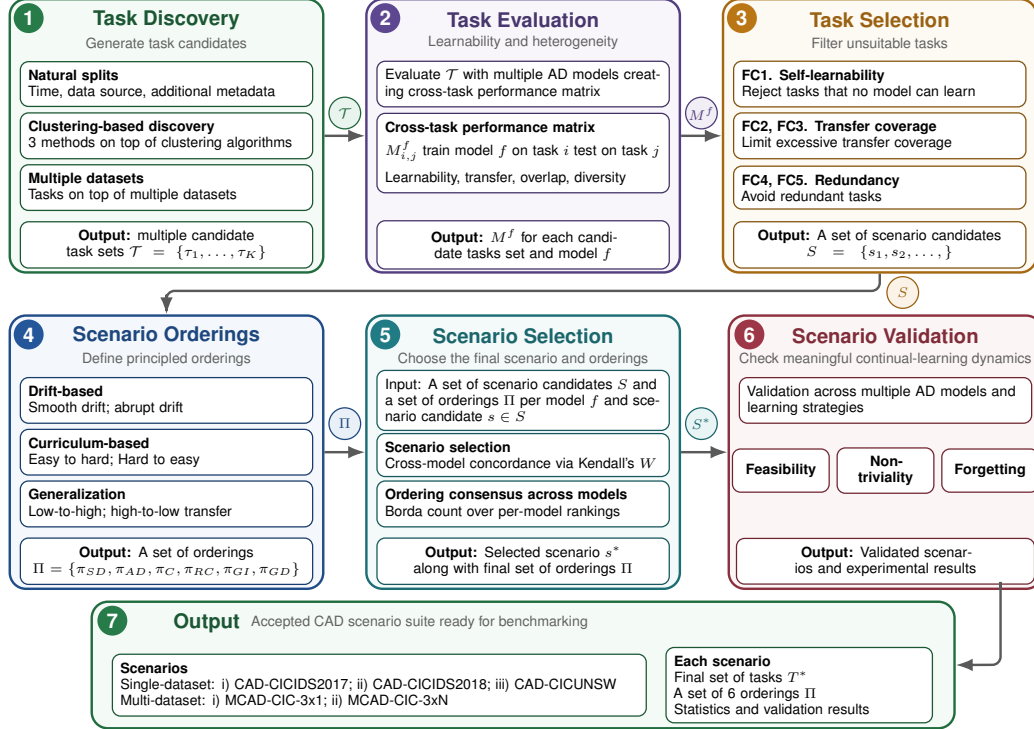


Figure 3: Detailed overview of the proposed framework for transforming anomaly detection datasets into validated continual anomaly detection scenarios.

B.4 Clustering algorithms for task discovery

We use three clustering-based strategies to derive candidate tasks:

- **Both Classes (BC):** the clustering algorithm c is applied to normal and anomalous samples jointly, so that candidate tasks reflect the structure of both classes.
- **Random Anomalies (RA):** the clustering algorithm c is applied only to normal samples, after which anomalous samples are distributed randomly across the resulting tasks.
- **Closest Anomalies (CA):** the clustering algorithm c is applied only to normal samples, after which each anomalous sample is assigned to the task with the nearest normal-cluster centroid.

In our experiments, we leverage three clustering algorithms c :

- **Gaussian Mixture Models**, which provide a soft probabilistic partition of the feature space and can capture ellipsoidal clusters with different covariance structures.
- **k -means**, which provides a simple centroid-based partition and acts as a scalable baseline for inducing compact normal-behavior regimes.
- **Spectral Clustering**, which uses graph structure to recover non-convex groupings that may not be well represented by centroid- or Gaussian-based assumptions.

Combining these algorithms with BC, RA, and CA yields a diverse pool of candidate task decompositions. The subsequent STE-based filtering stage then determines which decompositions produce learnable, non-redundant, and non-dominating tasks.

B.5 Leveraged models

We use different anomaly detectors during scenario construction and final scenario validation. During **task evaluation**, the goal is not to benchmark anomaly detection methods, but to obtain a robust signal

about task learnability and cross-task transfer. We therefore use a compact and heterogeneous pool: an Autoencoder, AE1SVM from PyOD, and Isolation Forest. This combines reconstruction-based, representation-plus-boundary, and tree-ensemble perspectives while keeping the construction phase computationally manageable.

During **scenario validation**, we evaluate the retained scenarios with a broader set of neural anomaly detectors: DAGMM, Deep SVDD, VAE, and NeutralAD. These models are intentionally distinct from the construction-stage pool, which reduces the risk that the final validation merely confirms biases of the models used to select tasks. Hyperparameters are reported in Appendix D.

B.6 Datasets

The five cybersecurity scenarios are derived from three public tabular network-intrusion datasets. **CICIDS2017** and **CSE-CIC-IDS2018** contain network-flow records collected by the Canadian Institute for Cybersecurity and include benign traffic together with multiple attack categories [Sharafaldin et al., 2018]. They provide large-scale intrusion-detection settings with heterogeneous attack behavior and substantial sample-size imbalance across candidate regimes. **CIC-UNSW-NB15**, used in our CAD-CICUNSW scenario, provides an additional network-intrusion dataset with different traffic-generation conditions and attack families [Moustafa and Slay, 2015, Mohammadian et al., 2024]. We also derive three scenarios from tabular datasets outside cybersecurity. **MiniBooNE** [Roe et al., 2005, Roe, 2005] comes from particle physics: its 50 features describe particle-identification events, and we treat electron-neutrino “signal” events as anomalies and muon-neutrino “background” events as normal data. **APS Failure at Scania Trucks** [Lindgren and Biteus, 2016] contains anonymized operational readouts (counters and histogram bins) of the air pressure system of heavy trucks, where anomalies are failures of that system. **TCM** [Jakubowski et al., 2024] is a synthetic predictive-maintenance dataset generated from a physics-based model of a five-stand tandem cold mill in steel manufacturing.

These datasets were selected because they are large enough to support multiple retained tasks after filtering, contain meaningful anomaly labels, and differ in size, anomaly prevalence, and distributional structure. The framework uses them as source assets, but the contribution of this paper is the resulting set of validated CAD scenarios, task splits, and orderings derived from them.

B.7 Evaluation protocol

We evaluate each final scenario by training a model sequentially over the ordered tasks and measuring performance on all tasks after each training step. This produces a task-by-time performance matrix P , where $P_{t,k}$ denotes the performance on task k after training has progressed through task t . Aggregate performance for metric is computed across the tasks in the scenario

$$L_{\text{AGG}} = \frac{2}{K(K+1)} \sum_{t=1}^K \sum_{j=1}^t P_{t,j}. \quad (9)$$

Where K corresponds to the number of tasks in the scenario.

We use ROC-AUC as the primary ranking-based anomaly detection metric. Because all scenarios are imbalanced, we also report normalized PR-AUC, which provides a complementary view of detection quality under skewed anomaly prevalence. Normalized PR-AUC rescales PR-AUC relative to the task-specific anomaly prior, making results more comparable across tasks with different class ratios.

$$\text{nPR-AUC} = \frac{\text{PR-AUC} - \pi}{1 - \pi},$$

where π is the anomaly prevalence in the corresponding test set. A value of 0 corresponds to the expected PR-AUC of a random ranking under the task-specific anomaly prior.

For continual-learning behavior, we report the Forgetting Measure (FM). For each task, forgetting is computed as the gap between the best performance achieved on that task after it was learned and the final performance on the same task after subsequent training. FM is then averaged across tasks. Higher FM indicates stronger loss of previously acquired performance, while lower FM indicates better retention. First, we assess FM' that corresponds to forgetting measured after learning a single

new task k :

$$\text{FM}'_k = \frac{1}{k-1} \sum_{j=1}^{k-1} \left(\max_{t \in \{j, \dots, k\}} P_{t,j} - P_{k,j} \right) \quad (10)$$

And then we compute the final measure FM as an average in induced forgetting in the scenario:

$$\text{FM} = \frac{1}{K} \sum_{k=2}^K \text{FM}'_k \quad (11)$$

B.8 Continual learning strategies

We evaluate four diagnostic training strategies. **Naive** trains sequentially on the current task without any explicit retention mechanism and therefore serves as the standard lower reference for forgetting. **Replay** augments sequential training with a fixed-size buffer of samples from previous tasks, providing a practical retention-based continual-learning strategy. **Cumulative** retrains or updates using all data observed so far, which is not a realistic continual setting but is useful as a strong reference with full past-data access. **MSTE** uses a pool of multiple single-task experts, one per task, and therefore separates task-specific learnability from forgetting. Together, these strategies allow us to distinguish scenarios that are unlearnable, scenarios that are trivially solved by naive adaptation, and scenarios that create genuine retention pressure.

C Filtering thresholds sensitivity analysis

Figures 4–6 report a one-at-a-time sensitivity analysis of the filtering thresholds used in the task-selection stage. For each candidate split, we vary one threshold while keeping the others fixed at their default values, and we measure how many tasks are removed by the corresponding filtering rule. The red curve shows the mean number of removed tasks across candidate splits, while the faint blue curves show split-specific behavior. The dashed green line marks the default threshold adopted in the main paper.

Across the three cybersecurity datasets, the chosen defaults consistently lie in moderate operating regions rather than at pathological extremes. In particular, the self-learnability threshold (shown in the plots as the minimum required ROC-AUC, $b_k + \gamma_l = 0.75$, i.e., $\gamma_l = 0.25$) is placed before the sharp rise in removed tasks that appears near the most restrictive values, which avoids discarding large portions of the scenario due to an overly strict learnability requirement. Similarly, the transfer thresholds $\gamma_t = \gamma_d = 0.9$ retain enough stringency to eliminate trivial or dominating tasks, but are not so severe as to collapse the candidate split space. The count thresholds $P_t = P_d = 2$ also fall near the transition between permissive and highly selective behavior, which is consistent with the intended goal of tolerating limited transfer while rejecting tasks that are broadly redundant or dominant.

The redundancy thresholds $\delta_r = 0.1$ for both row-wise and column-wise profiles show the expected monotonic behavior: larger values merge more tasks as increasingly dissimilar profiles are treated as redundant. Here again, the selected default remains in a conservative regime where redundancy filtering is active but not overwhelming. Overall, these plots support the threshold choices used in the main paper: they enforce meaningful filtering across datasets, yet avoid the unstable regions in which small parameter changes would lead to disproportionately large changes in the number of retained tasks.

Figures 7–9 repeat the analysis for MiniBooNE, Scania, and TCM with the same default thresholds. The count and redundancy thresholds behave as in the cybersecurity datasets. On Scania and TCM, the self-learnability threshold has almost no effect, as almost no task is removed below a minimum ROC-AUC of about 0.95, because the candidate tasks are easy to learn in isolation. On MiniBooNE, by contrast, removals start around a minimum ROC-AUC of 0.55 and the default lies on the rising part of the curve, so this threshold has a noticeable effect on the number of retained tasks. On Scania, the transfer thresholds γ_t and γ_d remove the same number of tasks for all values up to about 0.95, so the default of 0.9 lies on a flat plateau. For the count and transfer thresholds, the defaults do not fall in a region where a small change causes a large change in the number of retained tasks.

OAT sensitivity: filtering parameter vs. # tasks removed (10 splits)

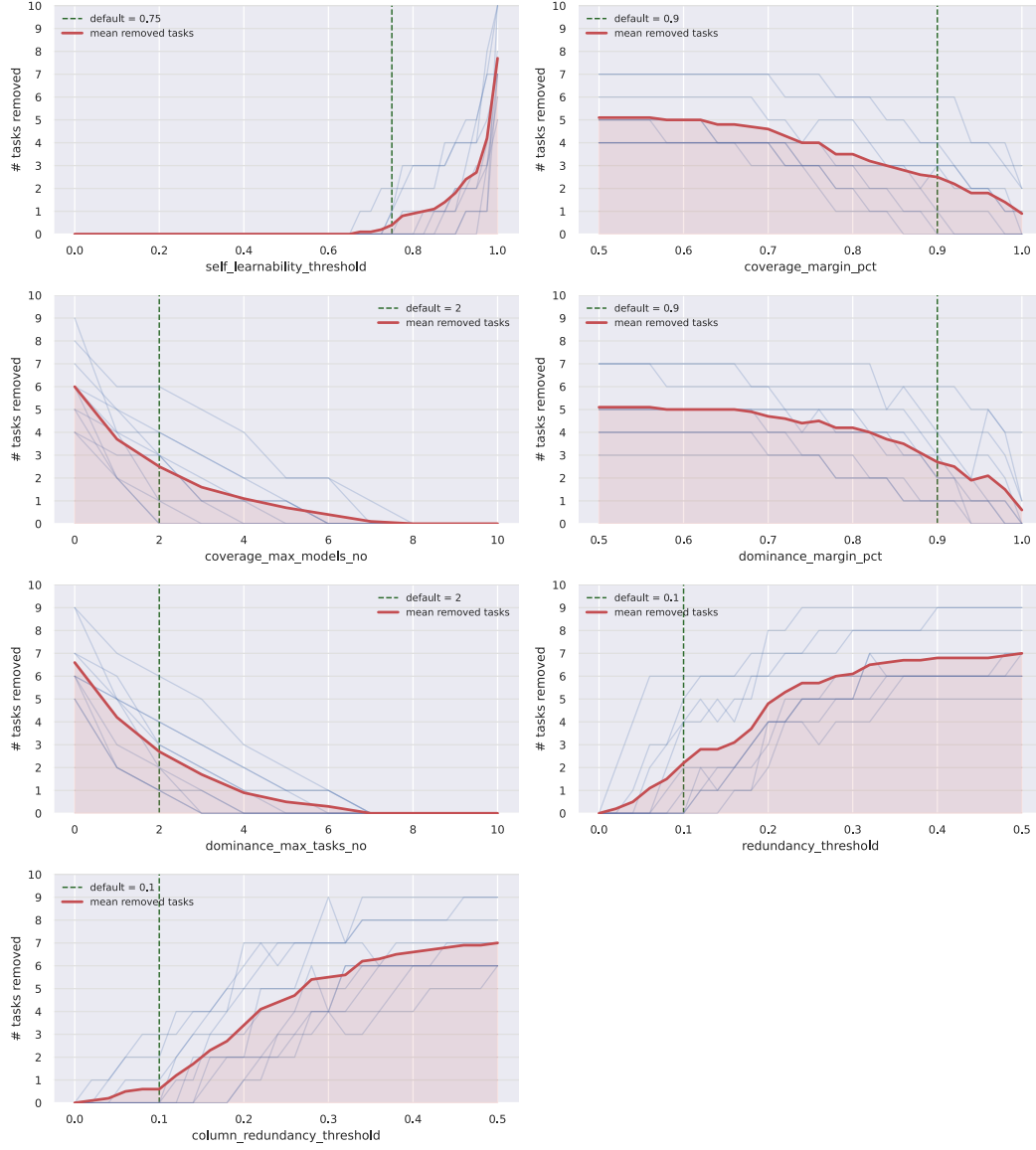


Figure 4: One-at-a-time sensitivity analysis of the filtering thresholds on CICIDS2017. The red curve reports the mean number of removed tasks across candidate splits, the blue curves show split-specific trajectories, and the dashed green line marks the default threshold used in the main paper.

OAT sensitivity: filtering parameter vs. # tasks removed (10 splits)

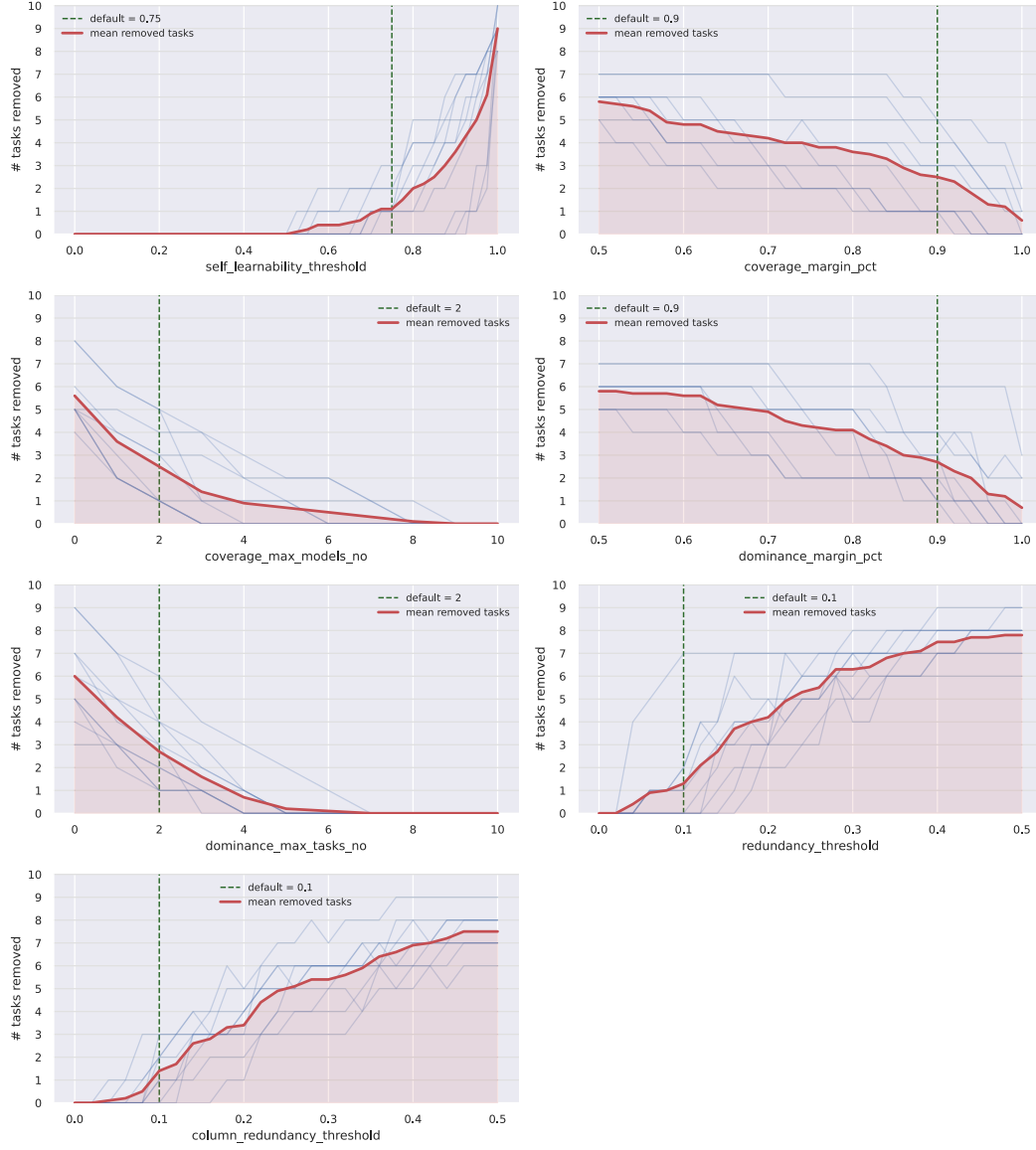


Figure 5: One-at-a-time sensitivity analysis of the filtering thresholds on CICIDS2018. The red curve reports the mean number of removed tasks across candidate splits, the blue curves show split-specific trajectories, and the dashed green line marks the default threshold used in the main paper.

OAT sensitivity: filtering parameter vs. # tasks removed (10 splits)

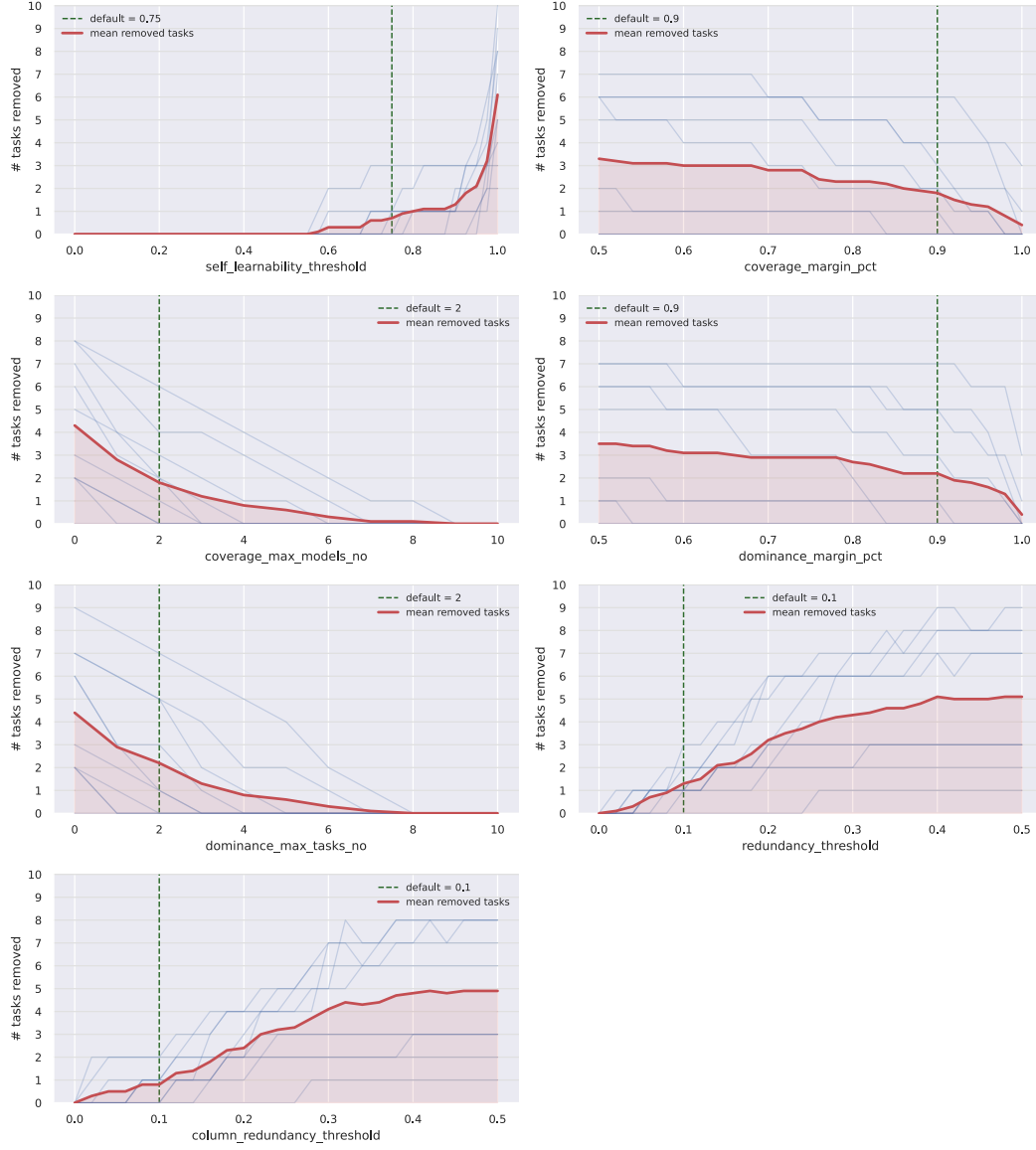


Figure 6: One-at-a-time sensitivity analysis of the filtering thresholds on CICUNSW. The red curve reports the mean number of removed tasks across candidate splits, the blue curves show split-specific trajectories, and the dashed green line marks the default threshold used in the main paper.

OAT sensitivity: filtering parameter vs. # tasks removed (9 splits)

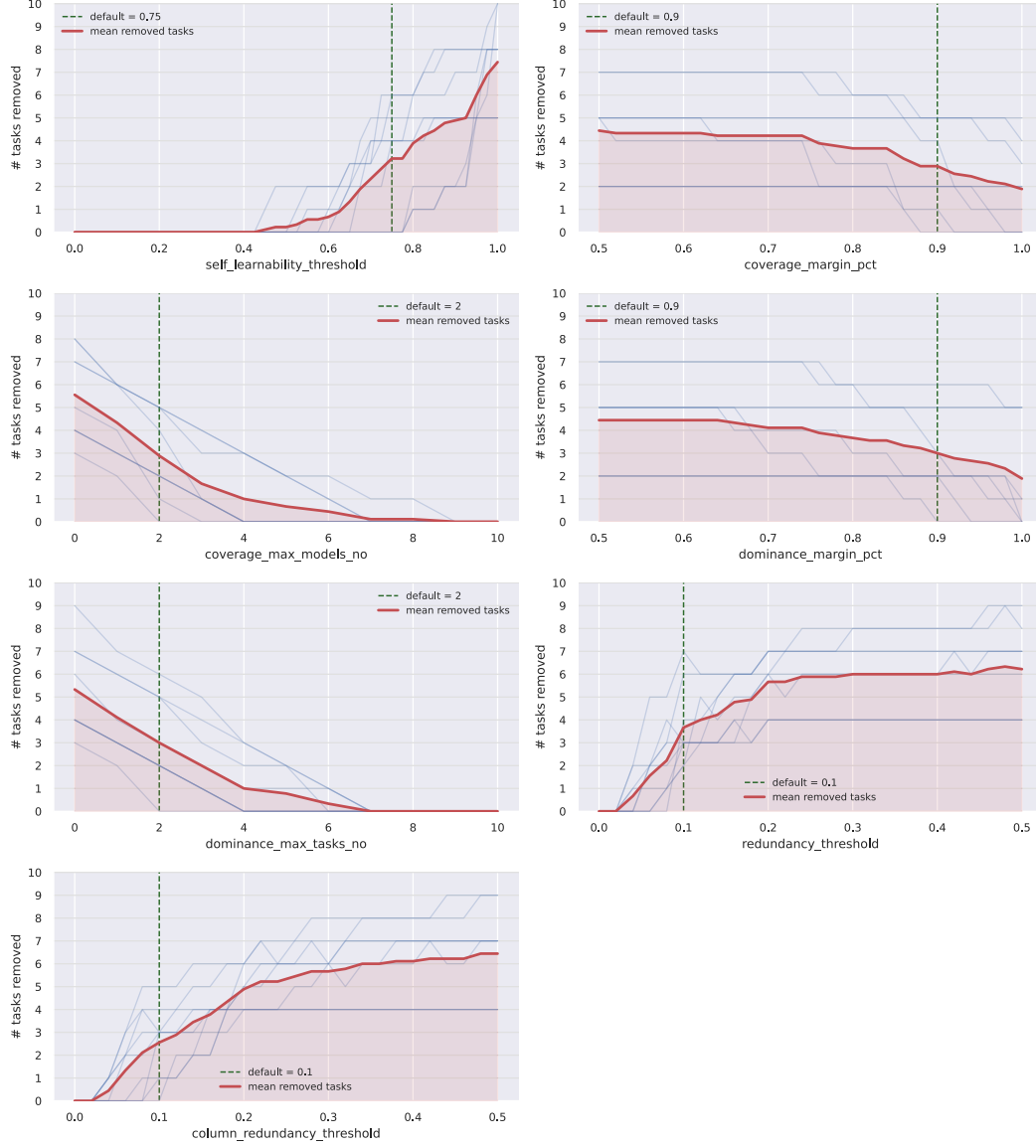


Figure 7: One-at-a-time sensitivity analysis of the filtering thresholds on MiniBooNE. The red curve reports the mean number of removed tasks across candidate splits, the blue curves show split-specific trajectories, and the dashed green line marks the default threshold used in the main paper.

OAT sensitivity: filtering parameter vs. # tasks removed (4 splits)

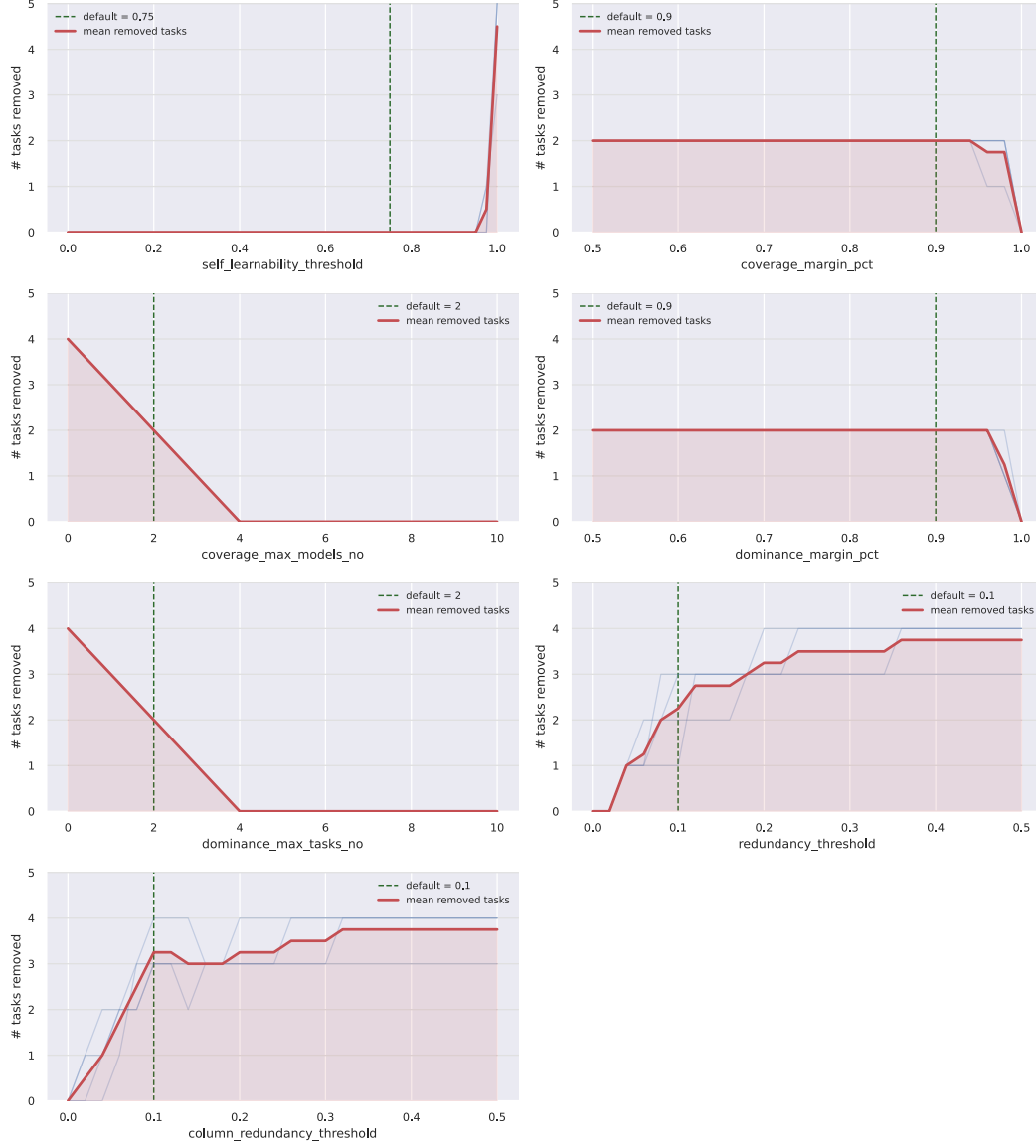


Figure 8: One-at-a-time sensitivity analysis of the filtering thresholds on Scania. The red curve reports the mean number of removed tasks across candidate splits, the blue curves show split-specific trajectories, and the dashed green line marks the default threshold used in the main paper.

OAT sensitivity: filtering parameter vs. # tasks removed (8 splits)

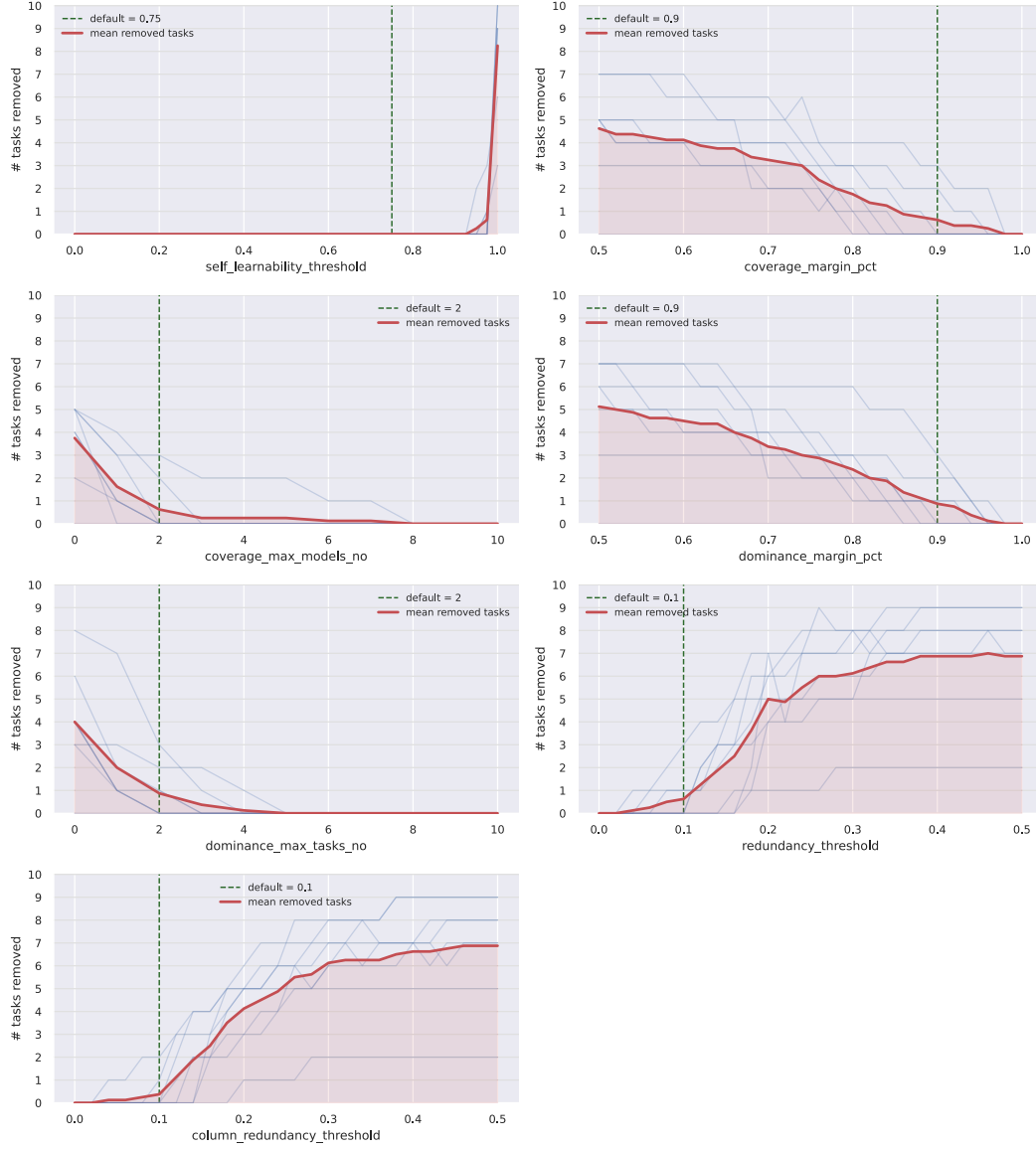


Figure 9: One-at-a-time sensitivity analysis of the filtering thresholds on TCM. The red curve reports the mean number of removed tasks across candidate splits, the blue curves show split-specific trajectories, and the dashed green line marks the default threshold used in the main paper.

C.1 Execution times

It is noteworthy that just our validation phase of the scenarios includes 768 full experimental phases (8 scenarios x 6 orderings x 4 AD models x 4 CL strategies), which corresponds to over 4500 trainings of individual tasks. The overall number of experiments is significantly higher, due to the initial phase involving task selection.

Dataset	Model	Time	Naive	Cumulative	MSTE	Replay
CAD-CICIDS2017	VAE	Train	26m 42s	101m 46s	26m 47s	22m 11s
		Eval	1m 41s	1m 42s	1m 2s	1m 41s
	DAGMM	Train	20m 46s	72m 58s	20m 36s	20m 5s
		Eval	23.0s	23.1s	14.5s	22.9s
	DeepSVDD	Train	9m 15s	32m 55s	9m 19s	5m 23s
		Eval	4.8s	4.8s	3.2s	4.8s
	NeutralAD	Train	21m 5s	52m 13s	16m 59s	19m 20s
		Eval	58.8s	58.9s	35.7s	59.3s
CAD-CICIDS2018	VAE	Train	28m 45s	90m 8s	28m 50s	28m 37s
		Eval	1m 44s	1m 41s	1m 7s	1m 57s
	DAGMM	Train	23m 14s	82m 10s	23m 37s	24m 9s
		Eval	24.7s	25.9s	17.2s	25.9s
	DeepSVDD	Train	8m 10s	28m 8s	8m 15s	6m 47s
		Eval	5.2s	5.2s	3.7s	5.5s
	NeutralAD	Train	44m 15s	62m 46s	43m 57s	45m 33s
		Eval	1m 9s	1m 8s	43.6s	1m 9s
CAD-CICUNSW	VAE	Train	12m 45s	41m 11s	12m 43s	12m 15s
		Eval	39.1s	38.7s	24.2s	42.9s
	DAGMM	Train	10m 55s	34m 50s	10m 57s	11m 16s
		Eval	10.4s	9.8s	6.4s	10.6s
	DeepSVDD	Train	5m 51s	19m 1s	5m 54s	2m 39s
		Eval	2.0s	2.0s	1.4s	1.9s
	NeutralAD	Train	23m 16s	54m 45s	23m 11s	25m 11s
		Eval	24.9s	25.0s	15.8s	25.0s
MCAD-CIC-3x1	VAE	Train	182m 5s	376m 29s	183m 23s	163m 17s
		Eval	9m 46s	9m 45s	6m 57s	9m 46s
	DAGMM	Train	188m 14s	339m 20s	176m 30s	163m 13s
		Eval	2m 40s	2m 25s	1m 51s	2m 38s
	DeepSVDD	Train	50m 39s	130m 60s	50m 54s	53m 45s
		Eval	29.3s	29.7s	21.6s	30.0s
	NeutralAD	Train	44m 40s	45m 5s	43m 44s	45m 33s
		Eval	6m 24s	6m 33s	4m 24s	6m 20s
MCAD-CIC-3xN	VAE	Train	48m 25s	280m 46s	49m 48s	37m 40s
		Eval	7m 21s	6m 10s	4m 8s	6m 13s
	DAGMM	Train	43m 16s	224m 41s	40m 30s	29m 53s
		Eval	1m 53s	1m 29s	59.0s	1m 30s
	DeepSVDD	Train	17m 9s	82m 43s	13m 11s	8m 27s
		Eval	18.9s	18.6s	11.1s	18.5s
	NeutralAD	Train	70m 1s	160m 5s	70m 25s	78m 44s
		Eval	4m 17s	4m 10s	2m 17s	4m 21s
CAD-MiniBooNE	VAE	Train	23.2s	1m 15s	22.8s	25.1s
		Eval	4.3s	4.3s	2.6s	4.3s
	DAGMM	Train	20.1s	1m 6s	19.9s	21.8s
		Eval	1.2s	1.3s	0.8s	1.2s
	DeepSVDD	Train	6.1s	18.7s	5.9s	6.1s
		Eval	0.2s	0.2s	0.1s	0.2s
	NeutralAD	Train	52.3s	2m 44s	52.2s	57.7s
		Eval	3.3s	3.2s	2.0s	3.2s
CAD-Scania	VAE	Train	29.0s	1m 7s	28.3s	30.8s
		Eval	1.0s	1.2s	0.7s	1.1s
	DAGMM	Train	28.2s	1m 4s	27.9s	30.2s
		Eval	0.4s	0.4s	0.3s	0.4s
	DeepSVDD	Train	8.0s	17.9s	7.7s	8.0s
		Eval	0.1s	0.1s	0.1s	0.1s
	NeutralAD	Train	1m 18s	2m 53s	1m 19s	1m 25s
		Eval	0.9s	1.0s	0.7s	1.0s
CAD-TCM	VAE	Train	56.7s	4m 26s	58.7s	59.8s
		Eval	5.6s	5.8s	3.1s	5.8s
	DAGMM	Train	49.1s	3m 49s	48.3s	51.9s
		Eval	1.7s	1.9s	1.0s	1.8s
	DeepSVDD	Train	14.3s	1m 8s	14.0s	14.5s
		Eval	0.3s	0.3s	0.2s	0.3s
	NeutralAD	Train	2m 8s	9m 16s	2m 6s	2m 16s
		Eval	4.4s	4.5s	2.5s	4.4s

Table 5: Execution times (train and eval time separately, mean across orderings) on *CAD-CICIDS2017*, *CAD-CICIDS2018*, *CAD-CICUNSW*, *MCAD-CIC-3x1*, *MCAD-CIC-3xN*, *CAD-MiniBooNE*, *CAD-Scania*, *CAD-TCM*.

D Reproducibility

D.1 Compute resources

All experiments were executed on a high-performance computing (HPC) system whose full peak capacity is approximately 36 PFLOPS, with AMD EPYC 9654 and NVIDIA Grace CPUs and about 300 TB of RAM. The experiments reported in this paper used only a subset of these resources. In particular, the GPU-based runs were executed on (in peak) 15 NVIDIA GH200 96GB GPUs, while the full system provides up to 440 nodes. This is reported to clarify both the scale of the underlying platform and the smaller portion of it effectively used for the results presented here.

D.2 Datasets licensing

The datasets used in this work are credited to their original creators and were used under the access conditions stated by their source providers. CICIDS2017 [Sharafaldin et al., 2018] is made publicly available by the Canadian Institute for Cybersecurity (University of New Brunswick) for research use. CSE-CIC-IDS2018 [Sharafaldin et al., 2018] is released by the same source with permission to redistribute, republish, and mirror the dataset. For the UNSW-NB15 dataset [Moustafa and Slay, 2015] (used here through our CICUNSW scenario), UNSW Canberra states that free use for academic research is granted in perpetuity, while commercial use requires agreement from the authors. The MiniBooNE Particle Identification dataset [Roe, 2005] and the APS Failure at Scania Trucks dataset [Lindgren and Biteus, 2016] are distributed by the UCI Machine Learning Repository under the CC BY 4.0 license. The TCM dataset [Jakubowski et al., 2024] is published on Zenodo.

D.3 Hyperparameters

Table 6: Hyperparameters of the anomaly detection models.

Hyperparameter	AE	VAE	Deep SVDD	DAGMM	NeuTraLAD	AE1SVM	IForest
<i>Training</i>							
Epochs	20	20	20	20	20	20	—
Batch size	128	128	128	128	128	128	—
Optimizer	Adam	Adam	Adam	Adam	Adam	Adam	—
Learning rate	1e-3	1e-3	1e-3	1e-3	1e-3	1e-3	—
Weight decay	—	1e-5	0.1	—	—	1e-5	—
<i>Architecture</i>							
Encoder hidden	[64, 16]	[128, 64, 32]	[64, 32]	auto	[100, 50]	[64, 32]	—
Decoder hidden	[16, 64]	[32, 64, 128]	—	auto	—	[32, 64]	—
Latent dim	16	2	—	$5 + \lfloor d/20 \rfloor$	128	32	—
Activation	ReLU	ReLU	ReLU	Tanh	LeakyReLU	ReLU	—
Batch norm	—	—	—	—	—	✓	—
Dropout	—	0.2	0.2	—	—	0.2	—
<i>Model-specific</i>							
β (KL weight)	—	1.0	—	—	—	—	—
GMM components K	—	—	—	3	—	—	—
λ_{energy}	—	—	—	0.1	—	—	—
λ_{cov}	—	—	—	0.005	—	—	—
Transformations K	—	—	—	—	11	—	—
Transform type	—	—	—	—	residual	—	—
Temperature τ	—	—	—	—	0.1	—	—
RFF features	—	—	—	—	—	512	—
σ (RBF kernel)	—	—	—	—	—	1.0	—
ν (one-class SVM)	—	—	—	—	—	0.1	—
Estimators	—	—	—	—	—	—	100
Max samples	—	—	—	—	—	—	auto

Additional hyperparameters across the framework execution:

- Minimal number of samples in clusters created during task discovery: i) normal = 1500; ii) anomalous = 1000
- Minimal number of tasks candidates in the split: i) CAD-CICIDS2017 = 5 tasks; ii) CAD-CICIDS2018 = 5 tasks; iii) CAD-CICUNSW = 3 tasks; iv) MCAD-CIC-3x1 = 3 tasks; v)

Table 7: Clustering algorithm hyperparameters. K is the number of clusters, set per dataset.

Algorithm	Hyperparameter	Value
KMeans	n_clusters	10
Gaussian Mixture	n_components	10
	covariance_type	diag
	reg_covar	10^{-3}
Spectral Clustering	n_clusters	10
	affinity	nearest_neighbors

Table 8: Hyperparameters of the STE filtering criteria. Each criterion iteratively removes concepts violating its condition; parameters define the decision boundaries. For FC1, γ_l is the minimum gain over the random baseline $b_k = 0.5$, so a task is retained only if its best single-task expert reaches a ROC-AUC of at least $b_k + \gamma_l = 0.75$.

Criterion	Parameter	Value
FC1. Self-learnability	γ_l	0.25
FC2. Limited incoming transfer coverage	γ_t	0.90
	P_t	2
FC3. Limited outgoing transfer coverage	γ_d	0.90
	P_d	2
FC4. Source profile redundancy	δ_r	0.10
FC5. Target profile redundancy	δ_r	0.10

MCAD-CIC-3xN = 5 tasks; vi) CAD-MiniBooNE = 5 tasks; vii) CAD-Scania = 3 tasks; viii) CAD-TCM = 3 tasks.

- Replay uses a fixed global buffer of 25,000 samples. At each training stage, the buffer budget is distributed uniformly over previously observed tasks. Samples are selected uniformly at random from each observed task.

E Scenario selection details

E.1 Scenario selection

For scenario selection, we start from a set of scenario candidates $\mathcal{S} = \{s_1, \dots, s_m\}$ and the orderings induced by each model f on each scenario candidate s_i : $\Pi^{f, s_i} = \{\pi_{SD}, \pi_{AD}, \pi_C, \pi_{RC}, \pi_{GI}, \pi_{GD}\}$. The rationale is that if multiple models with distinct inductive biases independently produce similar concept orderings, this agreement suggests the ordering reflects genuine structural properties of the concept space rather than model-specific behaviors.

We quantify cross-model concordance using Kendall’s W [Kendall, 1938], a coefficient of concordance for $|F|$ rankers (models $f \in F$). For each task t , W compares the sum of ranks assigned by all models f against what would be expected under independent random rankings. It is normalized to $[0, 1]$, where $W = 1$ indicates perfect agreement and $\mathbb{E}[W] = \frac{1}{|F|}$ under random rankings. Crucially, the normalization ensures W is comparable across scenarios with different numbers of tasks.

Kendall’s W only measures whether models agree with each other, not what direction they agree in. If all models consistently rank task t early in ascending order, they also consistently rank it late in descending order. Due to this, we use only four ordering families for split selection: abrupt drift, smooth drift, curriculum ascending, and generalization ascending, omitting curriculum descending and generalization descending.

Then, we select the scenario that maximizes the mean concordance:

$$s^* = \arg \max_{s \in \mathcal{S}} \frac{W_{\pi_{SD}} + W_{\pi_{AD}} + W_{\pi_C} + W_{\pi_G}}{4}. \quad (12)$$

E.2 Final ordering for selected scenario

For the final consensus ordering, given the selected scenario s^* , we aggregate per-model orderings Π^{f,s^*} into a single consensus Π^* using Borda count [Rothe, 2019], by summing the rank assigned to each task across all models and sorting concepts by their total rank. Borda count naturally complements Kendall’s W : while W identifies splits with high cross-model agreement, Borda extracts the ordering that best reflects this agreement. By integrating rankings across models, it mitigates the influence of any single model and produces a stable consensus under moderate variability. Concepts consistently ranked early receive low aggregate scores and appear first, yielding an ordering that captures shared structure while preserving the interpretability of each ordering family.

The final ordering $\pi_l^* \forall \pi_l \in \Pi^*$ (where Π^* consist of all orderings defined in the „Scenario orderings” step) is achieved by sorting concepts by ascending Borda score:

$$B_\ell(t) = \sum_{f \in F} r_{\ell,f}(t), \quad \pi_\ell^* = \text{argsort}_{t \in T_{s^*}} B_\ell(t). \quad (13)$$

where $r_{\ell,f}(t)$ denotes the rank assigned to task t by model f for ordering family ℓ .

E.3 Kendall W

Let $m = |F|$ be the number of construction models and let K be the number of tasks in scenario candidate s . For ordering family ℓ , let $r_{\ell,f}(t)$ be the rank assigned to task t by model f . The rank sum is

$$R_\ell(t) = \sum_{f \in F} r_{\ell,f}(t), \quad \bar{R} = \frac{m(K+1)}{2}.$$

Kendall’s coefficient of concordance is

$$W_\ell(s) = \frac{12 \sum_{t \in T_s} (R_\ell(t) - \bar{R})^2}{m^2(K^3 - K)}.$$

Under independent random rankings, the expected value is approximately $1/m$, where m is the number of rankers. Values substantially exceeding this baseline indicate concordance beyond chance. The denominator $m^2(K^3 - K)$ ensures that $W_\ell(s) \in [0, 1]$, making the coefficient comparable across scenario candidates with different numbers of tasks.

F Detailed Specifications of Final Benchmark Scenarios

This appendix provides detailed information about the final benchmark scenarios produced by our framework. For each scenario, we report the resulting sequence of tasks together with per-task statistics, including the number of normal and anomalous samples. We also provide the final task orderings used in the experiments. These details complement the main paper by making the constructed scenarios fully transparent and reproducible, and by allowing readers to inspect the scale, composition, and progression of each benchmark beyond the aggregate results reported in the main text.

F.1 CAD-CICIDS2017

Table 9: Task orderings for CAD-CICIDS2017

Ordering	Task sequence
Curriculum (asc.)	t5 → t2 → t0 → t3 → t4 → t1
Curriculum (desc.)	t1 → t4 → t3 → t0 → t2 → t5
Generalization (desc.)	t4 → t3 → t0 → t2 → t5 → t1
Generalization (asc.)	t1 → t5 → t2 → t0 → t3 → t4
Smooth drift	t5 → t1 → t4 → t0 → t2 → t3
Abrupt drift	t4 → t5 → t3 → t1 → t2 → t0

Table 10: Per-task statistics for CAD-CICIDS2017

Task ID	Task Name	Train Samples	Test Samples	Test Anomalies
0	cicids2017_0	77,385	38,694	19,347
1	cicids2017_1	1,434,082	411,738	53,217
2	cicids2017_2	19,006	9,504	4,752
3	cicids2017_3	8,392	4,196	2,098
4	cicids2017_4	47,520	23,760	11,880
5	cicids2017_5	1,713	858	429

Tables 9 and 10 specify the retained CAD-CICIDS2017 scenario. The task statistics show a strongly imbalanced task-size distribution, with one dominant concept and several much smaller retained concepts. This imbalance is useful for stress-testing cumulative and replay-based strategies, because good scenario-level performance requires preserving smaller tasks rather than only fitting the dominant regime. The six orderings expose the same task set under curriculum, generalization, and drift assumptions, allowing the scenario to be reused without tying conclusions to one arbitrary sequence.

E.2 CAD-CICIDS2018

Table 11: Task orderings for CAD-CICIDS2018

Ordering	Task sequence
Curriculum (asc.)	t3 → t4 → t2 → t1 → t0
Curriculum (desc.)	t0 → t1 → t2 → t4 → t3
Generalization (desc.)	t2 → t3 → t0 → t1 → t4
Generalization (asc.)	t4 → t1 → t0 → t3 → t2
Smooth drift	t2 → t0 → t1 → t3 → t4
Abrupt drift	t1 → t2 → t3 → t0 → t4

Table 12: Per-task statistics for CAD-CICIDS2018

Task ID	Task Name	Train Samples	Test Samples	Test Anomalies
0	cicids2018_0	251,873	67,807	4,855
1	cicids2018_1	652,959	176,660	13,421
2	cicids2018_2	642,521	316,606	158,303
3	cicids2018_3	211,855	62,074	9,161
4	cicids2018_4	166,025	42,391	907

Tables 11 and 12 summarize the final CAD-CICIDS2018 scenario. Compared with CAD-CICIDS2017, the retained tasks are larger and more balanced in sample size, but the anomaly ratios still vary substantially across tasks. This makes the scenario complementary: it tests CAD behavior under a different balance of scale, anomaly prevalence, and ordering sensitivity while preserving enough task heterogeneity to avoid a trivial sequence.

E.3 CAD-CICUNSW

Tables 13 and 14 describe the CAD-CICUNSW scenario. The retained split combines several large tasks with smaller high-anomaly tasks, yielding a setting where both task imbalance and distributional heterogeneity matter. This supports the role of CAD-CICUNSW as a complementary cybersecurity scenario rather than a simple variant of the CICIDS-derived tasks.

Table 13: Task orderings for CAD-CICUNSW

Ordering	Task sequence
Curriculum (asc.)	t4 → t1 → t3 → t2 → t0
Curriculum (desc.)	t0 → t2 → t3 → t1 → t4
Generalization (desc.)	t4 → t1 → t2 → t3 → t0
Generalization (asc.)	t0 → t3 → t2 → t1 → t4
Smooth drift	t4 → t2 → t1 → t3 → t0
Abrupt drift	t1 → t2 → t3 → t0 → t4

Table 14: Per-task statistics for CAD-CICUNSW

Task ID	Task Name	Train Samples	Test Samples	Test Anomalies
0	cicunsw_0	180,512	54,078	8,950
1	cicunsw_1	406,320	110,531	8,950
2	cicunsw_2	240,550	69,088	8,950
3	cicunsw_3	3,503	1,752	876
4	cicunsw_4	12,396	6,198	3,099

F.4 CAD-MiniBooNE

Table 15: Task orderings for CAD-MiniBooNE

Ordering	Task sequence
Curriculum (asc.)	t2 → t3 → t0 → t4 → t1
Curriculum (desc.)	t1 → t4 → t0 → t3 → t2
Generalization (desc.)	t2 → t0 → t3 → t4 → t1
Generalization (asc.)	t1 → t4 → t3 → t0 → t2
Smooth drift	t1 → t4 → t0 → t3 → t2
Abrupt drift	t0 → t1 → t2 → t4 → t3

Tables 15 and 16 specify the CAD-MiniBooNE scenario. It is the only scenario in which anomalies form the majority of the test data: every task has 4,561 test anomalies, so the anomaly share ranges from 50% in the largest task to 89% in the smallest. Task sizes are also uneven, with `miniboone_0` holding about half of the training data. Detection is harder here than in the other scenarios even for single-task experts (Table 2), which makes CAD-MiniBooNE a test of continual learning when the per-task problems themselves are difficult.

Table 16: Per-task statistics for CAD-MiniBooNE

Task ID	Task Name	Train Samples	Test Samples	Test Anomalies
0	miniboone_0	18,372	9,154	4,561
1	miniboone_1	2,591	5,209	4,561
2	miniboone_2	3,706	5,488	4,561
3	miniboone_3	2,246	5,123	4,561
4	miniboone_4	7,732	6,495	4,561

F.5 CAD-Scania

Table 17: Task orderings for CAD-Scania

Ordering	Task sequence
Curriculum (asc.)	t2 $\rightarrow$ t0 $\rightarrow$ t1
Curriculum (desc.)	t1 $\rightarrow$ t0 $\rightarrow$ t2
Generalization (desc.)	t1 $\rightarrow$ t2 $\rightarrow$ t0
Generalization (asc.)	t0 $\rightarrow$ t2 $\rightarrow$ t1
Smooth drift	t0 $\rightarrow$ t1 $\rightarrow$ t2
Abrupt drift	t1 $\rightarrow$ t0 $\rightarrow$ t2

Table 18: Per-task statistics for CAD-Scania

Task ID	Task Name	Train Samples	Test Samples	Test Anomalies
0	scania_0	6,284	1,847	275
1	scania_1	31,225	8,082	275
2	scania_2	2,696	950	275

Tables 17 and 18 specify the CAD-Scania scenario. With three tasks it is the shortest single-source scenario, and it is dominated by `scania_1`, which holds 78% of the training data. Each task has 275 test anomalies, so the test anomaly ratio ranges from 3.4% in the largest task to 29% in the smallest. The scenario therefore combines a short sequence with strong task-size imbalance and low anomaly prevalence.

F.6 CAD-TCM

Tables 19 and 20 specify the CAD-TCM scenario. With eight tasks it is the longest single-source scenario, and its tasks are more even in size than in the other scenarios (3,829 to 22,302 training samples). Each task has 597 test anomalies, giving test anomaly ratios between 9.7% and 38%. Because the underlying data are simulated sensor readings of a steel mill, CAD-TCM provides an industrial counterpart to the network-traffic scenarios.

Table 19: Task orderings for CAD-TCM

Ordering	Task sequence
Curriculum (asc.)	t2 → t5 → t7 → t6 → t4 → t0 → t1 → t3
Curriculum (desc.)	t3 → t1 → t0 → t4 → t6 → t7 → t5 → t2
Generalization (desc.)	t0 → t5 → t2 → t7 → t4 → t6 → t1 → t3
Generalization (asc.)	t3 → t1 → t6 → t4 → t7 → t2 → t5 → t0
Smooth drift	t5 → t7 → t0 → t2 → t1 → t6 → t4 → t3
Abrupt drift	t0 → t3 → t5 → t1 → t2 → t6 → t7 → t4

Table 20: Per-task statistics for CAD-TCM

Task ID	Task Name	Train Samples	Test Samples	Test Anomalies
0	tcm_0	7,898	2,572	597
1	tcm_1	22,302	6,173	597
2	tcm_2	3,829	1,555	597
3	tcm_3	18,480	5,217	597
4	tcm_4	10,120	3,127	597
5	tcm_5	4,304	1,673	597
6	tcm_6	11,204	3,398	597
7	tcm_7	5,138	1,882	597

E.7 MCAD-CIC-3x1

Table 21: Task orderings for MCAD-CIC-3x1

Ordering	Task sequence
Curriculum (asc.)	cicunsw → cicids2017 → cicids2018
Curriculum (desc.)	cicids2018 → cicids2017 → cicunsw
Generalization (desc.)	cicids2018 → cicunsw → cicids2017
Generalization (asc.)	cicids2017 → cicunsw → cicids2018
Smooth drift	cicids2018 → cicids2017 → cicunsw
Abrupt drift	cicids2017 → cicids2018 → cicunsw

Tables 21 and 22 define the coarse multi-dataset scenario in which each source dataset contributes one task. This setting is intentionally simple in number of tasks, but not necessarily easy: each task corresponds to a different dataset-level distribution, so the scenario emphasizes cross-dataset transfer and dataset-level domain shift.

Table 22: Per-task statistics for MCAD-CIC-3x1

Task ID	Task Name	Train Samples	Test Samples	Test Anomalies
0	cicids2017	1,571,292	949,506	425,741
1	cicids2018	7,923,578	3,992,560	1,351,367
2	cicunsw	2,541,843	936,790	89,508

F.8 MCAD-CIC-3xN

Table 23: Task orderings for MCAD-CIC-3xN: *c17* corresponds to tasks from CICIDS2017, *c18* corresponds to tasks from CICIDS2018, *cu* corresponds to tasks from CICUNSW

Ordering	Task sequence
Curriculum (asc.)	$cu_3 \rightarrow c17_1 \rightarrow c17_4 \rightarrow c17_0 \rightarrow cu_1 \rightarrow c17_2 \rightarrow c18_2 \rightarrow c17_3 \rightarrow cu_0 \rightarrow cu_2 \rightarrow c18_3 \rightarrow c18_1 \rightarrow c18_0$
Curriculum (desc.)	$c18_0 \rightarrow c18_1 \rightarrow c18_3 \rightarrow cu_2 \rightarrow cu_0 \rightarrow c17_3 \rightarrow c18_2 \rightarrow c17_2 \rightarrow cu_1 \rightarrow c17_0 \rightarrow c17_4 \rightarrow c17_1 \rightarrow cu_3$
Generalization (desc.)	$cu_3 \rightarrow cu_1 \rightarrow cu_0 \rightarrow c17_4 \rightarrow cu_2 \rightarrow c18_0 \rightarrow c18_2 \rightarrow c17_1 \rightarrow c18_1 \rightarrow c17_3 \rightarrow c17_2 \rightarrow c17_0 \rightarrow c18_3$
Generalization (asc.)	$c18_3 \rightarrow c17_0 \rightarrow c17_2 \rightarrow c17_3 \rightarrow c18_1 \rightarrow c17_1 \rightarrow c18_2 \rightarrow c18_0 \rightarrow cu_2 \rightarrow c17_4 \rightarrow cu_0 \rightarrow cu_1 \rightarrow cu_3$
Smooth drift	$c18_2 \rightarrow c18_3 \rightarrow cu_0 \rightarrow c18_1 \rightarrow c17_4 \rightarrow cu_1 \rightarrow cu_3 \rightarrow cu_2 \rightarrow c18_0 \rightarrow c17_0 \rightarrow c17_3 \rightarrow c17_1 \rightarrow c17_2$
Abrupt drift	$cu_0 \rightarrow c17_2 \rightarrow cu_3 \rightarrow c17_1 \rightarrow cu_2 \rightarrow c18_2 \rightarrow c17_4 \rightarrow c17_0 \rightarrow cu_1 \rightarrow c17_3 \rightarrow c18_1 \rightarrow c18_3 \rightarrow c18_0$

Table 24: Per-task statistics for MCAD-CIC-3xN

Task ID	Task Name	Train Samples	Test Samples	Test Anomalies
0	cicids2017_0	77,385	38,694	19,347
1	cicids2017_1	19,006	9,504	4,752
2	cicids2017_2	8,392	4,196	2,098
3	cicids2017_3	47,520	23,760	11,880
4	cicids2017_4	1,713	858	429
5	cicids2018_0	652,959	176,660	13,421
6	cicids2018_1	642,521	316,606	158,303
7	cicids2018_2	211,855	62,074	9,161
8	cicids2018_3	166,025	42,391	907
9	cicunsw_0	180,512	54,078	8,950
10	cicunsw_1	406,320	110,531	8,950
11	cicunsw_2	240,550	69,088	8,950
12	cicunsw_3	12,396	6,198	3,099

Tables 23 and 24 specify the fine-grained multi-dataset scenario. Unlike MCAD-CIC-3x1, this scenario retains multiple concepts from each source dataset and interleaves them through principled orderings. It is therefore the most heterogeneous scenario in the suite, combining within-dataset task variation with cross-dataset distribution shift.

G Validation results in terms of PR-AUC

Table 25: PR-AUC (mean $\pm$ std across orderings) on *CAD-CICIDS2017*, *CAD-CICIDS2018*, *CAD-CICUNSW*, *MCAD-CIC-3x1*, *MCAD-CIC-3xN*, *CAD-MiniBooNE*, *CAD-Scania*, *CAD-TCM*. FM $\downarrow$: lower is better. Best per row in **bold**.

Dataset	Model	Naive		Replay		Cumulative		MSTE	
		PR $\uparrow$	FM $\downarrow$	PR $\uparrow$	FM $\downarrow$	PR $\uparrow$	FM $\downarrow$	PR $\uparrow$	FM $\downarrow$
CAD-CICIDS2017	VAE	64.7 $\pm$ 3.3	34.6 $\pm$ 2.9	83.1 $\pm$ 2.9	10.0 $\pm$ 2.5	79.1 $\pm$ 6.9	6.0 $\pm$ 4.1	94.7$\pm$2.8	0.0$\pm$0.0
	Deep SVDD	61.0 $\pm$ 5.3	34.5 $\pm$ 5.0	68.7 $\pm$ 9.2	12.7 $\pm$ 2.9	72.9 $\pm$ 9.2	11.1 $\pm$ 3.6	89.9$\pm$3.0	0.0$\pm$0.0
	DAGMM	59.9 $\pm$ 5.1	30.6 $\pm$ 3.6	71.2 $\pm$ 7.1	11.1 $\pm$ 4.2	60.0 $\pm$ 11.9	13.4 $\pm$ 4.6	83.3$\pm$9.0	0.0$\pm$0.0
	NeutralAD	62.8 $\pm$ 5.4	32.2 $\pm$ 5.2	86.4 $\pm$ 2.8	6.3 $\pm$ 3.3	78.3 $\pm$ 11.2	7.5 $\pm$ 5.8	91.1$\pm$5.9	0.0$\pm$0.0
CAD-CICIDS2018	VAE	35.2 $\pm$ 5.4	34.6 $\pm$ 10.1	43.0 $\pm$ 10.6	26.3 $\pm$ 11.3	46.3 $\pm$ 9.6	13.4 $\pm$ 9.3	64.8$\pm$8.5	0.0$\pm$0.0
	Deep SVDD	32.5 $\pm$ 4.6	34.8 $\pm$ 6.3	38.3 $\pm$ 8.4	25.4 $\pm$ 6.3	34.6 $\pm$ 10.9	16.4 $\pm$ 9.4	60.1$\pm$6.9	0.0$\pm$0.0
	DAGMM	32.1 $\pm$ 7.4	19.9 $\pm$ 8.4	39.0$\pm$8.0	13.5 $\pm$ 10.9	35.3 $\pm$ 6.9	13.2 $\pm$ 6.6	36.2 $\pm$ 3.4	0.0$\pm$0.0
	NeutralAD	35.4 $\pm$ 3.3	40.7 $\pm$ 3.8	53.9 $\pm$ 12.9	16.9 $\pm$ 11.1	50.3 $\pm$ 7.6	15.2 $\pm$ 2.8	65.9$\pm$3.4	0.0$\pm$0.0
CAD-CICUNSW	VAE	49.5 $\pm$ 3.9	45.1 $\pm$ 8.0	64.3 $\pm$ 2.0	23.0 $\pm$ 3.9	68.0 $\pm$ 7.4	7.9 $\pm$ 2.0	86.8$\pm$4.1	0.0$\pm$0.0
	Deep SVDD	43.6 $\pm$ 2.7	28.1 $\pm$ 4.7	41.4 $\pm$ 9.8	18.0 $\pm$ 4.4	39.3 $\pm$ 5.6	15.2 $\pm$ 3.3	62.8$\pm$5.6	0.0$\pm$0.0
	DAGMM	40.3 $\pm$ 5.9	32.7 $\pm$ 9.4	43.4 $\pm$ 8.3	23.6 $\pm$ 7.4	44.1 $\pm$ 10.1	17.5 $\pm$ 7.4	72.5$\pm$11.8	0.0$\pm$0.0
	NeutralAD	48.9 $\pm$ 5.3	41.7 $\pm$ 7.0	66.4 $\pm$ 3.5	17.7 $\pm$ 4.6	56.2 $\pm$ 9.9	17.3 $\pm$ 4.2	84.6$\pm$3.3	0.0$\pm$0.0
CAD-MiniBooNE	VAE	89.3 $\pm$ 1.6	6.5 $\pm$ 4.3	87.8 $\pm$ 3.3	4.6 $\pm$ 1.3	88.1 $\pm$ 3.9	2.4 $\pm$ 1.3	95.9$\pm$0.7	0.0$\pm$0.0
	Deep SVDD	86.8 $\pm$ 3.8	9.1 $\pm$ 4.5	86.5 $\pm$ 3.6	9.2 $\pm$ 4.2	87.3$\pm$4.9	5.7 $\pm$ 3.3	86.1 $\pm$ 2.7	0.0$\pm$0.0
	DAGMM	81.0 $\pm$ 5.7	14.1 $\pm$ 7.1	80.4 $\pm$ 8.0	10.5 $\pm$ 4.6	88.5$\pm$2.2	4.5 $\pm$ 2.4	83.8 $\pm$ 3.3	0.0$\pm$0.0
	NeutralAD	83.4 $\pm$ 8.0	10.0 $\pm$ 7.5	81.8 $\pm$ 9.2	9.6 $\pm$ 7.3	81.5 $\pm$ 8.8	7.8 $\pm$ 5.6	87.1$\pm$2.2	0.0$\pm$0.0
CAD-Scania	VAE	54.8 $\pm$ 6.1	0.4 $\pm$ 0.4	54.7 $\pm$ 5.9	0.4 $\pm$ 0.5	57.1 $\pm$ 8.9	0.1 $\pm$ 0.1	84.4$\pm$6.9	0.0$\pm$0.0
	Deep SVDD	37.8 $\pm$ 5.2	19.6 $\pm$ 8.1	41.7 $\pm$ 8.1	11.5 $\pm$ 8.1	39.8 $\pm$ 13.7	5.1 $\pm$ 9.0	70.8$\pm$12.9	0.0$\pm$0.0
	DAGMM	35.7 $\pm$ 5.8	28.2 $\pm$ 10.8	46.3 $\pm$ 13.7	17.4 $\pm$ 18.1	47.1 $\pm$ 13.1	14.8 $\pm$ 19.2	56.6$\pm$10.7	0.0$\pm$0.0
	NeutralAD	42.2 $\pm$ 12.2	22.3 $\pm$ 18.5	45.9 $\pm$ 13.5	16.5 $\pm$ 21.4	43.8 $\pm$ 10.6	18.3 $\pm$ 20.2	57.4$\pm$11.8	0.0$\pm$0.0
CAD-TCM	VAE	59.6 $\pm$ 3.0	39.6 $\pm$ 3.7	78.0 $\pm$ 3.6	14.8 $\pm$ 3.8	85.7 $\pm$ 5.3	3.0 $\pm$ 1.6	97.1$\pm$1.1	0.0$\pm$0.0
	Deep SVDD	54.3 $\pm$ 3.0	38.6 $\pm$ 7.5	57.8 $\pm$ 2.8	31.5 $\pm$ 4.2	55.8 $\pm$ 7.7	21.3 $\pm$ 3.7	78.7$\pm$5.8	0.0$\pm$0.0
	DAGMM	38.3 $\pm$ 3.4	48.0 $\pm$ 7.3	47.6 $\pm$ 4.7	33.5 $\pm$ 5.5	49.5 $\pm$ 5.9	19.3 $\pm$ 3.5	80.2$\pm$8.8	0.0$\pm$0.0
	NeutralAD	54.3 $\pm$ 4.5	46.4 $\pm$ 6.1	73.2 $\pm$ 8.6	19.8 $\pm$ 7.3	75.7 $\pm$ 6.8	13.1 $\pm$ 5.0	95.1$\pm$1.8	0.0$\pm$0.0
MCAD-CIC-3x1	VAE	52.9 $\pm$ 5.2	16.3 $\pm$ 2.7	56.5 $\pm$ 6.0	10.3 $\pm$ 2.4	57.9 $\pm$ 3.4	3.7 $\pm$ 1.9	64.8$\pm$3.9	0.0$\pm$0.0
	Deep SVDD	47.8 $\pm$ 7.4	1.9 $\pm$ 2.1	47.5 $\pm$ 5.0	3.8 $\pm$ 1.3	47.3 $\pm$ 4.4	2.2 $\pm$ 0.7	47.9$\pm$4.7	0.0$\pm$0.0
	DAGMM	45.5 $\pm$ 8.0	6.6 $\pm$ 2.5	46.1 $\pm$ 5.3	5.7 $\pm$ 5.9	48.4$\pm$2.8	4.9 $\pm$ 4.4	44.0 $\pm$ 5.6	0.0$\pm$0.0
	NeutralAD	56.1 $\pm$ 4.1	27.6 $\pm$ 4.3	61.8 $\pm$ 9.1	11.2 $\pm$ 6.3	62.8 $\pm$ 6.8	6.3 $\pm$ 4.9	68.3$\pm$9.6	0.0$\pm$0.0
MCAD-CIC-3xN	VAE	42.6 $\pm$ 4.4	50.2 $\pm$ 4.0	67.1 $\pm$ 7.9	18.8 $\pm$ 5.3	63.5 $\pm$ 11.8	13.1 $\pm$ 4.3	88.3$\pm$4.4	0.0$\pm$0.0
	Deep SVDD	43.0 $\pm$ 6.4	35.6 $\pm$ 4.6	56.8 $\pm$ 10.1	18.6 $\pm$ 4.2	56.4 $\pm$ 11.1	14.0 $\pm$ 3.2	73.6$\pm$4.7	0.0$\pm$0.0
	DAGMM	38.5 $\pm$ 5.5	38.7 $\pm$ 3.7	57.3 $\pm$ 10.9	18.1 $\pm$ 3.3	54.7 $\pm$ 8.5	20.0 $\pm$ 5.5	68.8$\pm$7.9	0.0$\pm$0.0
	NeutralAD	45.7 $\pm$ 5.7	45.5 $\pm$ 1.2	74.9 $\pm$ 7.5	11.7 $\pm$ 3.3	70.0 $\pm$ 10.5	11.8 $\pm$ 4.7	86.4$\pm$4.3	0.0$\pm$0.0

H Per-Dataset Validation Results for Final Scenarios

This appendix provides additional validation results for the final benchmark scenarios constructed by our framework. For each dataset, we report the performance obtained under the considered task orderings using ROC-AUC and normalized PR-AUC, enabling a more detailed comparison of how ordering affects anomaly detection performance. We further include heatmaps for each ordering and continual learning strategy, reported for both evaluation metrics. These results complement the aggregate analysis in the main paper by exposing the full per-dataset validation evidence, including task-level transfer patterns and ordering-specific performance differences.

H.1 CAD-CICIDS2017

The ordering-specific results for CAD-CICIDS2017 confirm the aggregate findings from the main paper. Across the six retained orderings, Naive remains consistently below Replay, Cumulative, and MSTE, while forgetting remains clearly visible for sequential fine-tuning alone. This indicates that the continual-learning difficulty of the scenario is not tied to a single ordering, but persists across curriculum-, generalization-, and drift-based sequences. Figures 10 and 11 visualize this pattern under ROC-AUC and normalized PR-AUC.

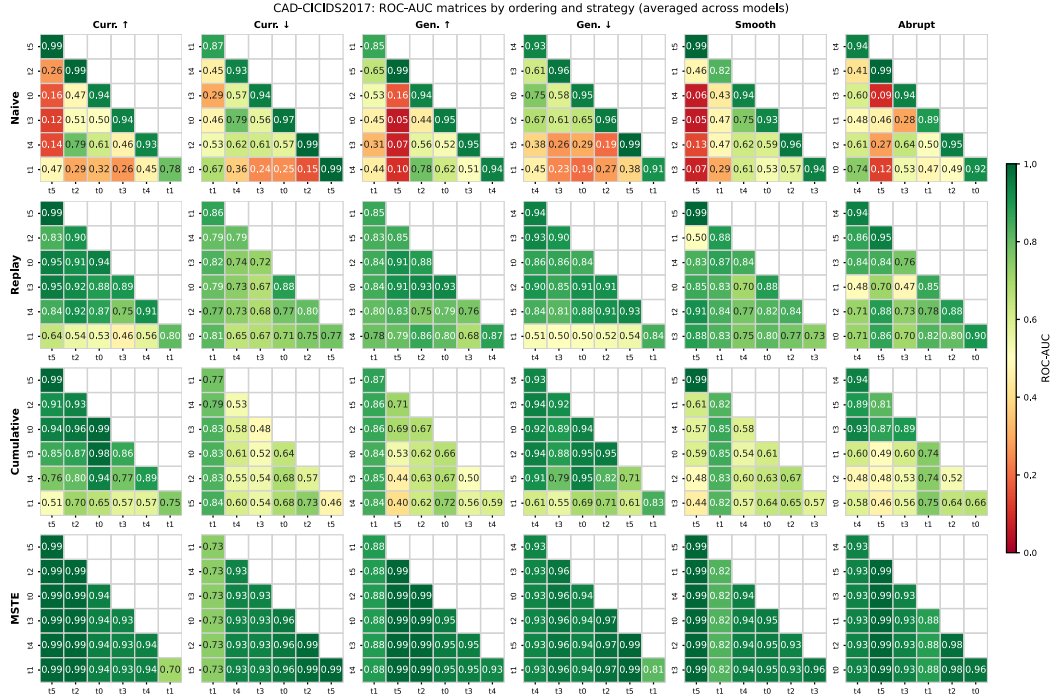


Figure 10: Ordering-specific validation results on CAD-CICIDS2017 measured with ROC-AUC. The heatmaps compare Naive, Replay, Cumulative, and MSTE across the six retained orderings, highlighting the persistent performance gap between Naive and the stronger reference strategies.

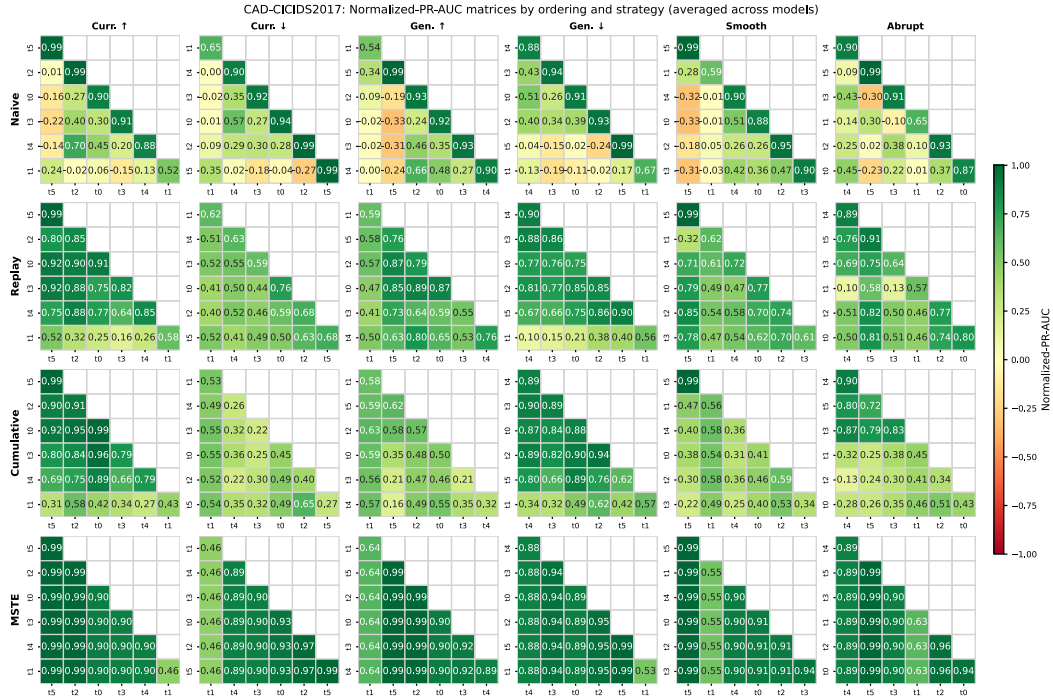


Figure 11: Ordering-specific validation results on CAD-CICIDS2017 measured with normalized PR-AUC. The qualitative pattern matches ROC-AUC and confirms that the scenario remains non-trivial under an imbalance-aware metric.

Table 26: ROC-AUC (mean $\pm$ std across models) per ordering on *CAD-CICIDS2017*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	ROC-AUC $\uparrow$	54.7 $\pm$ 5.5	82.3 $\pm$ 9.9	96.1$\pm$3.7	81.5 $\pm$ 6.4
	FM $\downarrow$	51.4 $\pm$ 11.2	12.0 $\pm$ 7.7	0.0$\pm$0.0	12.4 $\pm$ 5.3
Curriculum (desc.)	ROC-AUC $\uparrow$	61.5 $\pm$ 2.1	65.2 $\pm$ 10.5	89.3$\pm$8.1	76.3 $\pm$ 15.0
	FM $\downarrow$	36.2 $\pm$ 4.5	5.4 $\pm$ 5.6	0.0$\pm$0.0	7.6 $\pm$ 2.9
Generalization (asc.)	ROC-AUC $\uparrow$	56.7 $\pm$ 4.5	67.7 $\pm$ 9.0	95.3$\pm$2.3	83.7 $\pm$ 9.7
	FM $\downarrow$	40.4 $\pm$ 6.5	6.6 $\pm$ 4.5	0.0$\pm$0.0	6.5 $\pm$ 2.8
Generalization (desc.)	ROC-AUC $\uparrow$	58.7 $\pm$ 1.1	83.5 $\pm$ 8.0	95.0$\pm$4.5	79.9 $\pm$ 6.8
	FM $\downarrow$	37.2 $\pm$ 3.1	6.7 $\pm$ 2.8	0.0$\pm$0.0	10.2 $\pm$ 1.0
Smooth drift	ROC-AUC $\uparrow$	56.0 $\pm$ 2.5	66.8 $\pm$ 8.2	93.4$\pm$5.7	81.9 $\pm$ 9.9
	FM $\downarrow$	44.8 $\pm$ 5.4	18.2 $\pm$ 2.7	0.0$\pm$0.0	14.7 $\pm$ 6.9
Abrupt drift	ROC-AUC $\uparrow$	59.1 $\pm$ 3.3	67.9 $\pm$ 6.9	95.0$\pm$4.0	79.0 $\pm$ 9.4
	FM $\downarrow$	43.6 $\pm$ 5.0	16.1 $\pm$ 3.2	0.0$\pm$0.0	12.9 $\pm$ 2.7

Table 27: Normalized-PR-AUC (mean $\pm$ std across models) per ordering on *CAD-CICIDS2017*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	Normalized-PR-AUC $\uparrow$	35.1 $\pm$ 5.5	72.8 $\pm$ 14.6	93.3$\pm$5.7	70.7 $\pm$ 9.3
	FM $\downarrow$	70.9 $\pm$ 13.5	16.7 $\pm$ 10.0	0.0$\pm$0.0	18.1 $\pm$ 6.1
Curriculum (desc.)	Normalized-PR-AUC $\uparrow$	35.8 $\pm$ 4.6	41.1 $\pm$ 19.1	79.2$\pm$12.3	54.9 $\pm$ 24.6
	FM $\downarrow$	55.5 $\pm$ 3.2	9.8 $\pm$ 10.5	0.0$\pm$0.0	14.2 $\pm$ 5.7
Generalization (asc.)	Normalized-PR-AUC $\uparrow$	33.8 $\pm$ 7.5	47.3 $\pm$ 16.8	87.1$\pm$4.8	67.3 $\pm$ 17.0
	FM $\downarrow$	54.5 $\pm$ 10.5	9.8 $\pm$ 6.0	0.0$\pm$0.0	11.1 $\pm$ 4.9
Generalization (desc.)	Normalized-PR-AUC $\uparrow$	35.3 $\pm$ 4.3	73.5 $\pm$ 13.0	90.8$\pm$7.4	66.4 $\pm$ 11.3
	FM $\downarrow$	57.7 $\pm$ 3.3	10.9 $\pm$ 5.1	0.0$\pm$0.0	16.9 $\pm$ 1.8
Smooth drift	Normalized-PR-AUC $\uparrow$	32.1 $\pm$ 3.7	46.0 $\pm$ 18.0	85.4$\pm$10.5	65.4 $\pm$ 16.5
	FM $\downarrow$	64.1 $\pm$ 4.9	26.3 $\pm$ 6.9	0.0$\pm$0.0	22.5 $\pm$ 8.9
Abrupt drift	Normalized-PR-AUC $\uparrow$	35.5 $\pm$ 3.9	48.1 $\pm$ 15.1	89.3$\pm$6.6	62.0 $\pm$ 17.5
	FM $\downarrow$	64.9 $\pm$ 5.5	25.3 $\pm$ 7.5	0.0$\pm$0.0	21.4 $\pm$ 6.3

H.2 CAD-CICIDS2018

CAD-CICIDS2018 exhibits a similar but somewhat more variable picture. Stronger strategies still dominate Naive overall, but the margins depend more visibly on the detector and on the ordering, which makes this dataset a useful complement to CICIDS2017. This reinforces the motivation for evaluating multiple principled sequences rather than relying on a single task order. Figures 12 and 13 summarize these effects under both metrics.

H.3 CAD-CICUNSW

The CICUNSW scenario shows one of the clearest separations between naive sequential adaptation and stronger continual references. Across orderings, Naive exhibits both lower final performance and larger forgetting, while MSTE and, in many cases, Replay or Cumulative preserve substantially better performance. This makes CICUNSW particularly useful for stressing the framework under harder transfer and retention conditions. Figures 14 and 15 provide the corresponding ordering-wise summaries.

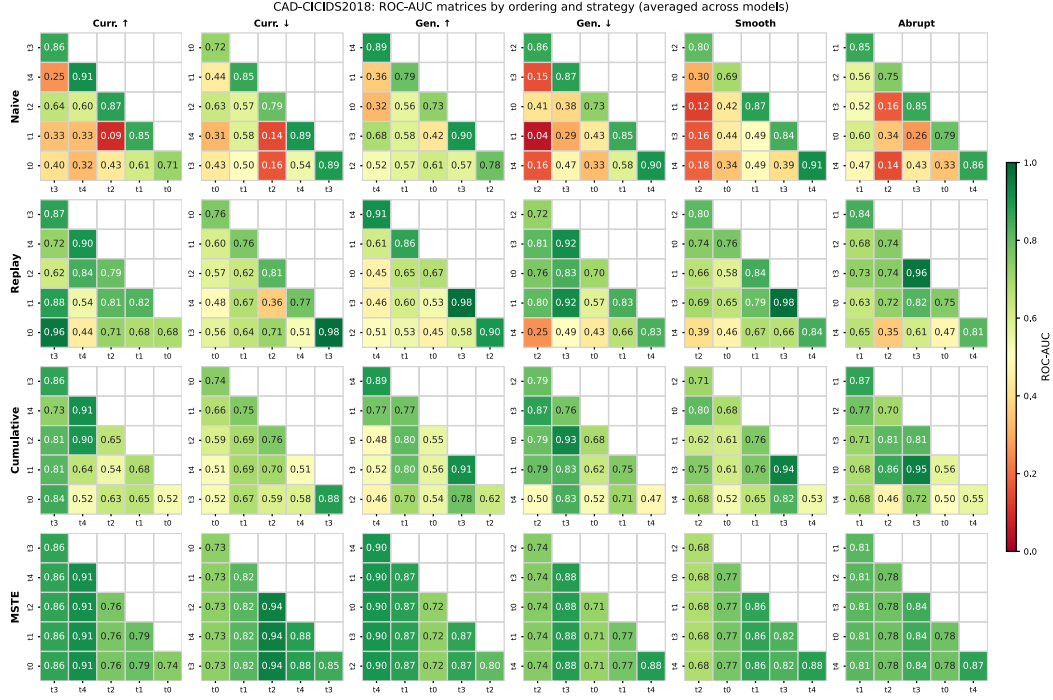


Figure 12: Ordering-specific validation results on CAD-CICIDS2018 measured with ROC-AUC. Compared with CAD-CICIDS2017, the results show somewhat stronger ordering and model sensitivity while preserving the overall advantage of stronger continual references over Naive.

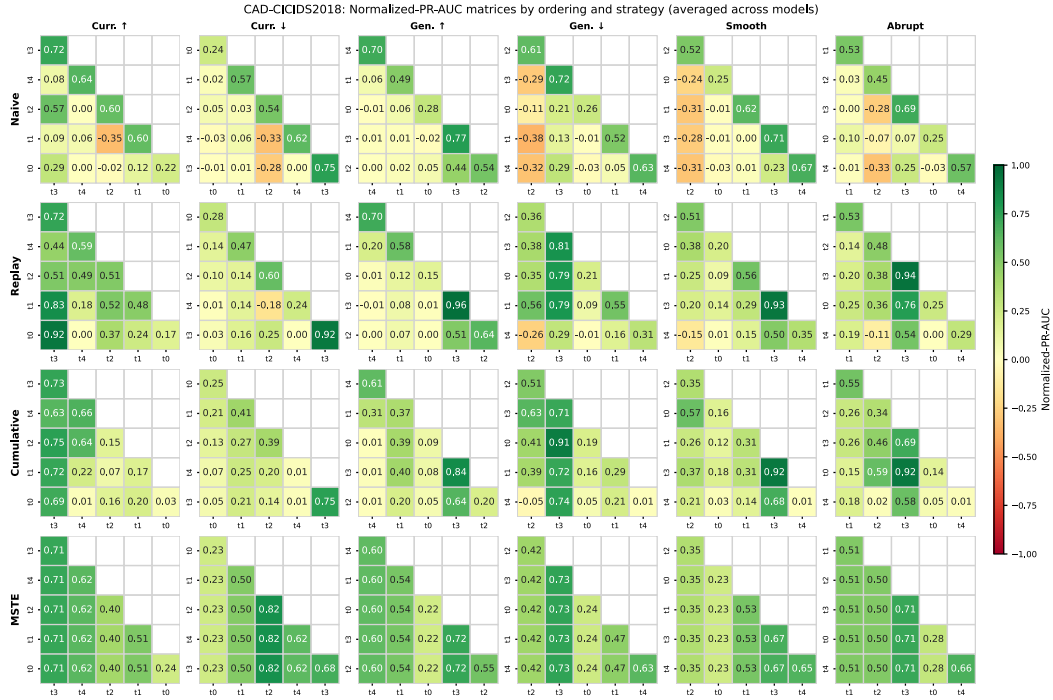


Figure 13: Ordering-specific validation results on CAD-CICIDS2018 measured with normalized PR-AUC. The figure confirms that the scenario remains challenging under imbalance-aware evaluation and that ordering effects remain visible at the task-sequence level.



Figure 14: Ordering-specific validation results on CAD-CICUNSW measured with ROC-AUC. The figure highlights the strong performance gap between Naive and stronger reference strategies across the retained orderings.

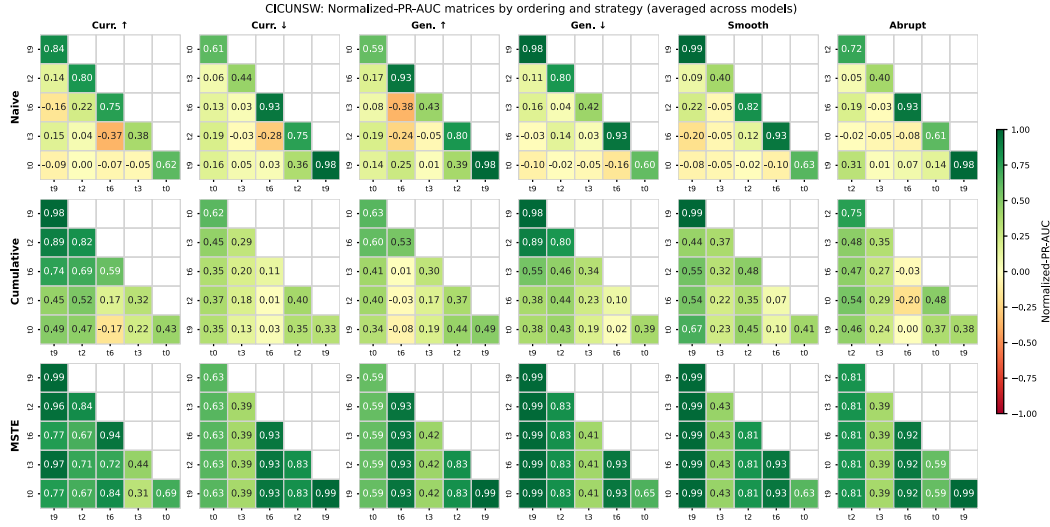


Figure 15: Ordering-specific validation results on CAD-CICUNSW measured with normalized PR-AUC. The same general pattern as in ROC-AUC is preserved, confirming that the scenario remains challenging under imbalance-aware evaluation.

Table 28: ROC-AUC (mean $\pm$ std across models) per ordering on *CAD-CICIDS2018*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	ROC-AUC $\uparrow$	55.2 $\pm$ 4.0	71.8 $\pm$ 12.3	84.2$\pm$9.5	75.5 $\pm$ 6.4
	FM $\downarrow$	39.9 $\pm$ 9.9	12.3 $\pm$ 7.6	0.0$\pm$0.0	15.1 $\pm$ 6.8
Curriculum (desc.)	ROC-AUC $\uparrow$	56.6 $\pm$ 2.1	66.2 $\pm$ 7.3	82.8$\pm$8.3	65.9 $\pm$ 4.4
	FM $\downarrow$	26.8 $\pm$ 5.8	8.8 $\pm$ 3.5	0.0$\pm$0.0	17.8 $\pm$ 6.4
Generalization (asc.)	ROC-AUC $\uparrow$	62.2 $\pm$ 4.9	68.2 $\pm$ 5.9	85.1$\pm$5.8	65.0 $\pm$ 6.0
	FM $\downarrow$	29.4 $\pm$ 3.8	13.2 $\pm$ 1.7	0.0$\pm$0.0	25.7 $\pm$ 4.9
Generalization (desc.)	ROC-AUC $\uparrow$	50.0 $\pm$ 3.6	72.8 $\pm$ 9.2	79.3$\pm$13.7	70.6 $\pm$ 2.3
	FM $\downarrow$	44.4 $\pm$ 14.4	7.2 $\pm$ 2.3	0.0$\pm$0.0	13.2 $\pm$ 5.4
Smooth drift	ROC-AUC $\uparrow$	50.1 $\pm$ 5.0	70.1 $\pm$ 7.3	77.7$\pm$10.5	70.6 $\pm$ 5.8
	FM $\downarrow$	37.4 $\pm$ 14.3	11.3 $\pm$ 5.5	0.0$\pm$0.0	16.0 $\pm$ 2.1
Abrupt drift	ROC-AUC $\uparrow$	53.3 $\pm$ 3.7	71.4 $\pm$ 10.7	81.3$\pm$13.2	70.4 $\pm$ 6.8
	FM $\downarrow$	33.5 $\pm$ 6.5	10.3 $\pm$ 6.1	0.0$\pm$0.0	16.2 $\pm$ 9.7

Table 29: Normalized-PR-AUC (mean $\pm$ std across models) per ordering on *CAD-CICIDS2018*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	Normalized-PR-AUC $\uparrow$	24.5 $\pm$ 5.4	39.3 $\pm$ 17.3	57.3$\pm$20.0	47.1 $\pm$ 14.2
	FM $\downarrow$	49.3 $\pm$ 13.8	17.4 $\pm$ 9.7	0.5$\pm$1.0	23.7 $\pm$ 12.2
Curriculum (desc.)	Normalized-PR-AUC $\uparrow$	15.4 $\pm$ 4.6	22.9 $\pm$ 10.8	50.7$\pm$13.5	22.4 $\pm$ 5.4
	FM $\downarrow$	34.8 $\pm$ 11.0	10.9 $\pm$ 5.6	0.0$\pm$0.0	25.3 $\pm$ 9.3
Generalization (asc.)	Normalized-PR-AUC $\uparrow$	23.3 $\pm$ 6.8	28.5 $\pm$ 10.9	52.8$\pm$15.0	27.2 $\pm$ 5.9
	FM $\downarrow$	43.2 $\pm$ 8.6	22.4 $\pm$ 5.8	0.0$\pm$0.0	39.4 $\pm$ 4.9
Generalization (desc.)	Normalized-PR-AUC $\uparrow$	15.7 $\pm$ 3.7	39.6 $\pm$ 12.4	49.1$\pm$23.1	36.5 $\pm$ 5.8
	FM $\downarrow$	55.3 $\pm$ 21.2	13.6 $\pm$ 6.0	0.0$\pm$0.0	20.1 $\pm$ 10.9
Smooth drift	Normalized-PR-AUC $\uparrow$	12.7 $\pm$ 3.1	31.2 $\pm$ 9.4	42.4$\pm$19.5	29.9 $\pm$ 8.9
	FM $\downarrow$	48.5 $\pm$ 22.2	18.7 $\pm$ 9.5	0.0$\pm$0.0	24.1 $\pm$ 6.2
Abrupt drift	Normalized-PR-AUC $\uparrow$	15.6 $\pm$ 2.3	35.2 $\pm$ 11.3	53.4$\pm$24.2	35.1 $\pm$ 11.2
	FM $\downarrow$	45.9 $\pm$ 16.1	23.6 $\pm$ 6.9	0.9$\pm$2.0	28.4 $\pm$ 15.1

Table 30: ROC-AUC (mean $\pm$ std across models) per ordering on *CAD-CICUNSW*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	ROC-AUC $\uparrow$	48.7 $\pm$ 4.3	74.4 $\pm$ 16.7	92.8$\pm$7.5	70.9 $\pm$ 8.0
	FM $\downarrow$	51.1 $\pm$ 10.1	12.9 $\pm$ 4.8	0.0$\pm$0.0	19.6 $\pm$ 3.0
Curriculum (desc.)	ROC-AUC $\uparrow$	62.8 $\pm$ 4.3	64.5 $\pm$ 15.4	87.5$\pm$10.8	72.8 $\pm$ 19.5
	FM $\downarrow$	32.2 $\pm$ 5.8	10.2 $\pm$ 5.9	0.0$\pm$0.0	13.6 $\pm$ 5.9
Generalization (asc.)	ROC-AUC $\uparrow$	61.1 $\pm$ 4.7	67.3 $\pm$ 12.9	88.1$\pm$9.7	68.4 $\pm$ 20.1
	FM $\downarrow$	34.2 $\pm$ 5.2	11.7 $\pm$ 4.7	0.0$\pm$0.0	15.5 $\pm$ 4.1
Generalization (desc.)	ROC-AUC $\uparrow$	49.5 $\pm$ 3.5	72.3 $\pm$ 14.7	91.0$\pm$9.4	73.3 $\pm$ 10.8
	FM $\downarrow$	49.4 $\pm$ 9.3	12.3 $\pm$ 9.7	0.0$\pm$0.0	18.8 $\pm$ 4.9
Smooth drift	ROC-AUC $\uparrow$	49.3 $\pm$ 4.7	70.8 $\pm$ 18.5	91.6$\pm$8.7	71.9 $\pm$ 15.3
	FM $\downarrow$	48.3 $\pm$ 7.5	16.3 $\pm$ 13.5	0.0$\pm$0.0	19.5 $\pm$ 10.9
Abrupt drift	ROC-AUC $\uparrow$	55.1 $\pm$ 2.3	68.3 $\pm$ 16.2	87.5$\pm$11.9	72.5 $\pm$ 18.2
	FM $\downarrow$	43.5 $\pm$ 13.0	8.1 $\pm$ 5.1	0.0$\pm$0.0	16.0 $\pm$ 2.2

Table 31: Normalized-PR-AUC (mean $\pm$ std across models) per ordering on *CAD-CICUNSW*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	Normalized-PR-AUC $\uparrow$	25.0 $\pm$ 5.7	51.2 $\pm$ 20.0	83.5$\pm$10.7	47.6 $\pm$ 10.4
	FM $\downarrow$	70.5 $\pm$ 12.9	23.2 $\pm$ 7.0	0.0$\pm$0.0	31.7 $\pm$ 5.4
Curriculum (desc.)	Normalized-PR-AUC $\uparrow$	30.0 $\pm$ 9.1	28.3 $\pm$ 14.7	68.0$\pm$15.8	39.5 $\pm$ 20.7
	FM $\downarrow$	45.5 $\pm$ 10.1	14.9 $\pm$ 6.6	0.0$\pm$0.0	23.9 $\pm$ 4.4
Generalization (asc.)	Normalized-PR-AUC $\uparrow$	29.1 $\pm$ 6.9	32.3 $\pm$ 11.6	71.2$\pm$13.9	35.4 $\pm$ 22.7
	FM $\downarrow$	50.4 $\pm$ 10.2	20.2 $\pm$ 7.4	0.0$\pm$0.0	27.9 $\pm$ 5.5
Generalization (desc.)	Normalized-PR-AUC $\uparrow$	26.1 $\pm$ 6.1	44.2 $\pm$ 17.8	80.3$\pm$12.0	48.0 $\pm$ 12.5
	FM $\downarrow$	65.0 $\pm$ 12.4	23.2 $\pm$ 12.8	0.0$\pm$0.0	31.4 $\pm$ 6.4
Smooth drift	Normalized-PR-AUC $\uparrow$	24.8 $\pm$ 3.5	41.7 $\pm$ 22.9	77.7$\pm$11.6	44.3 $\pm$ 20.4
	FM $\downarrow$	63.3 $\pm$ 9.6	24.9 $\pm$ 15.9	0.0$\pm$0.0	31.0 $\pm$ 10.3
Abrupt drift	Normalized-PR-AUC $\uparrow$	28.8 $\pm$ 7.7	32.9 $\pm$ 18.6	70.9$\pm$16.2	39.8 $\pm$ 21.7
	FM $\downarrow$	50.0 $\pm$ 18.0	16.9 $\pm$ 6.2	0.0$\pm$0.0	30.6 $\pm$ 10.4

H.4 CAD-MiniBooNE

Table 32: ROC-AUC (mean $\pm$ std across models) per ordering on *CAD-MiniBooNE*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	ROC-AUC $\uparrow$	53.7 $\pm$ 9.2	62.7 $\pm$ 12.8	78.6$\pm$9.0	55.0 $\pm$ 15.8
	FM $\downarrow$	37.5 $\pm$ 9.0	21.7 $\pm$ 9.9	0.0$\pm$0.0	31.1 $\pm$ 13.2
Curriculum (desc.)	ROC-AUC $\uparrow$	66.7 $\pm$ 3.8	65.6 $\pm$ 1.8	69.2$\pm$12.3	65.0 $\pm$ 1.3
	FM $\downarrow$	14.8 $\pm$ 2.6	8.8 $\pm$ 2.1	0.0$\pm$0.0	14.0 $\pm$ 3.7
Generalization (asc.)	ROC-AUC $\uparrow$	66.1 $\pm$ 3.3	64.9 $\pm$ 2.1	71.6$\pm$10.7	65.3 $\pm$ 0.5
	FM $\downarrow$	14.2 $\pm$ 3.1	7.7 $\pm$ 3.0	0.0$\pm$0.0	13.0 $\pm$ 4.0
Generalization (desc.)	ROC-AUC $\uparrow$	53.0 $\pm$ 10.1	61.6 $\pm$ 15.3	77.3$\pm$10.5	55.2 $\pm$ 16.0
	FM $\downarrow$	39.7 $\pm$ 9.0	19.2 $\pm$ 10.4	0.0$\pm$0.0	30.2 $\pm$ 12.0
Smooth drift	ROC-AUC $\uparrow$	66.8 $\pm$ 3.8	65.6 $\pm$ 1.8	69.9$\pm$11.4	65.1 $\pm$ 1.2
	FM $\downarrow$	14.5 $\pm$ 2.8	8.8 $\pm$ 2.1	0.0$\pm$0.0	13.9 $\pm$ 3.7
Abrupt drift	ROC-AUC $\uparrow$	59.9 $\pm$ 6.4	58.9 $\pm$ 8.0	74.7$\pm$10.0	56.9 $\pm$ 12.4
	FM $\downarrow$	28.7 $\pm$ 2.2	9.1 $\pm$ 4.2	0.0$\pm$0.0	19.3 $\pm$ 6.3

Table 33: Normalized-PR-AUC (mean $\pm$ std across models) per ordering on *CAD-MiniBooNE*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	Normalized-PR-AUC $\uparrow$	18.2 $\pm$ 25.0	34.7 $\pm$ 30.6	65.8$\pm$14.7	17.8 $\pm$ 37.2
	FM $\downarrow$	73.7 $\pm$ 26.0	45.3 $\pm$ 25.9	0.0$\pm$0.0	64.4 $\pm$ 33.1
Curriculum (desc.)	Normalized-PR-AUC $\uparrow$	46.6 $\pm$ 10.4	44.5 $\pm$ 7.4	47.1$\pm$21.5	41.5 $\pm$ 5.3
	FM $\downarrow$	25.4 $\pm$ 10.1	15.6 $\pm$ 3.1	0.3$\pm$0.5	26.9 $\pm$ 9.6
Generalization (asc.)	Normalized-PR-AUC $\uparrow$	45.8 $\pm$ 10.6	44.5 $\pm$ 5.2	52.9$\pm$18.4	42.9 $\pm$ 3.8
	FM $\downarrow$	25.9 $\pm$ 12.1	14.0 $\pm$ 6.5	0.1$\pm$0.1	24.2 $\pm$ 10.4
Generalization (desc.)	Normalized-PR-AUC $\uparrow$	16.5 $\pm$ 26.3	34.2 $\pm$ 34.7	63.6$\pm$18.1	17.0 $\pm$ 37.8
	FM $\downarrow$	76.0 $\pm$ 27.4	38.3 $\pm$ 26.5	0.4$\pm$0.8	57.8 $\pm$ 29.2
Smooth drift	Normalized-PR-AUC $\uparrow$	46.9 $\pm$ 10.4	44.5 $\pm$ 7.5	48.6$\pm$20.0	42.0 $\pm$ 5.5
	FM $\downarrow$	25.5 $\pm$ 10.1	15.6 $\pm$ 3.1	0.3$\pm$0.5	26.5 $\pm$ 9.6
Abrupt drift	Normalized-PR-AUC $\uparrow$	31.2 $\pm$ 15.9	27.8 $\pm$ 20.1	57.3$\pm$17.0	26.8 $\pm$ 26.7
	FM $\downarrow$	51.2 $\pm$ 8.4	18.9 $\pm$ 8.7	0.0$\pm$0.0	33.9 $\pm$ 11.1

CAD-MiniBooNE is difficult already for single-task experts: in Table 2, MSTE reaches 88.6 ROC-AUC with VAE but only 64.3–71.9 with the other models. Per ordering, MSTE is still the best strategy everywhere, but in Curriculum (desc.), Generalization (asc.), and Smooth drift it leads by only 2.5–5.5 points, and Replay gives no consistent gain over Naive. The ordering has a strong effect on forgetting: Curriculum (asc.) and Generalization (desc.) lead to an FM of 37–40 under Naive, while the other orderings stay around 14–29. Improvements on this scenario therefore require better detectors as well as better continual strategies. Figures 16 and 17 show the corresponding task-level matrices.

H.5 CAD-Scania

On CAD-Scania, the models behave very differently: VAE barely forgets, whereas DAGMM and NeutralAD lose much of their performance on earlier tasks. In Table 2, VAE and Deep SVDD forget little under Naive (FM 0.5 and 8.0) yet stay far below MSTE (69.6 vs. 98.8 and 68.8 vs. 89.2), which points to limited plasticity rather than forgetting. Generalization (desc.) is the easiest ordering for Naive (79.6), and Generalization (asc.) and Smooth drift are the hardest. With only three tasks, Curriculum (desc.) and Abrupt drift produce the same task sequence, which is why their results are almost identical. Figures 18 and 19 show the corresponding task-level matrices.

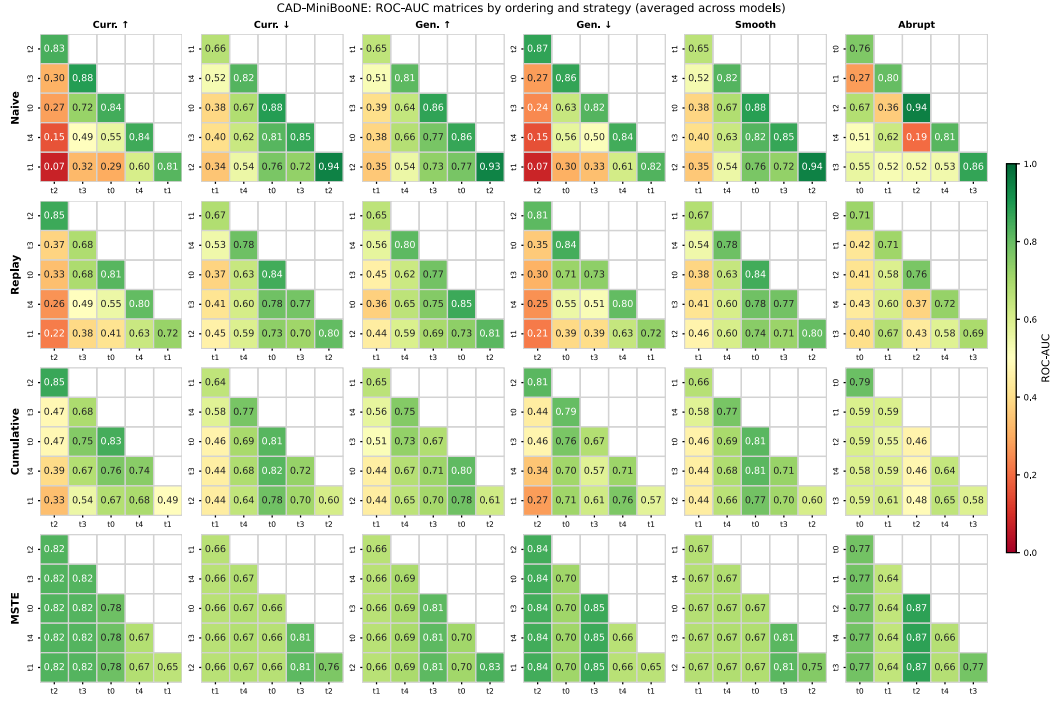


Figure 16: Ordering-specific validation results on CAD-MiniBooNE measured with ROC-AUC. MSTE remains the best strategy in every ordering



Figure 17: Ordering-specific validation results on CAD-MiniBooNE measured with normalized PR-AUC. Normalized PR-AUC accounts for the high anomaly prevalence of this scenario, in which anomalies are the majority of the test data.

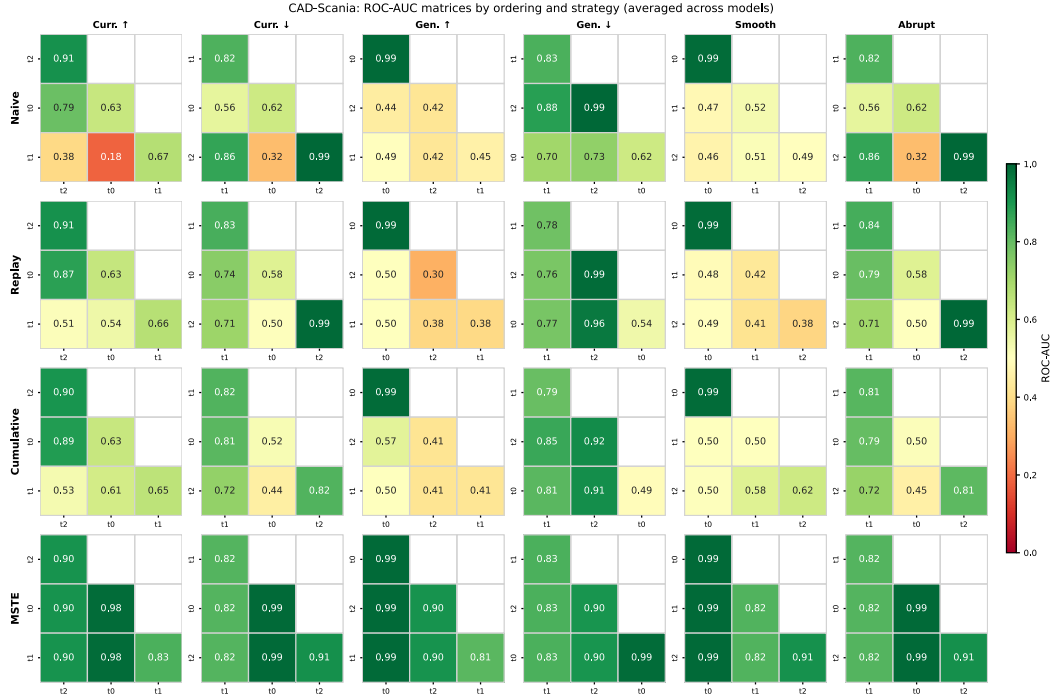


Figure 18: Ordering-specific validation results on CAD-Scania measured with ROC-AUC. With three tasks the matrices are small, but the drop of the sequential strategies on earlier tasks is visible in several orderings.

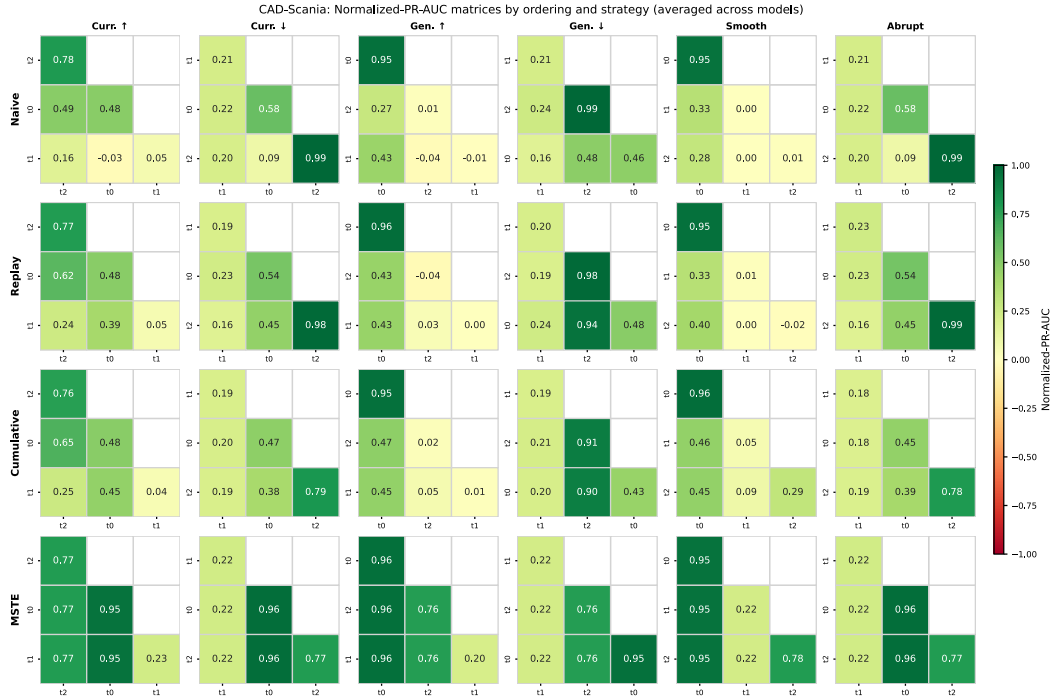


Figure 19: Ordering-specific validation results on CAD-Scania measured with normalized PR-AUC. The ordering-dependent gap between MSTE and the sequential strategies is preserved under imbalance-aware evaluation.

Table 34: ROC-AUC (mean $\pm$ std across models) per ordering on *CAD-Scania*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	ROC-AUC $\uparrow$	59.7 $\pm$ 7.4	70.4 $\pm$ 4.3	92.3$\pm$6.0	69.1 $\pm$ 2.4
	FM $\downarrow$	21.4 $\pm$ 17.8	8.6 $\pm$ 6.0	0.0$\pm$0.0	10.1 $\pm$ 7.6
Curriculum (desc.)	ROC-AUC $\uparrow$	70.2 $\pm$ 4.0	69.5 $\pm$ 16.9	89.5$\pm$6.4	73.0 $\pm$ 19.5
	FM $\downarrow$	14.4 $\pm$ 14.7	3.4 $\pm$ 5.3	0.0$\pm$0.0	6.6 $\pm$ 8.9
Generalization (asc.)	ROC-AUC $\uparrow$	53.9 $\pm$ 16.8	55.3 $\pm$ 12.7	93.4$\pm$4.4	51.3 $\pm$ 16.3
	FM $\downarrow$	28.1 $\pm$ 27.8	22.2 $\pm$ 25.7	0.0$\pm$0.0	25.2 $\pm$ 27.8
Generalization (desc.)	ROC-AUC $\uparrow$	79.6 $\pm$ 10.0	80.0 $\pm$ 14.2	88.4$\pm$7.0	80.3 $\pm$ 13.5
	FM $\downarrow$	9.1 $\pm$ 9.2	1.0 $\pm$ 1.4	0.0$\pm$0.0	4.0 $\pm$ 5.9
Smooth drift	ROC-AUC $\uparrow$	57.9 $\pm$ 16.9	61.8 $\pm$ 18.9	92.4$\pm$4.5	53.3 $\pm$ 14.7
	FM $\downarrow$	26.3 $\pm$ 26.9	24.5 $\pm$ 28.2	0.0$\pm$0.0	25.6 $\pm$ 26.7
Abrupt drift	ROC-AUC $\uparrow$	70.2 $\pm$ 4.0	68.5 $\pm$ 18.3	89.5$\pm$6.4	73.9 $\pm$ 18.1
	FM $\downarrow$	14.4 $\pm$ 14.7	3.5 $\pm$ 5.2	0.0$\pm$0.0	5.2 $\pm$ 6.3

Table 35: Normalized-PR-AUC (mean $\pm$ std across models) per ordering on *CAD-Scania*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	Normalized-PR-AUC $\uparrow$	32.5 $\pm$ 13.4	44.4 $\pm$ 11.0	74.4$\pm$17.6	43.0 $\pm$ 10.4
	FM $\downarrow$	29.7 $\pm$ 20.5	15.5 $\pm$ 13.4	0.0$\pm$0.0	16.5 $\pm$ 13.4
Curriculum (desc.)	Normalized-PR-AUC $\uparrow$	38.6 $\pm$ 5.1	37.7 $\pm$ 13.2	56.6$\pm$15.6	43.2 $\pm$ 11.6
	FM $\downarrow$	11.8 $\pm$ 8.4	2.0 $\pm$ 2.5	0.0$\pm$0.0	4.0 $\pm$ 4.2
Generalization (asc.)	Normalized-PR-AUC $\uparrow$	27.4 $\pm$ 15.8	33.0 $\pm$ 14.5	77.3$\pm$13.2	30.7 $\pm$ 16.6
	FM $\downarrow$	32.8 $\pm$ 23.8	25.3 $\pm$ 25.7	0.0$\pm$0.0	27.4 $\pm$ 26.2
Generalization (desc.)	Normalized-PR-AUC $\uparrow$	42.8 $\pm$ 18.2	47.8 $\pm$ 13.5	52.7$\pm$19.0	51.1 $\pm$ 9.4
	FM $\downarrow$	11.2 $\pm$ 10.7	1.2 $\pm$ 1.3	0.0$\pm$0.0	1.5 $\pm$ 1.6
Smooth drift	Normalized-PR-AUC $\uparrow$	26.6 $\pm$ 16.5	38.7 $\pm$ 24.2	68.1$\pm$13.1	28.4 $\pm$ 15.8
	FM $\downarrow$	31.7 $\pm$ 22.5	25.6 $\pm$ 28.6	0.0$\pm$0.0	30.1 $\pm$ 23.2
Abrupt drift	Normalized-PR-AUC $\uparrow$	38.6 $\pm$ 5.1	36.7 $\pm$ 15.0	56.6$\pm$15.6	43.7 $\pm$ 10.6
	FM $\downarrow$	11.8 $\pm$ 8.4	1.8 $\pm$ 2.7	0.0$\pm$0.0	4.1 $\pm$ 4.4

H.6 CAD-TCM

CAD-TCM shows the clearest continual-learning pressure of the non-cybersecurity scenarios and behaves most like the cybersecurity ones. Naive stays between 48 and 63 ROC-AUC with an FM of 35–52 in every ordering, while MSTE reaches 91–95. Replay and Cumulative recover part of the gap (60–78), and Smooth drift and Curriculum (asc.) are the hardest orderings. Figures 20 and 21 show that Naive loses performance on earlier tasks in every ordering.

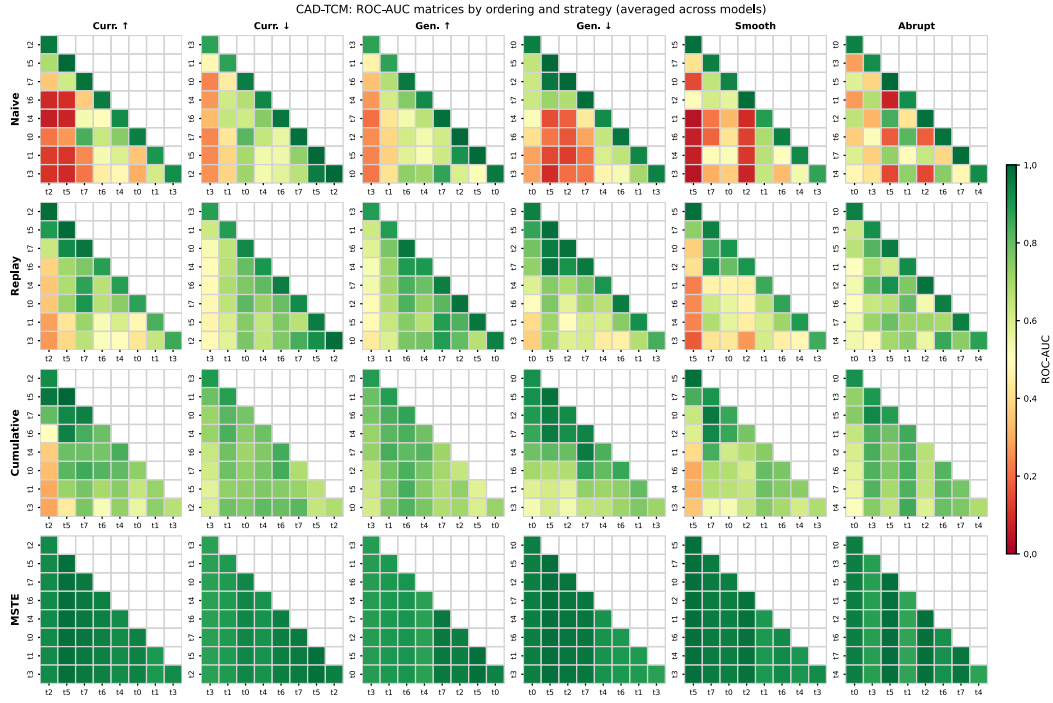


Figure 20: Ordering-specific validation results on CAD-TCM measured with ROC-AUC. Naive degrades on earlier tasks in every ordering, while MSTE keeps high performance on all eight tasks.

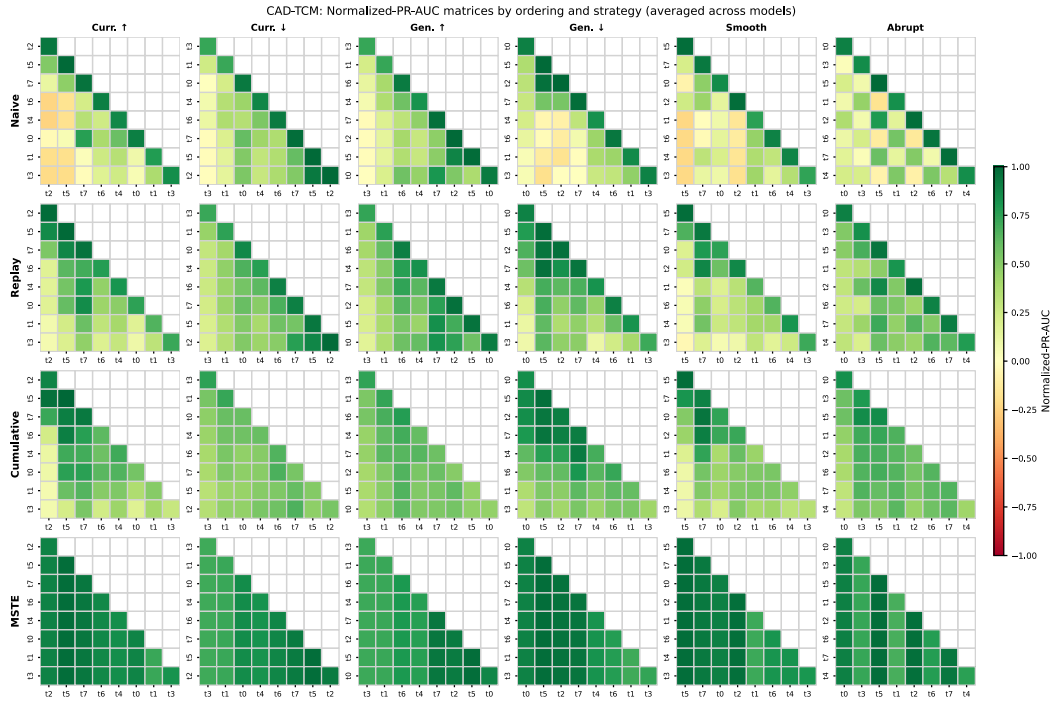


Figure 21: Ordering-specific validation results on CAD-TCM measured with normalized PR-AUC. The same pattern holds under imbalance-aware evaluation.

Table 36: ROC-AUC (mean $\pm$ std across models) per ordering on *CAD-TCM*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	ROC-AUC $\uparrow$	49.5 $\pm$ 3.4	71.9 $\pm$ 14.6	94.6$\pm$6.6	66.4 $\pm$ 13.8
	FM $\downarrow$	49.3 $\pm$ 4.4	16.8 $\pm$ 10.3	0.0$\pm$0.0	26.5 $\pm$ 13.8
Curriculum (desc.)	ROC-AUC $\uparrow$	62.5 $\pm$ 6.4	74.8 $\pm$ 13.1	91.7$\pm$7.4	73.0 $\pm$ 16.4
	FM $\downarrow$	34.9 $\pm$ 2.3	9.3 $\pm$ 5.6	0.0$\pm$0.0	21.7 $\pm$ 13.2
Generalization (asc.)	ROC-AUC $\uparrow$	61.6 $\pm$ 5.6	75.3 $\pm$ 14.7	91.1$\pm$8.1	75.5 $\pm$ 16.9
	FM $\downarrow$	35.0 $\pm$ 2.0	8.2 $\pm$ 5.8	0.0$\pm$0.0	19.2 $\pm$ 12.8
Generalization (desc.)	ROC-AUC $\uparrow$	54.0 $\pm$ 3.9	77.6 $\pm$ 17.1	94.8$\pm$4.9	72.5 $\pm$ 19.2
	FM $\downarrow$	41.5 $\pm$ 3.6	11.4 $\pm$ 8.4	0.0$\pm$0.0	20.3 $\pm$ 14.4
Smooth drift	ROC-AUC $\uparrow$	47.6 $\pm$ 3.6	68.8 $\pm$ 18.3	95.2$\pm$4.3	59.6 $\pm$ 13.0
	FM $\downarrow$	52.2 $\pm$ 7.0	18.8 $\pm$ 12.6	0.0$\pm$0.0	33.4 $\pm$ 15.0
Abrupt drift	ROC-AUC $\uparrow$	56.9 $\pm$ 3.3	75.2 $\pm$ 18.8	93.0$\pm$6.5	73.1 $\pm$ 18.3
	FM $\downarrow$	45.1 $\pm$ 2.0	12.2 $\pm$ 9.9	0.0$\pm$0.0	23.7 $\pm$ 16.4

Table 37: Normalized-PR-AUC (mean $\pm$ std across models) per ordering on *CAD-TCM*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	Normalized-PR-AUC $\uparrow$	29.5 $\pm$ 8.6	54.3 $\pm$ 19.0	89.2$\pm$10.4	50.2 $\pm$ 16.4
	FM $\downarrow$	66.6 $\pm$ 9.0	25.0 $\pm$ 13.8	0.0$\pm$0.0	36.5 $\pm$ 17.2
Curriculum (desc.)	Normalized-PR-AUC $\uparrow$	44.1 $\pm$ 17.0	52.1 $\pm$ 23.0	80.3$\pm$15.1	54.7 $\pm$ 18.3
	FM $\downarrow$	45.2 $\pm$ 7.1	16.3 $\pm$ 9.6	0.0$\pm$0.0	29.2 $\pm$ 7.9
Generalization (asc.)	Normalized-PR-AUC $\uparrow$	44.2 $\pm$ 13.2	55.6 $\pm$ 25.0	79.3$\pm$16.6	59.8 $\pm$ 17.6
	FM $\downarrow$	44.5 $\pm$ 7.2	14.3 $\pm$ 9.6	0.0$\pm$0.0	25.8 $\pm$ 8.2
Generalization (desc.)	Normalized-PR-AUC $\uparrow$	35.0 $\pm$ 12.1	66.3 $\pm$ 22.2	89.0$\pm$9.0	59.1 $\pm$ 22.6
	FM $\downarrow$	58.7 $\pm$ 10.1	15.4 $\pm$ 10.0	0.0$\pm$0.0	28.4 $\pm$ 15.5
Smooth drift	Normalized-PR-AUC $\uparrow$	27.1 $\pm$ 10.2	50.6 $\pm$ 22.3	89.2$\pm$8.1	41.2 $\pm$ 18.7
	FM $\downarrow$	69.7 $\pm$ 13.5	25.9 $\pm$ 14.9	0.0$\pm$0.0	43.6 $\pm$ 18.6
Abrupt drift	Normalized-PR-AUC $\uparrow$	36.2 $\pm$ 10.1	59.7 $\pm$ 25.5	84.1$\pm$12.8	56.3 $\pm$ 21.6
	FM $\downarrow$	63.7 $\pm$ 3.8	17.6 $\pm$ 12.3	0.0$\pm$0.0	34.8 $\pm$ 16.2

H.7 MCAD-CIC-3x1

On MCAD-CIC-3x1, the strategies are much closer to each other than in the other scenarios: all of them stay between 64 and 80 ROC-AUC, and Naive forgets less (FM 9–18). MSTE is the best strategy in four of the six orderings, while Replay is best in Generalization (asc.) and Naive in Abrupt drift. This agrees with the observation that the scenario is difficult already for single-task experts. Curriculum (desc.) and Smooth drift produce the same task sequence here; their slightly different results reflect run-to-run variation. Figures 22 and 23 show that Naive still forgets in several orderings, for example cicunsw drops from 0.79 to 0.44 ROC-AUC in Curriculum (asc.) and cids2017 from 0.82 to 0.41 in Generalization (desc.).

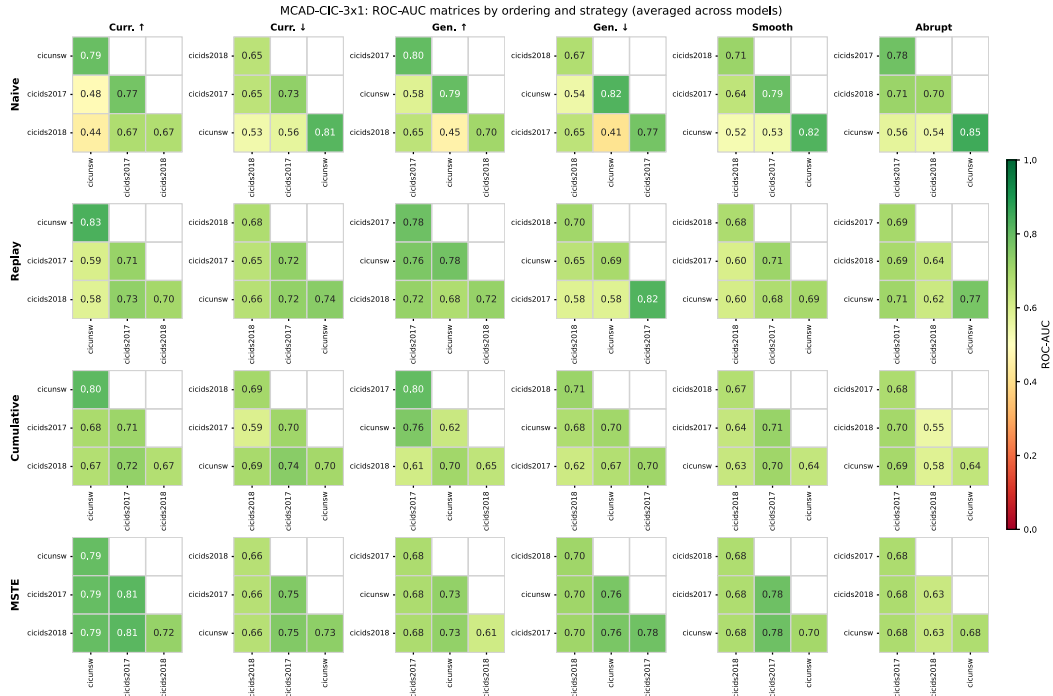


Figure 22: Ordering-specific validation results on MCAD-CIC-3x1 measured with ROC-AUC. Each task corresponds to one source dataset.

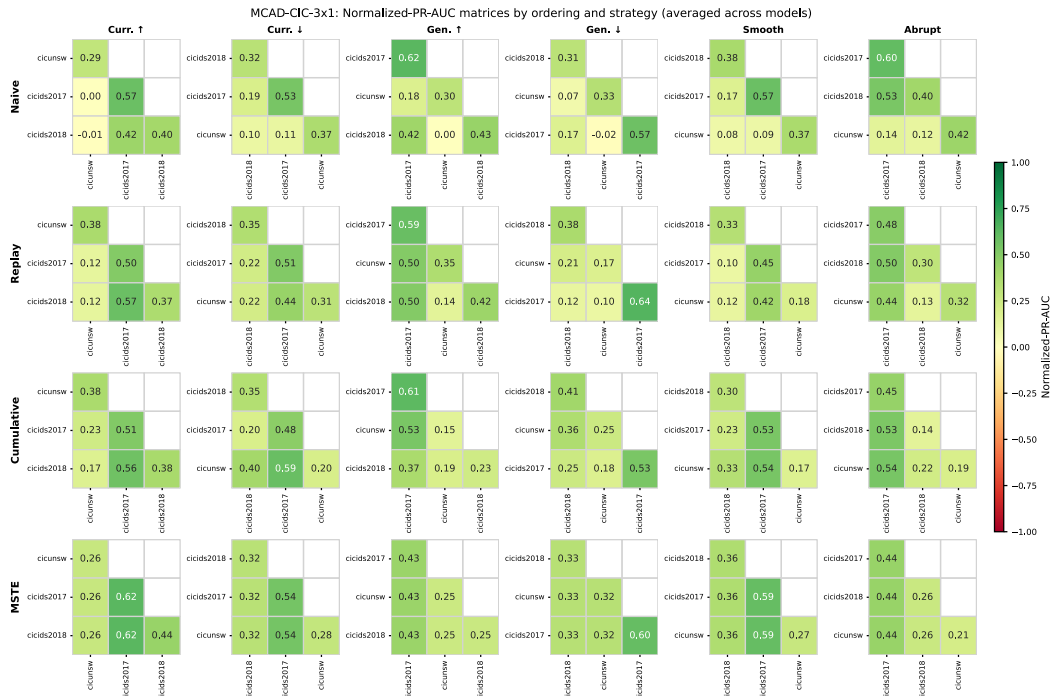


Figure 23: Ordering-specific validation results on MCAD-CIC-3x1 measured with normalized PR-AUC.

Table 38: ROC-AUC (mean $\pm$ std across models) per ordering on *MCAD-CIC-3xI*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	ROC-AUC $\uparrow$	64.3 $\pm$ 6.5	71.5 $\pm$ 9.7	79.1$\pm$9.6	69.4 $\pm$ 9.4
	FM $\downarrow$	17.8 $\pm$ 7.6	7.2 $\pm$ 4.3	0.0$\pm$0.0	12.8 $\pm$ 4.4
Curriculum (desc.)	ROC-AUC $\uparrow$	66.1 $\pm$ 5.3	68.9 $\pm$ 13.0	70.7$\pm$13.8	70.1 $\pm$ 8.3
	FM $\downarrow$	9.3 $\pm$ 7.7	4.7 $\pm$ 2.2	0.0$\pm$0.0	3.0 $\pm$ 4.4
Generalization (asc.)	ROC-AUC $\uparrow$	66.7 $\pm$ 5.2	69.4 $\pm$ 8.0	68.9 $\pm$ 11.0	74.4$\pm$9.4
	FM $\downarrow$	15.4 $\pm$ 9.4	4.7 $\pm$ 3.9	0.0$\pm$0.0	5.3 $\pm$ 2.7
Generalization (desc.)	ROC-AUC $\uparrow$	64.8 $\pm$ 5.3	68.5 $\pm$ 11.0	74.1$\pm$16.4	67.5 $\pm$ 5.1
	FM $\downarrow$	13.7 $\pm$ 9.6	4.7 $\pm$ 5.7	0.0$\pm$0.0	7.7 $\pm$ 6.3
Smooth drift	ROC-AUC $\uparrow$	67.4 $\pm$ 5.5	67.0 $\pm$ 11.0	72.0$\pm$11.0	66.4 $\pm$ 9.8
	FM $\downarrow$	11.6 $\pm$ 9.3	2.6 $\pm$ 1.3	0.0$\pm$0.0	7.2 $\pm$ 6.6
Abrupt drift	ROC-AUC $\uparrow$	69.5$\pm$4.6	64.5 $\pm$ 6.0	66.8 $\pm$ 12.7	69.0 $\pm$ 7.9
	FM $\downarrow$	9.8 $\pm$ 9.3	3.2 $\pm$ 3.6	0.0$\pm$0.0	5.0 $\pm$ 3.5

Table 39: Normalized-PR-AUC (mean $\pm$ std across models) per ordering on *MCAD-CIC-3xI*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	Normalized-PR-AUC $\uparrow$	28.4 $\pm$ 12.2	37.7 $\pm$ 10.1	41.7$\pm$15.1	34.9 $\pm$ 12.0
	FM $\downarrow$	17.7 $\pm$ 12.0	8.7 $\pm$ 5.8	0.0$\pm$0.0	13.7 $\pm$ 4.9
Curriculum (desc.)	Normalized-PR-AUC $\uparrow$	27.4 $\pm$ 4.8	37.3 $\pm$ 18.6	39.2$\pm$23.6	34.7 $\pm$ 12.1
	FM $\downarrow$	19.5 $\pm$ 17.1	6.8 $\pm$ 3.8	0.0$\pm$0.0	9.1 $\pm$ 4.2
Generalization (asc.)	Normalized-PR-AUC $\uparrow$	33.1 $\pm$ 4.3	35.2 $\pm$ 8.9	34.5 $\pm$ 14.2	42.3$\pm$14.1
	FM $\downarrow$	23.2 $\pm$ 20.9	7.1 $\pm$ 7.2	0.0$\pm$0.0	9.0 $\pm$ 5.3
Generalization (desc.)	Normalized-PR-AUC $\uparrow$	24.7 $\pm$ 8.1	33.3 $\pm$ 9.0	37.7$\pm$26.1	27.5 $\pm$ 2.5
	FM $\downarrow$	18.1 $\pm$ 20.4	6.6 $\pm$ 5.5	0.0$\pm$0.0	11.6 $\pm$ 8.0
Smooth drift	Normalized-PR-AUC $\uparrow$	28.3 $\pm$ 6.9	35.7 $\pm$ 14.2	42.6$\pm$14.7	27.1 $\pm$ 14.2
	FM $\downarrow$	21.3 $\pm$ 19.1	3.9 $\pm$ 3.2	0.0$\pm$0.0	15.0 $\pm$ 14.7
Abrupt drift	Normalized-PR-AUC $\uparrow$	37.3$\pm$4.3	34.9 $\pm$ 7.0	34.6 $\pm$ 16.6	36.7 $\pm$ 10.8
	FM $\downarrow$	17.6 $\pm$ 14.3	3.4 $\pm$ 1.6	0.6$\pm$1.2	7.9 $\pm$ 4.9

H.8 MCAD-CIC-3xN

MCAD-CIC-3xN puts the strongest continual learning pressure on Naive of all scenarios: it stays between 43 and 53 ROC-AUC with an FM of 38–56 in every ordering, while MSTE reaches 86–92 and Replay and Cumulative recover part of the gap (66–81). This is consistent with the intended role of multi-dataset scenarios as a stress test for continual anomaly detection. Figures 24 and 25 illustrate this effect.

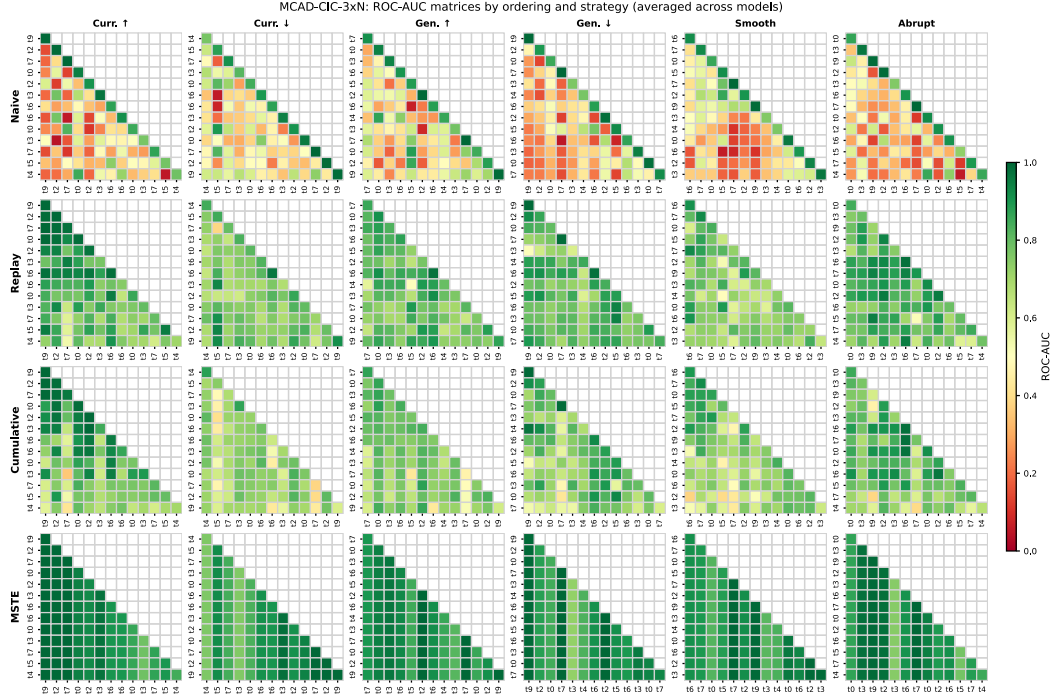


Figure 24: Ordering-specific validation results on MCAD-CIC-3xN measured with ROC-AUC. Compared with the single-dataset scenarios, the multi-dataset setting induces substantially harsher continual-learning conditions.

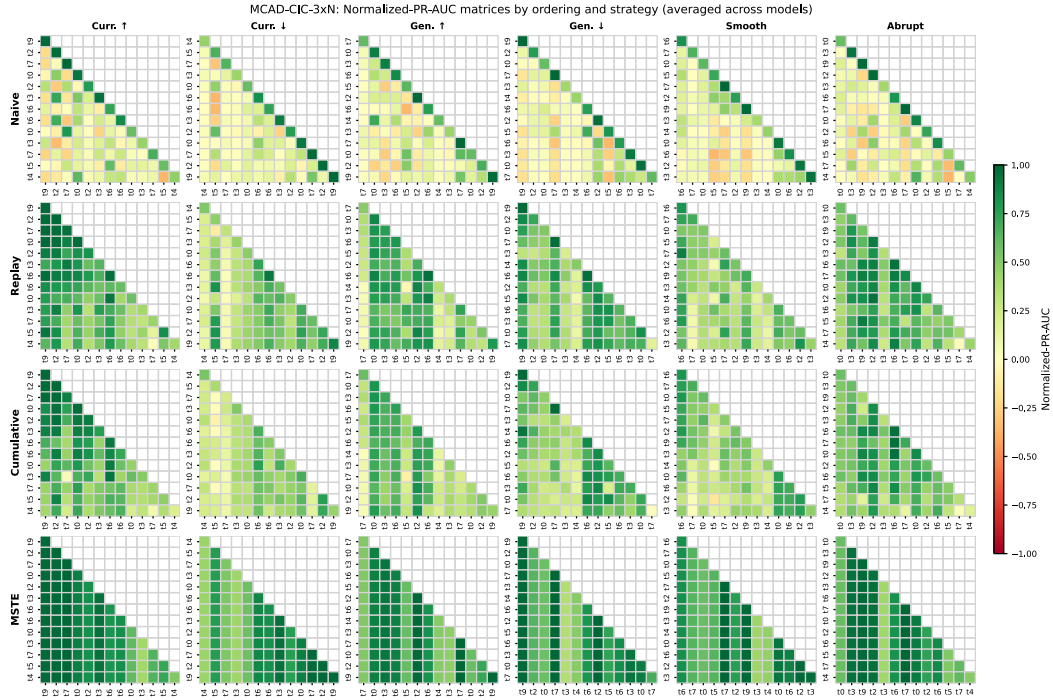


Figure 25: Ordering-specific validation results on MCAD-CIC-3xN measured with normalized PR-AUC. The stronger degradation of Naive persists under imbalance-aware evaluation, reinforcing the difficulty of the multi-dataset setting.

Table 40: ROC-AUC (mean $\pm$ std across models) per ordering on *MCAD-CIC-3xN*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	ROC-AUC $\uparrow$	46.1 $\pm$ 6.1	78.3 $\pm$ 9.4	92.0$\pm$6.5	80.7 $\pm$ 8.0
	FM $\downarrow$	56.2 $\pm$ 10.1	12.0 $\pm$ 7.4	0.0$\pm$0.0	12.0 $\pm$ 5.5
Curriculum (desc.)	ROC-AUC $\uparrow$	50.6 $\pm$ 1.1	66.5 $\pm$ 8.4	87.3$\pm$6.9	70.7 $\pm$ 13.0
	FM $\downarrow$	38.0 $\pm$ 6.1	15.1 $\pm$ 3.4	0.0$\pm$0.0	16.0 $\pm$ 5.4
Generalization (asc.)	ROC-AUC $\uparrow$	53.4 $\pm$ 1.4	70.7 $\pm$ 6.6	88.9$\pm$5.2	78.2 $\pm$ 7.7
	FM $\downarrow$	41.7 $\pm$ 6.7	12.4 $\pm$ 3.8	0.0$\pm$0.0	12.1 $\pm$ 3.4
Generalization (desc.)	ROC-AUC $\uparrow$	42.9 $\pm$ 5.9	71.7 $\pm$ 12.9	88.9$\pm$8.0	75.9 $\pm$ 12.0
	FM $\downarrow$	53.7 $\pm$ 10.1	15.5 $\pm$ 9.0	0.0$\pm$0.0	15.1 $\pm$ 6.9
Smooth drift	ROC-AUC $\uparrow$	46.7 $\pm$ 3.4	68.4 $\pm$ 11.0	86.0$\pm$9.1	74.8 $\pm$ 13.3
	FM $\downarrow$	42.9 $\pm$ 7.3	16.7 $\pm$ 7.4	0.0$\pm$0.0	13.8 $\pm$ 7.2
Abrupt drift	ROC-AUC $\uparrow$	45.2 $\pm$ 5.0	73.7 $\pm$ 11.1	90.5$\pm$7.7	79.9 $\pm$ 8.9
	FM $\downarrow$	48.5 $\pm$ 11.3	12.9 $\pm$ 5.3	0.0$\pm$0.0	12.2 $\pm$ 4.9

Table 41: Normalized-PR-AUC (mean $\pm$ std across models) per ordering on *MCAD-CIC-3xN*. FM $\downarrow$: lower is better. Best strategy per ordering in **bold**.

Ordering	Metric	Naive	Cumulative	MSTE	Replay
Curriculum (asc.)	Normalized-PR-AUC $\uparrow$	19.4 $\pm$ 4.8	61.1 $\pm$ 13.1	83.4$\pm$10.8	64.3 $\pm$ 10.7
	FM $\downarrow$	78.2 $\pm$ 9.6	18.0 $\pm$ 9.8	0.0$\pm$0.1	18.5 $\pm$ 6.7
Curriculum (desc.)	Normalized-PR-AUC $\uparrow$	15.6 $\pm$ 3.8	31.9 $\pm$ 8.1	67.8$\pm$14.5	36.1 $\pm$ 14.9
	FM $\downarrow$	54.2 $\pm$ 8.2	24.2 $\pm$ 3.2	0.8$\pm$1.5	27.9 $\pm$ 6.7
Generalization (asc.)	Normalized-PR-AUC $\uparrow$	19.2 $\pm$ 6.6	45.5 $\pm$ 8.2	73.7$\pm$11.2	54.7 $\pm$ 10.0
	FM $\downarrow$	61.9 $\pm$ 10.3	18.8 $\pm$ 4.0	0.4$\pm$0.9	22.6 $\pm$ 4.5
Generalization (desc.)	Normalized-PR-AUC $\uparrow$	14.7 $\pm$ 7.0	46.8 $\pm$ 16.9	73.1$\pm$13.4	51.1 $\pm$ 14.7
	FM $\downarrow$	70.5 $\pm$ 8.2	24.6 $\pm$ 11.6	0.0$\pm$0.0	25.9 $\pm$ 5.5
Smooth drift	Normalized-PR-AUC $\uparrow$	16.1 $\pm$ 3.2	39.3 $\pm$ 13.1	67.4$\pm$15.4	48.7 $\pm$ 15.7
	FM $\downarrow$	58.1 $\pm$ 6.9	24.3 $\pm$ 7.5	0.9$\pm$1.7	23.0 $\pm$ 6.4
Abrupt drift	Normalized-PR-AUC $\uparrow$	16.5 $\pm$ 5.5	52.6 $\pm$ 14.8	78.2$\pm$12.3	60.8 $\pm$ 11.2
	FM $\downarrow$	67.1 $\pm$ 13.3	17.8 $\pm$ 5.5	0.0$\pm$0.1	19.0 $\pm$ 4.6

I Continual learning methods: details and per-model results

This appendix describes the four continual learning methods evaluated in Section 4.4 and reports their results for each anomaly detection model.

Methods. **EWC** [Kirkpatrick et al., 2017] adds a quadratic penalty that keeps the parameters important for previous tasks, as measured by the Fisher information, close to their earlier values. **LwF** [Li and Hoiem, 2017] keeps a frozen copy of the model from before the current task and adds a distillation term that pulls the outputs of the current model towards those of the copy on the new data. **DER++** [Buzzega et al., 2020] stores past inputs together with the model’s outputs on them in a reservoir buffer; during training, it penalizes deviations from the stored outputs and also trains on the stored inputs with the model’s own loss. **VLAD** [Faber et al., 2023] is a lifelong anomaly detector built on the VAE: it detects distribution changes, keeps a hierarchical memory of past concepts, and retrain the model on a mix of recent and remembered data. Because it is tied to the VAE, VLAD is evaluated only with this model.

Results. Tables 42 and 43 report ROC-AUC and normalized PR-AUC for each scenario and model. Across the 32 scenario–model combinations, DER++ achieves the highest ROC-AUC of the four methods in 17 combinations, EWC in 8, VLAD in 5 (out of the 8 combinations in which it is evaluated), and LwF in 2. We can observe similar patterns for normalized PR-AUC. LwF has the

Table 42: ROC-AUC (mean $\pm$ std across orderings) and Forgetting Measure of the continual learning methods on all eight scenarios, per anomaly detection model. VLAD is defined only for the VAE model (-). FM $\downarrow$: lower is better. Best per row in **bold**.

Dataset	Model	EWC		LwF		DER++		VLAD	
		ROC $\uparrow$	FM $\downarrow$	ROC $\uparrow$	FM $\downarrow$	ROC $\uparrow$	FM $\downarrow$	ROC $\uparrow$	FM $\downarrow$
CAD-CICIDS2017	VAE	63.2 $\pm$ 8.0	21.8 $\pm$ 5.3	59.6 $\pm$ 4.8	33.8 $\pm$ 10.3	65.0 $\pm$ 13.1	9.3 $\pm$ 4.1	80.6$\pm$4.7	6.5$\pm$2.9
	Deep SVDD	73.7$\pm$8.7	9.5$\pm$4.4	55.1 $\pm$ 4.9	26.4 $\pm$ 12.2	67.8 $\pm$ 9.0	26.8 $\pm$ 12.6	–	–
	DAGMM	57.7 $\pm$ 4.8	36.4$\pm$9.9	58.5$\pm$3.8	37.4 $\pm$ 10.7	58.5 $\pm$ 4.1	37.2 $\pm$ 10.5	–	–
	NeutralAD	77.0 $\pm$ 12.5	2.0$\pm$2.0	67.9 $\pm$ 13.3	36.0 $\pm$ 17.6	85.8$\pm$10.6	14.1 $\pm$ 14.6	–	–
CAD-CICIDS2018	VAE	54.8 $\pm$ 3.5	24.4 $\pm$ 5.6	52.7 $\pm$ 7.1	23.0 $\pm$ 9.5	59.9 $\pm$ 7.5	12.1$\pm$4.9	61.2$\pm$8.4	16.1 $\pm$ 8.2
	Deep SVDD	57.4 $\pm$ 8.5	13.4$\pm$5.6	50.8 $\pm$ 2.2	18.8 $\pm$ 9.9	59.8$\pm$8.5	19.4 $\pm$ 7.4	–	–
	DAGMM	56.4 $\pm$ 3.7	28.4 $\pm$ 9.4	59.6 $\pm$ 3.3	26.2 $\pm$ 5.6	59.7$\pm$2.4	23.7$\pm$7.0	–	–
	NeutralAD	52.4 $\pm$ 8.5	7.0$\pm$8.6	48.4 $\pm$ 3.5	31.8 $\pm$ 12.7	70.9$\pm$6.6	7.3 $\pm$ 6.9	–	–
CAD-CICUNSW	VAE	67.7 $\pm$ 1.0	30.1 $\pm$ 3.0	76.8 $\pm$ 10.7	15.0 $\pm$ 17.6	78.5 $\pm$ 9.3	4.4 $\pm$ 5.7	84.4$\pm$5.8	4.4$\pm$4.2
	Deep SVDD	75.7$\pm$6.3	9.0$\pm$3.8	44.9 $\pm$ 11.7	29.1 $\pm$ 11.1	59.9 $\pm$ 13.8	36.8 $\pm$ 19.4	–	–
	DAGMM	57.2 $\pm$ 9.9	37.6$\pm$15.4	53.4 $\pm$ 7.9	43.8 $\pm$ 17.3	58.3$\pm$9.6	39.5 $\pm$ 14.4	–	–
	NeutralAD	72.5 $\pm$ 13.1	0.7$\pm$0.4	59.5 $\pm$ 11.3	42.3 $\pm$ 14.5	83.2$\pm$7.8	15.5 $\pm$ 8.3	–	–
CAD-MiniBooNE	VAE	70.8 $\pm$ 1.3	18.5 $\pm$ 7.9	68.8 $\pm$ 2.4	20.4 $\pm$ 8.2	71.6$\pm$4.8	8.4$\pm$3.9	68.7 $\pm$ 7.7	9.7 $\pm$ 2.4
	Deep SVDD	57.9 $\pm$ 11.6	1.2$\pm$1.0	59.4 $\pm$ 8.7	20.8 $\pm$ 13.6	62.3$\pm$8.1	19.4 $\pm$ 11.9	–	–
	DAGMM	62.2 $\pm$ 10.9	11.7$\pm$5.9	62.0 $\pm$ 5.8	16.9 $\pm$ 6.8	63.6$\pm$7.8	12.4 $\pm$ 8.4	–	–
	NeutralAD	51.9 $\pm$ 5.7	0.4$\pm$0.4	57.8$\pm$14.0	19.4 $\pm$ 11.1	55.4 $\pm$ 12.1	22.3 $\pm$ 10.3	–	–
CAD-Scania	VAE	89.4$\pm$8.8	2.7 $\pm$ 3.6	80.5 $\pm$ 14.6	3.8 $\pm$ 5.9	81.1 $\pm$ 13.8	0.6 $\pm$ 0.5	81.1 $\pm$ 12.9	0.5$\pm$0.8
	Deep SVDD	67.4$\pm$10.0	6.5$\pm$4.6	62.5 $\pm$ 7.3	21.2 $\pm$ 17.0	65.4 $\pm$ 6.9	12.4 $\pm$ 10.0	–	–
	DAGMM	58.2$\pm$14.6	19.6$\pm$21.8	48.8 $\pm$ 5.9	27.3 $\pm$ 17.1	44.0 $\pm$ 3.0	24.6 $\pm$ 17.3	–	–
	NeutralAD	68.2$\pm$10.7	1.0$\pm$0.7	58.0 $\pm$ 17.9	26.1 $\pm$ 21.5	66.5 $\pm$ 19.4	22.0 $\pm$ 24.2	–	–
CAD-TCM	VAE	64.1 $\pm$ 4.7	35.6 $\pm$ 5.4	64.0 $\pm$ 7.3	32.4 $\pm$ 15.8	85.0 $\pm$ 7.0	6.0 $\pm$ 3.2	89.9$\pm$8.2	2.2$\pm$1.0
	Deep SVDD	64.2$\pm$4.6	21.0$\pm$7.0	54.2 $\pm$ 5.7	38.9 $\pm$ 8.5	63.3 $\pm$ 4.0	27.0 $\pm$ 4.3	–	–
	DAGMM	55.2 $\pm$ 4.6	22.1$\pm$6.0	57.6 $\pm$ 6.4	27.7 $\pm$ 5.3	57.8$\pm$1.8	25.3 $\pm$ 3.6	–	–
	NeutralAD	64.7 $\pm$ 6.8	2.4$\pm$2.8	57.1 $\pm$ 8.4	43.5 $\pm$ 9.5	81.4$\pm$5.8	15.8 $\pm$ 5.9	–	–
MCAD-CIC-3x1	VAE	72.6 $\pm$ 3.3	13.7 $\pm$ 4.0	72.8 $\pm$ 3.9	17.3 $\pm$ 2.9	76.9$\pm$3.2	0.7$\pm$0.5	75.8 $\pm$ 3.9	4.6 $\pm$ 2.5
	Deep SVDD	62.4 $\pm$ 3.0	2.4$\pm$1.9	57.2 $\pm$ 4.2	8.8 $\pm$ 8.5	62.4$\pm$4.0	10.1 $\pm$ 7.5	–	–
	DAGMM	53.0 $\pm$ 1.5	5.2$\pm$4.2	55.8 $\pm$ 4.8	9.3 $\pm$ 4.8	57.4$\pm$3.1	7.0 $\pm$ 4.4	–	–
	NeutralAD	72.4 $\pm$ 8.4	1.9$\pm$2.6	65.1 $\pm$ 4.5	16.4 $\pm$ 9.7	86.1$\pm$2.4	3.2 $\pm$ 0.9	–	–
MCAD-CIC-3xN	VAE	51.6 $\pm$ 6.1	34.6 $\pm$ 3.4	55.5 $\pm$ 16.8	24.7 $\pm$ 20.7	67.6 $\pm$ 16.5	5.6$\pm$4.8	72.8$\pm$13.3	9.8 $\pm$ 4.3
	Deep SVDD	61.9 $\pm$ 5.7	14.7$\pm$8.8	49.5 $\pm$ 4.2	22.0 $\pm$ 12.2	62.1$\pm$10.1	23.4 $\pm$ 11.2	–	–
	DAGMM	52.2 $\pm$ 6.1	40.1$\pm$7.8	53.1 $\pm$ 4.0	40.5 $\pm$ 6.4	53.7$\pm$5.7	40.8 $\pm$ 7.3	–	–
	NeutralAD	63.3 $\pm$ 12.8	4.0$\pm$3.2	50.8 $\pm$ 6.6	41.0 $\pm$ 10.6	81.2$\pm$11.8	9.4 $\pm$ 5.8	–	–

lowest average ROC-AUC (58.7) and the highest average FM (26.3). EWC has the lowest FM in 23 of the 32 combinations, but low forgetting does not always come with good detection: on CAD-MiniBooNE, NeutralAD with EWC has an FM of only 0.4 but a ROC-AUC of 51.9, which indicates that the model barely adapts to new tasks rather than retaining what it has learned. No method is best across all scenarios and models.

Table 43: Normalized PR-AUC (mean $\pm$ std across orderings) and Forgetting Measure of the continual learning methods on all eight scenarios, per anomaly detection model. VLAD is defined only for the VAE model (–). FM $\downarrow$: lower is better. Best per row in **bold**.

Dataset	Model	EWC		LwF		DER++		VLAD	
		N-PR-AUC $\uparrow$	FM $\downarrow$	N-PR-AUC $\uparrow$	FM $\downarrow$	N-PR-AUC $\uparrow$	FM $\downarrow$	N-PR-AUC $\uparrow$	FM $\downarrow$
CAD-CICIDS2017	VAE	40.0 $\pm$ 10.4	32.5 $\pm$ 8.0	38.2 $\pm$ 6.2	47.2 $\pm$ 11.9	40.6 $\pm$ 19.4	16.6 $\pm$ 6.2	58.2$\pm$13.1	12.2$\pm$3.3
	Deep SVDD	52.8$\pm$18.1	13.3$\pm$6.7	24.0 $\pm$ 12.8	36.0 $\pm$ 14.6	45.4 $\pm$ 16.3	38.2 $\pm$ 16.6	–	–
	DAGMM	25.3 $\pm$ 5.4	58.0 $\pm$ 17.0	29.0$\pm$6.5	57.3 $\pm$ 21.9	28.5 $\pm$ 6.4	56.7$\pm$18.3	–	–
	NeutralAD	59.3 $\pm$ 22.7	4.6$\pm$4.9	53.5 $\pm$ 19.7	47.2 $\pm$ 23.9	73.9$\pm$16.2	23.5 $\pm$ 20.2	–	–
CAD-CICIDS2018	VAE	16.2 $\pm$ 8.6	20.7 $\pm$ 7.4	15.0 $\pm$ 11.1	24.2 $\pm$ 9.6	20.4 $\pm$ 6.3	16.5$\pm$8.0	21.3$\pm$4.9	24.1 $\pm$ 10.3
	Deep SVDD	17.3 $\pm$ 12.4	15.2$\pm$10.2	9.2 $\pm$ 6.9	22.8 $\pm$ 11.0	20.1$\pm$7.8	26.1 $\pm$ 10.6	–	–
	DAGMM	12.7 $\pm$ 3.4	33.9$\pm$12.8	16.8 $\pm$ 5.8	37.0 $\pm$ 14.5	17.2$\pm$3.3	34.6 $\pm$ 10.5	–	–
	NeutralAD	19.4 $\pm$ 14.1	11.1$\pm$11.9	10.9 $\pm$ 6.3	41.3 $\pm$ 13.5	34.7$\pm$10.7	11.9 $\pm$ 8.1	–	–
CAD-CICUNSW	VAE	32.5 $\pm$ 5.1	39.7 $\pm$ 10.3	51.1 $\pm$ 19.3	20.6 $\pm$ 23.6	49.9 $\pm$ 20.5	7.1$\pm$8.2	59.6$\pm$14.8	8.6 $\pm$ 7.8
	Deep SVDD	42.2$\pm$4.5	10.9$\pm$5.2	17.2 $\pm$ 10.5	34.2 $\pm$ 12.3	27.8 $\pm$ 21.2	47.0 $\pm$ 26.9	–	–
	DAGMM	20.3 $\pm$ 8.6	55.8 $\pm$ 19.7	19.0 $\pm$ 5.8	56.3 $\pm$ 19.6	24.9$\pm$9.7	54.7$\pm$18.0	–	–
	NeutralAD	48.6 $\pm$ 16.6	1.3$\pm$0.6	30.4 $\pm$ 11.3	61.0 $\pm$ 17.7	53.1$\pm$13.0	40.8 $\pm$ 14.9	–	–
CAD-MiniBooNE	VAE	53.8$\pm$3.7	28.8 $\pm$ 14.5	49.8 $\pm$ 6.3	32.2 $\pm$ 15.8	48.8 $\pm$ 12.2	16.3$\pm$4.7	43.5 $\pm$ 18.9	19.2 $\pm$ 3.5
	Deep SVDD	24.4 $\pm$ 31.1	2.8$\pm$1.7	32.6 $\pm$ 17.6	39.8 $\pm$ 29.5	38.0$\pm$18.0	38.5 $\pm$ 27.0	–	–
	DAGMM	28.8 $\pm$ 23.0	25.4 $\pm$ 12.5	31.6 $\pm$ 16.3	35.1 $\pm$ 16.5	38.0$\pm$19.0	24.8$\pm$17.1	–	–
	NeutralAD	11.2 $\pm$ 8.5	3.2$\pm$1.3	24.7$\pm$32.2	39.8 $\pm$ 27.7	18.4 $\pm$ 28.8	45.4 $\pm$ 26.0	–	–
CAD-Scania	VAE	57.5$\pm$5.0	5.3 $\pm$ 7.1	51.7 $\pm$ 7.1	6.5 $\pm$ 9.2	49.9 $\pm$ 3.7	1.6 $\pm$ 1.0	50.4 $\pm$ 4.1	1.2$\pm$1.7
	Deep SVDD	29.9 $\pm$ 15.7	10.6$\pm$12.8	29.5 $\pm$ 3.4	24.8 $\pm$ 18.0	31.0$\pm$9.5	17.1 $\pm$ 14.5	–	–
	DAGMM	17.4$\pm$6.7	19.0$\pm$21.3	12.1 $\pm$ 3.4	22.8 $\pm$ 19.4	8.0 $\pm$ 6.1	19.9 $\pm$ 19.0	–	–
	NeutralAD	29.0$\pm$14.0	6.5$\pm$6.9	22.7 $\pm$ 12.4	27.8 $\pm$ 22.5	29.0 $\pm$ 14.5	25.4 $\pm$ 26.4	–	–
CAD-TCM	VAE	47.1 $\pm$ 5.0	49.6 $\pm$ 7.1	51.6 $\pm$ 8.7	41.0 $\pm$ 19.5	68.7 $\pm$ 6.0	13.9 $\pm$ 7.9	77.9$\pm$8.4	6.1$\pm$2.9
	Deep SVDD	43.0 $\pm$ 6.9	30.1$\pm$9.0	38.6 $\pm$ 8.4	46.0 $\pm$ 12.2	43.0$\pm$6.6	38.5 $\pm$ 6.8	–	–
	DAGMM	28.3 $\pm$ 7.9	25.2$\pm$9.4	37.6 $\pm$ 7.3	33.1 $\pm$ 8.8	38.7$\pm$3.6	30.6 $\pm$ 6.7	–	–
	NeutralAD	39.8 $\pm$ 10.8	4.3$\pm$3.7	38.3 $\pm$ 9.1	57.3 $\pm$ 17.6	66.1$\pm$9.3	26.8 $\pm$ 8.8	–	–
MCAD-CIC-3x1	VAE	27.8 $\pm$ 4.7	21.8 $\pm$ 7.2	31.6 $\pm$ 6.6	26.8 $\pm$ 3.5	32.2$\pm$7.3	2.2$\pm$1.3	31.3 $\pm$ 6.6	8.1 $\pm$ 3.9
	Deep SVDD	16.5 $\pm$ 3.3	1.6$\pm$1.2	11.3 $\pm$ 7.8	10.1 $\pm$ 9.9	16.6$\pm$6.9	14.1 $\pm$ 11.0	–	–
	DAGMM	8.6$\pm$3.6	4.7$\pm$2.9	7.2 $\pm$ 1.7	9.7 $\pm$ 4.2	7.6 $\pm$ 4.1	7.4 $\pm$ 4.0	–	–
	NeutralAD	33.3 $\pm$ 15.5	5.1$\pm$8.7	23.6 $\pm$ 10.1	26.2 $\pm$ 15.0	58.3$\pm$5.6	11.7 $\pm$ 2.4	–	–
MCAD-CIC-3xN	VAE	19.7 $\pm$ 6.9	43.9 $\pm$ 1.5	28.4 $\pm$ 19.7	31.7 $\pm$ 23.7	41.7 $\pm$ 19.8	10.8$\pm$6.9	46.1$\pm$17.8	16.3 $\pm$ 6.7
	Deep SVDD	25.6 $\pm$ 10.5	21.6$\pm$10.7	9.3 $\pm$ 6.9	32.1 $\pm$ 10.9	30.9$\pm$14.9	32.5 $\pm$ 12.8	–	–
	DAGMM	12.4 $\pm$ 6.7	59.4$\pm$13.8	13.9 $\pm$ 2.7	61.2 $\pm$ 9.5	16.3$\pm$6.3	60.5 $\pm$ 12.2	–	–
	NeutralAD	36.5 $\pm$ 17.5	9.4$\pm$6.8	24.3 $\pm$ 8.3	54.5 $\pm$ 11.6	56.5$\pm$15.8	21.5 $\pm$ 11.4	–	–

J Additional results discussion

J.1 Why can Cumulative underperform?

The CICIDS2017 scenario provides a concrete example of why full access to past data does not necessarily make Cumulative the strongest continual reference. The cumulative strategy retrains on the union of all data observed so far. While this removes forgetting in principle, it also means that the effective optimization objective is heavily shaped by the empirical distribution of the retained tasks. If that distribution is strongly imbalanced, the model can become disproportionately tuned to a small number of dominant concepts, at the expense of smaller ones.

The per-concept statistics (see Table 10) show that this imbalance is substantial in the final CICIDS2017 scenario. In particular, concept `cicids2017_1` contains 1,434,082 training samples and 411,738 test samples, whereas the smallest concepts contain only 1,713 or 8,392 training samples. Aggregated over the final task set, `cicids2017_1` alone contributes roughly 88% of the training data and about 81% of the test data. As a result, cumulative retraining without explicit task balancing or reweighting is dominated by the largest concept, while the optimization signal associated with smaller concepts becomes comparatively weak. Similar observations can be drawn for other datasets, as shown in Tables 12 and 14.

This matters because our evaluation is scenario-oriented rather than distribution-oriented: we care about performance across all tasks, not only about performance on the dominant regime. A detector that improves strongly on the largest concept can still yield disappointing average task-level performance if it underfits smaller concepts or fails to preserve their specific decision boundaries. In this sense, more data is not automatically better when that data is concentrated in a single regime. The additional samples can bias the learned representation toward the dominant concept instead of improving balanced generalization across the scenario.

This observation also clarifies why MSTE and Replay can behave more favorably. MSTE is unaffected by cross-task imbalance because each expert is trained independently on a single task. Replay, while still imperfect, partially limits domination by the largest concept because the replay buffer caps the amount of data that any one task can contribute during training. Cumulative, by contrast, inherits the raw imbalance of the scenario directly.

The broader implication is methodological. In CAD, cumulative should not always be interpreted as a clean upper bound solely because it has access to all past data. Its behavior also depends on how the scenario distributes samples across tasks. This is precisely why explicit scenario characterization is necessary: baseline performance is shaped not only by the continual strategy itself, but also by structural properties such as task size imbalance.

K Per-Dataset Scenario Construction Reports

This appendix provides a dataset-level trace of how each final benchmark scenario was constructed using our framework. For each dataset, we report the intermediate outputs produced at successive stages of the pipeline, including the generation of candidate tasks, the results of the filtering criteria, and the scenario task evaluation (STE) scores obtained across different candidate task configurations. These reports complement the aggregate description in the main paper by making the scenario-construction process transparent, auditable, and reproducible for each dataset individually.

For clarity, task identifiers used during framework execution may differ from the human-readable identifiers reported in the final scenario tables, as we rename them to keep the consistent incremental task ids (0, 1, 2)

K.1 CAD-CICIDS2017

Table 44: Scenario CAD-CICIDS2017: Task filtering results per task discovery configuration. BC = BothClasses, CA = Closest Anomalies, RA = Random Anomalies, GM = Gaussian Mixture, KM = K-Means, SC = Spectral Clustering. By Day = Split by day/part of the day

Config	IT	FC1	FC2	FC3	FC4	FC5	FT	MT	Final tasks.
By Day	7	t1	t1, t2, t3, t4	t1, t3, t4	t0, t1, t2, t3, t4, t5	t0, t2, t3, t4	1	✗	—
BC / GM	7	—	t4	—	—	—	6	✓	t0, t1, t2, t3, t5, t6
BC / KM	7	—	t1, t5	t1, t5	—	—	5	✓	t0, t2, t3, t4, t6
BC / SC	10	—	t1, t4, t6, t7	t0, t7	t1, t6, t7	—	5	✓	t2, t3, t5, t8, t9
CA / GM	8	t2, t6	t0, t2, t5, t6	t5	—	—	4	✓	t1, t3, t4, t7
CA / KM	8	—	t0, t2, t4	t4	t0, t4	—	5	✓	t1, t3, t5, t6, t7
CA / SC	10	t2	t0, t1, t2, t3, t8	t0, t1, t2, t3, t6	t1, t6	—	4	✓	t4, t5, t7, t9
RA / GM	9	—	—	t4	t7	—	7	✓	t0, t1, t2, t3, t5, t6, t8
RA / KM	8	—	—	t5	t4	t4	6	✓	t0, t1, t2, t3, t6, t7
RA / SC	10	—	t1	t6	t1, t4, t5, t6, t8, t9	t1, t5, t6	4	✓	t0, t2, t3, t7

The CICIDS2017 construction report shows that the final split is not the outcome of a single heuristic, but of sequential filtering based on learnability, transfer coverage, and redundancy. The filtering table identifies which candidate task configurations are rejected and why, while Figures 26 and 27 visualize the corresponding STE transfer structure. The retained split exhibits the intended pattern of strong self-performance together with non-trivial off-diagonal variation, indicating that tasks are both learnable and sufficiently heterogeneous.

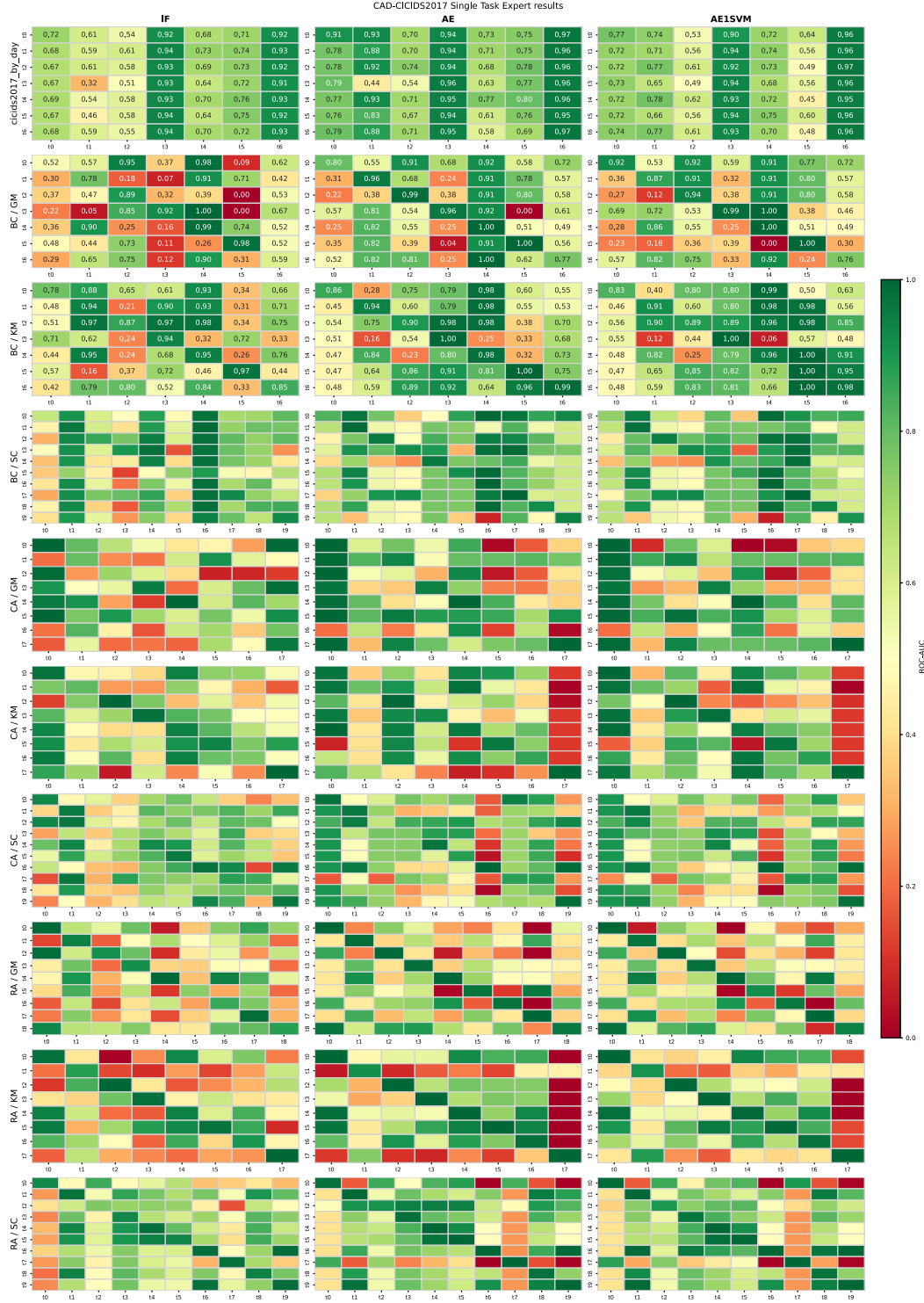


Figure 26: STE evaluation heatmaps for the CICIDS2017 candidate splits measured with ROC-AUC. Strong diagonal values indicate self-learnability, while the off-diagonal structure reveals transfer, dominance, and redundancy patterns used during filtering.



Figure 27: STE evaluation heatmaps for the CICIDS2017 candidate splits measured with normalized PR-AUC. The same split-level structure remains visible under an imbalance-aware metric, supporting the robustness of the filtering decisions.

K.2 CAD-CICIDS2018

Table 45: Scenario CAD-CICIDS2018: Task filtering results per task discovery configuration. BC = BothClasses, CA = Closest Anomalies, RA = Random Anomalies, GM = Gaussian Mixture, KM = K-Means, SC = Spectral Clustering. By Day = Split by day/part of the day

Config	IT	FC1	FC2	FC3	FC4	FC5	FT	MT	Final tasks.
By Day	10	t4, t9	t2, t4, t5, t6, t9	t0, t2, t4, t5, t6	t0, t2, t4, t5, t6, t7, t9	t2, t4, t5	3	✓	t1, t3, t8
BC / GM	9	t6	t8	t6	—	—	7	✓	t0, t1, t2, t3, t4, t5, t7
BC / KM	9	t3, t6	t2, t3, t5, t6	t2, t3	t3	t3	5	✓	t0, t1, t4, t7, t8
BC / SC	8	t2	t2	t2, t3	—	—	6	✓	t0, t1, t4, t5, t6, t7
CA / GM	9	t0	t2, t3, t8	t2, t3	—	—	5	✓	t1, t4, t5, t6, t7
CA / KM	9	t3	t2, t3, t4, t7	t3, t4	t3	t3, t7	5	✓	t0, t1, t5, t6, t8
CA / SC	10	t0, t1, t7	t0, t1, t9	t0, t4, t9	t4, t9	t1	5	✓	t2, t3, t5, t6, t8
RA / GM	9	—	t3	t2	—	t3	7	✓	t0, t1, t4, t5, t6, t7, t8
RA / KM	9	—	t3	t3	—	t3	8	✓	t0, t1, t2, t4, t5, t6, t7, t8
RA / SC	10	—	t4	t4, t8	t4, t8	t4, t8	8	✓	t0, t1, t2, t3, t5, t6, t7, t9

For CICIDS2018, the STE reports again show that not all candidate decompositions are equally suitable for CAD. Some splits exhibit stronger cross-task coverage or weaker self-learnability, whereas the selected configuration preserves a better balance between task solvability and heterogeneity. Figures 28 and 29 therefore provide direct evidence that the final scenario is grounded in observed task behavior rather than in arbitrary partitioning.



Figure 28: STE evaluation heatmaps for the CICIDS2018 candidate splits measured with ROC-AUC. The selected split retains clear diagonal dominance without collapsing into uniformly high transfer across tasks.

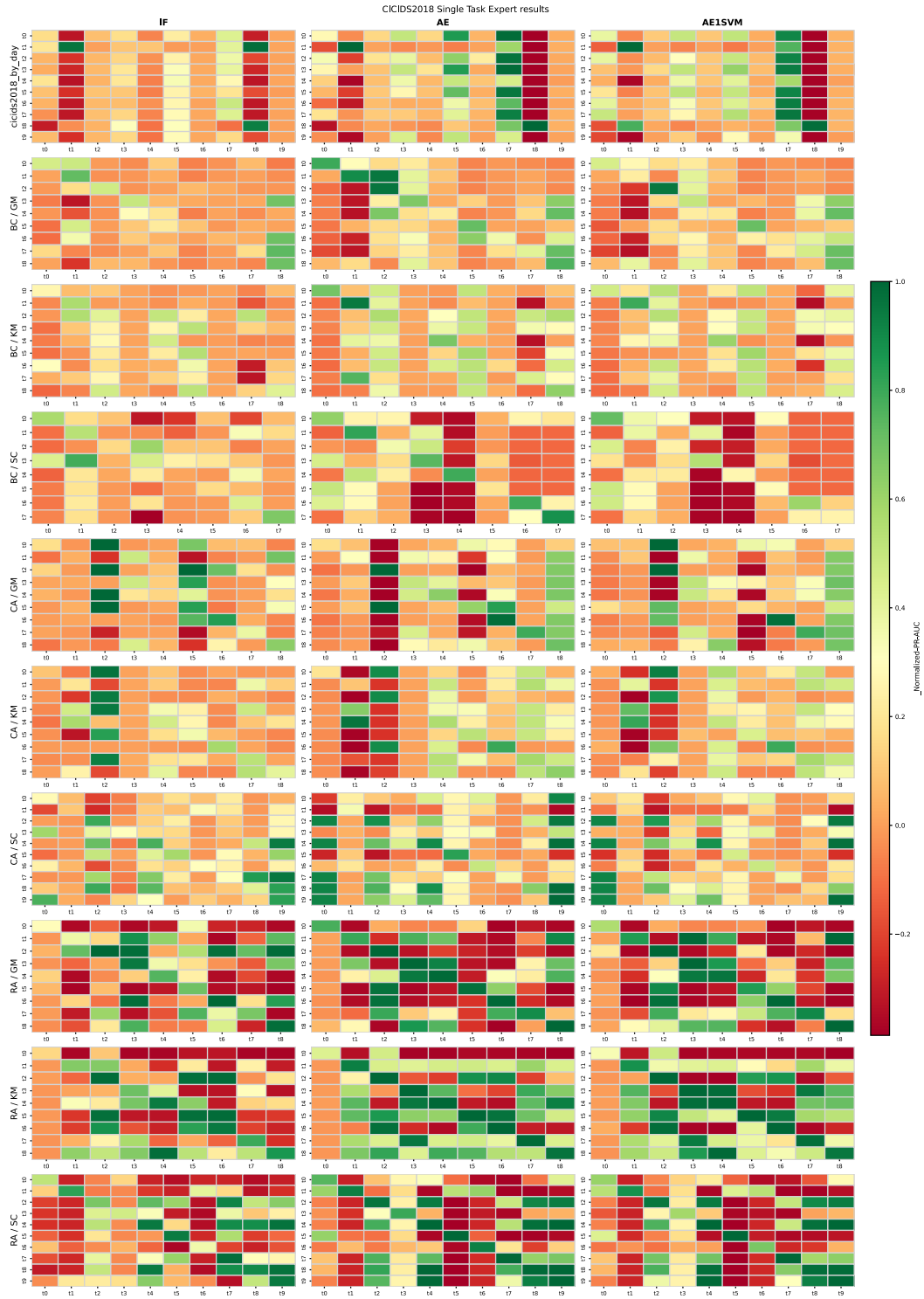


Figure 29: STE evaluation heatmaps for the CICIDS2018 candidate splits measured with normalized PR-AUC. The retained split remains differentiated under imbalance-aware evaluation, which supports the consistency of the filtering stage.

K.3 CAD-CICUNSW

Table 46: Scenario CAD-CICUNSW: Task filtering results per task discovery configuration. BC = BothClasses, CA = Closest Anomalies, RA = Random Anomalies, GM = Gaussian Mixture, KM = K-Means, SC = Spectral Clustering. By Day = Split by day/part of the day

Config	IT.	FC1	FC2	FC3	FC4	FC5	FT	MT	Final tasks.
By Day	3	—	—	—	t0, t1	t0, t1	1	✗	—
BC / GM	4	t0, t1, t2	t1	—	t1	t1	1	✗	—
BC / SC	8	t7	t1, t4	t1	t1, t4	t4	5	✓	t0, t2, t3, t5, t6
CA / GM	9	t6	t2, t5, t6	t2, t6	t6	—	6	✓	t0, t1, t3, t4, t7, t8
CA / KM	5	—	—	—	t3	—	4	✓	t0, t1, t2, t4
CA / SC	10	t6	t2, t4, t5, t6, t7, t8	t2, t4, t5, t6, t7	t5	t5	4	✓	t0, t1, t3, t9
RA / GM	9	—	t5, t7	t5	t7	—	7	✓	t0, t1, t2, t3, t4, t6, t8
RA / KM	5	—	—	—	t3	t3	4	✓	t0, t1, t2, t4
RA / SC	10	—	t4, t5, t7, t8	t1, t4	t4, t5	t4, t5, t7	5	✓	t0, t2, t3, t6, t9

The CICUNSW construction report highlights a more demanding transfer structure. Compared with the CICIDS scenarios, several candidate tasks are more weakly connected, which increases the importance of excluding dead or overly dominant tasks before ordering. Figures 30 and 31 show that the retained split preserves learnability while still exhibiting the heterogeneity needed for meaningful continual evaluation.

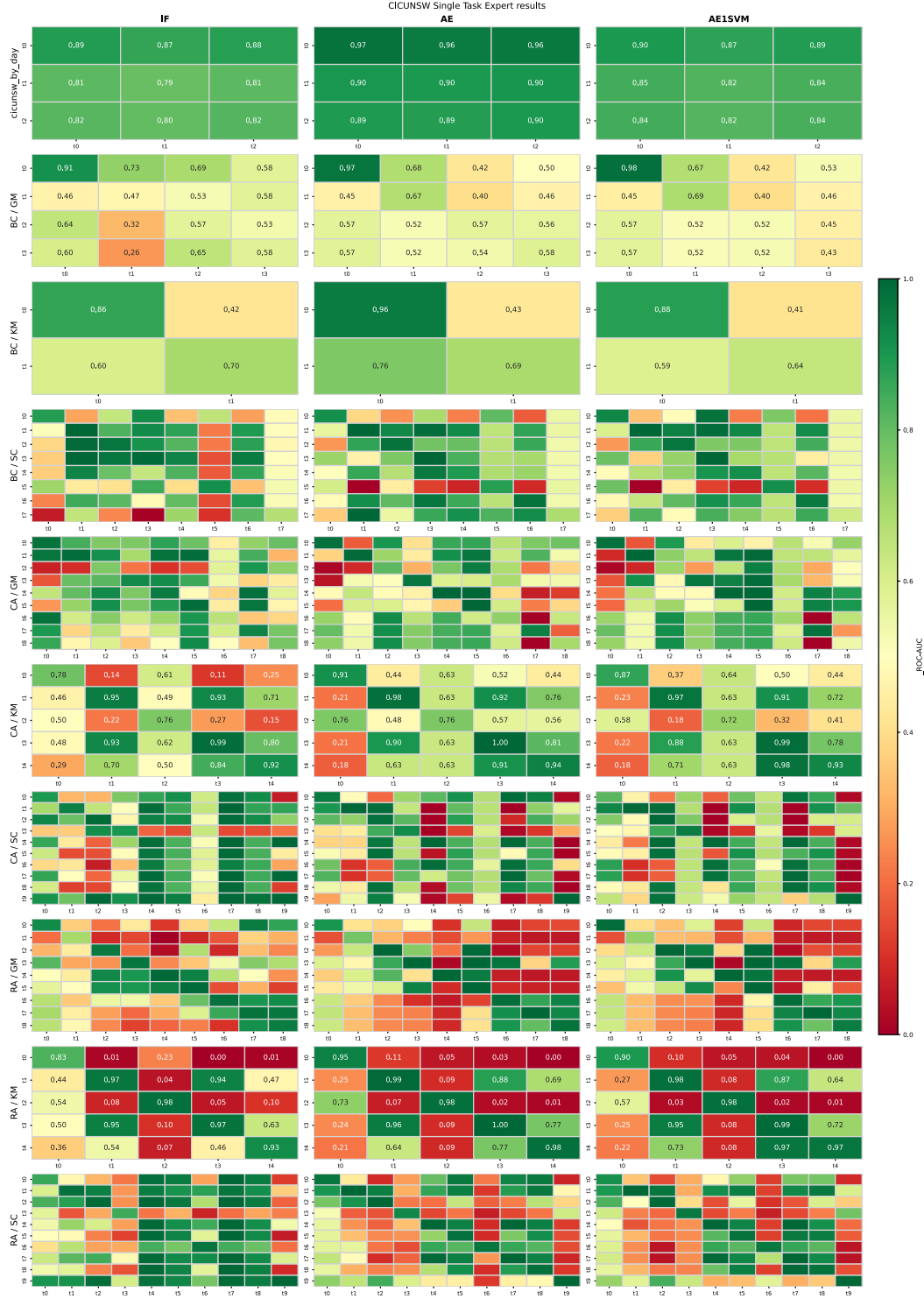


Figure 30: STE evaluation heatmaps for the CICUNSW candidate splits measured with ROC-AUC. The retained split combines diagonal learnability with sufficiently irregular off-diagonal structure to support non-trivial CAD dynamics.

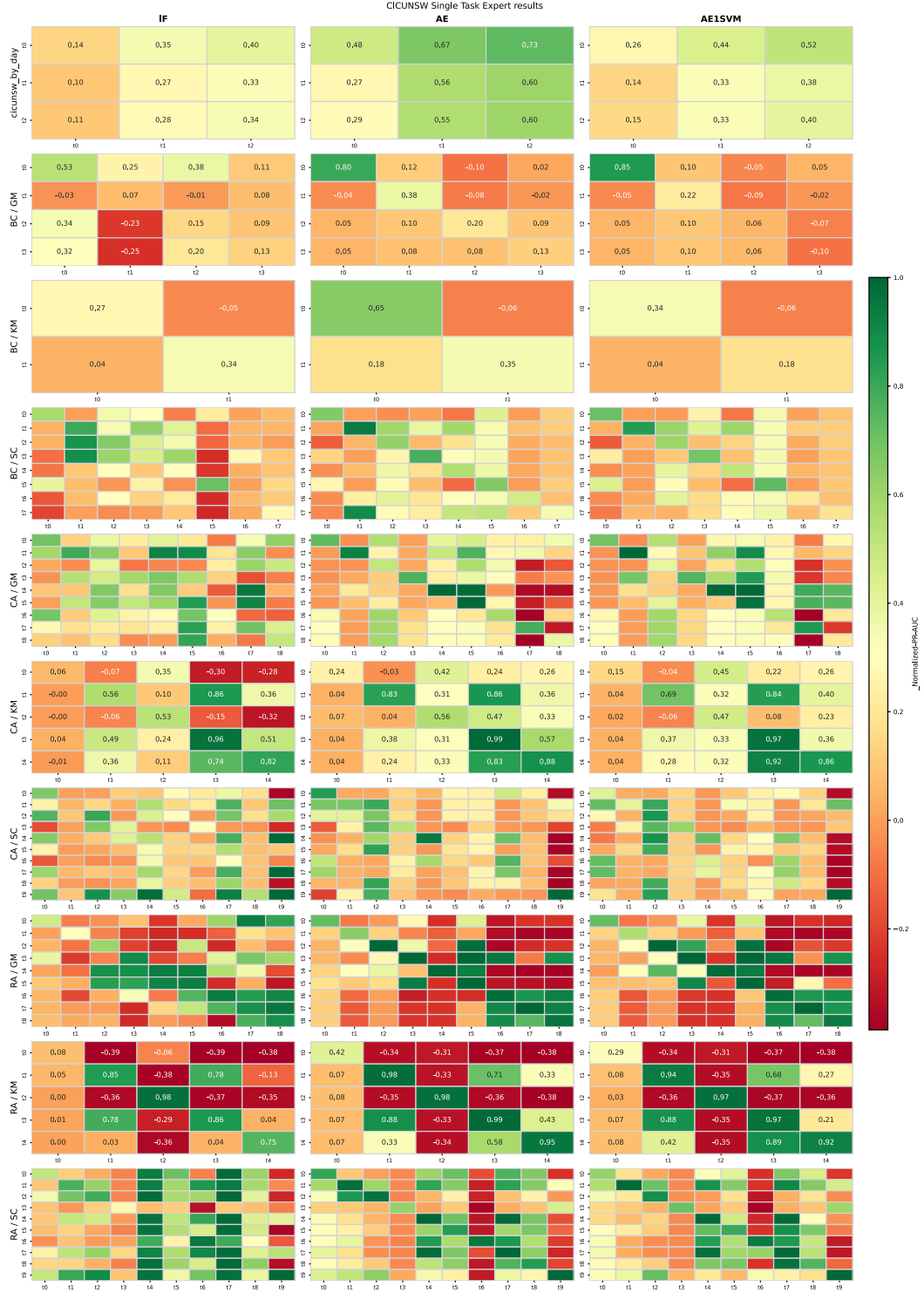


Figure 31: STE evaluation heatmaps for the CICUNSW candidate splits measured with normalized PR-AUC. The qualitative structure is preserved, indicating that the filtering decisions are not an artifact of a single metric.

K.4 CAD-MiniBooNE

Table 47: Scenario CAD-MiniBooNE: Task filtering results per task discovery configuration. BC = BothClasses, CA = Closest Anomalies, RA = Random Anomalies, GM = Gaussian Mixture, KM = K-Means, SC = Spectral Clustering.

Config	IT.	FC1	FC2	FC3	FC4	FC5	FT	MT	Final tasks.
BC / GM	5	t0, t1, t2, t3, t4	t0	t1	t0, t1, t2	t0, t1, t2	0	✗	—
BC / KM	5	t0, t1, t2, t3	t1	t3	t1, t2, t3	t0, t1, t2, t3	1	✗	—
BC / SC	5	t0, t1, t2, t3	t0	—	t0, t2, t3	t0, t1, t2	1	✗	—
CA / GM	10	t1, t2, t4, t9	t1, t2, t3, t4, t9	t9	t4	—	5	✓	t0, t5, t6, t7, t8
CA / KM	8	t0, t1, t2, t3, t4, t6	t0, t1, t3, t4	t0, t4, t5	t0, t1, t2, t3, t4, t5	t0, t1, t3, t4, t5	1	✗	—
CA / SC	8	t0, t1, t2, t3, t5, t6	t0, t1, t5, t6	t0, t2	t0, t2, t6	t0, t2, t4, t6	1	✗	—
RA / GM	10	—	t1, t2	t2, t5, t8	t5	t2	6	✓	t0, t3, t4, t6, t7, t9
RA / KM	8	—	t5	t5	t2, t5	t5	6	✓	t0, t1, t3, t4, t6, t7
RA / SC	8	—	—	—	t2, t3, t5	t2	5	✓	t0, t1, t4, t6, t7

For MiniBooNE, four of the candidate splits retain the minimum number of tasks after filtering, and all four assign anomalies at random or by the closest normal cluster. No Both Classes split survives, because most of their tasks fail the self-learnability criterion. The selected split (RA / SC) keeps five of its eight initial tasks. Figures 32 and 33 show the corresponding STE matrices.

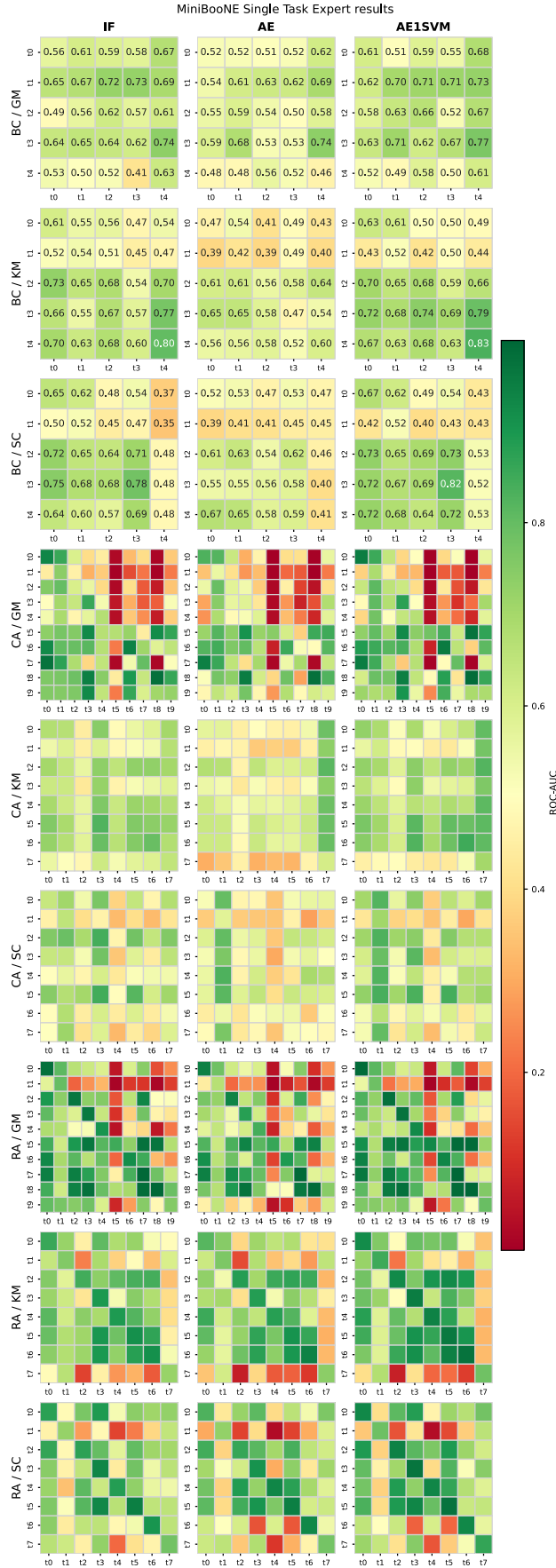


Figure 32: STE evaluation heatmaps for the MiniBooNE candidate splits measured with ROC-AUC. Few values approach 1, and the Both Classes splits stay close to chance, consistent with MiniBooNE being hard even for single-task experts.

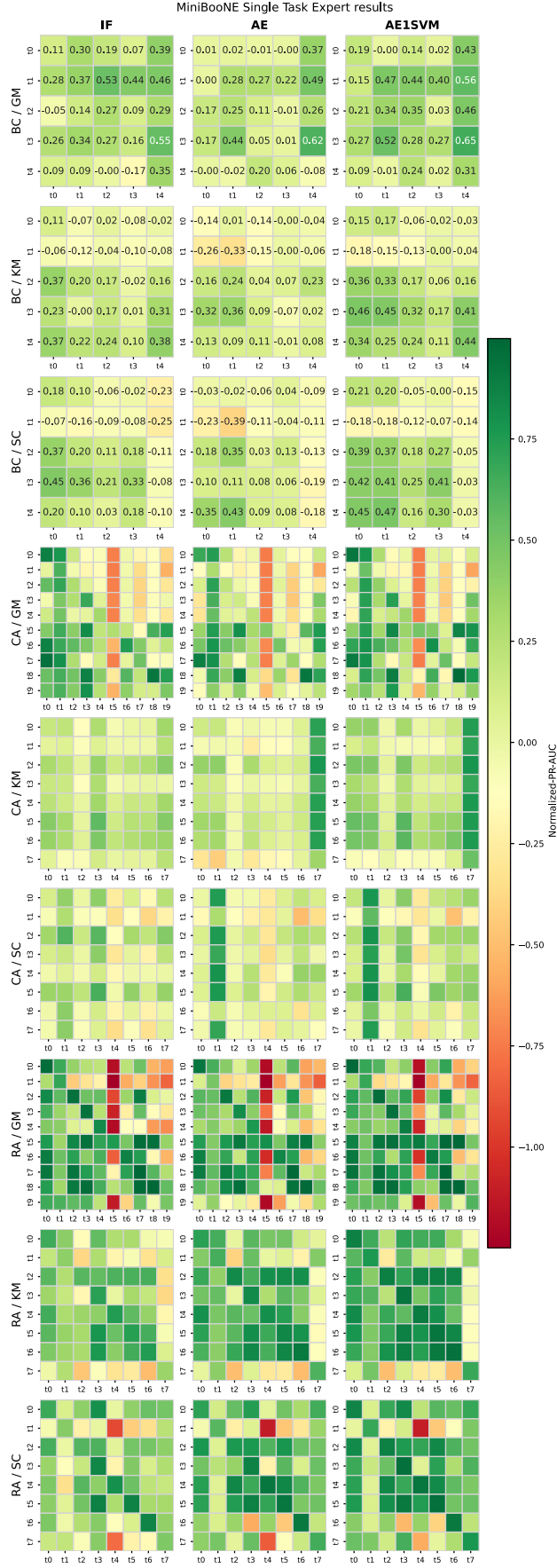


Figure 33: STE evaluation heatmaps for the MiniBooNE candidate splits measured with normalized PR-AUC. The same structure is visible under imbalance-aware evaluation.

K.5 CAD-Scania

Table 48: Scenario CAD-Scania: Task filtering results per task discovery configuration. BC = Both-Classes, CA = Closest Anomalies, RA = Random Anomalies, GM = Gaussian Mixture, KM = K-Means, SC = Spectral Clustering.

Config	IT.	FC2	FC3	FC4	FC5	FT	MT	Final tasks.
CA / GM	5	t0, t1	t0, t1	t0, t2	t0, t1	2	✗	—
CA / SC	5	t0, t1	t0	t0, t1, t2	t0, t1, t2	2	✗	—
RA / GM	5	t0, t1	t0, t1	t0, t1, t2	t0, t1	2	✗	—
RA / SC	5	t2, t4	t2	t2	t2, t4	3	✓	t0, t1, t3

Scania has the smallest candidate pool. Of the nine clustering configurations, five yielded fewer than three initial tasks (Both Classes with k-means or spectral clustering, and k-means with Closest or Random anomaly assignment) and were not evaluated further, so they do not appear in Table 48. Of the remaining four splits, only RA / SC keeps the minimum number of tasks after filtering, so the split selection step has a single candidate. In the other three splits, most tasks are removed because other tasks already cover them, because they dominate other tasks, or because they are redundant. Figures 34 and 35 show that many candidate tasks transfer almost perfectly to each other, which explains why so few survive.

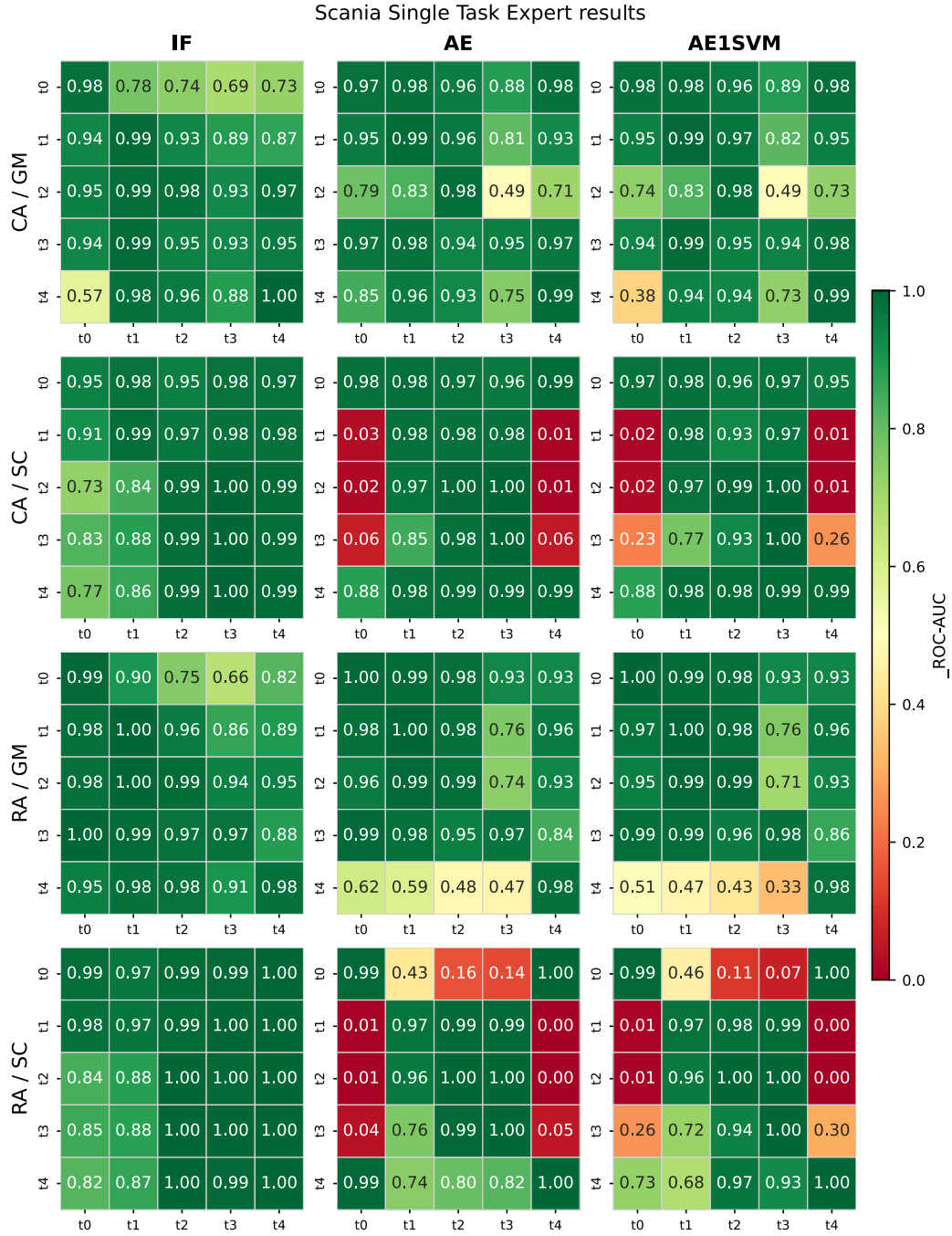


Figure 34: STE evaluation heatmaps for the Scania candidate splits measured with ROC-AUC.

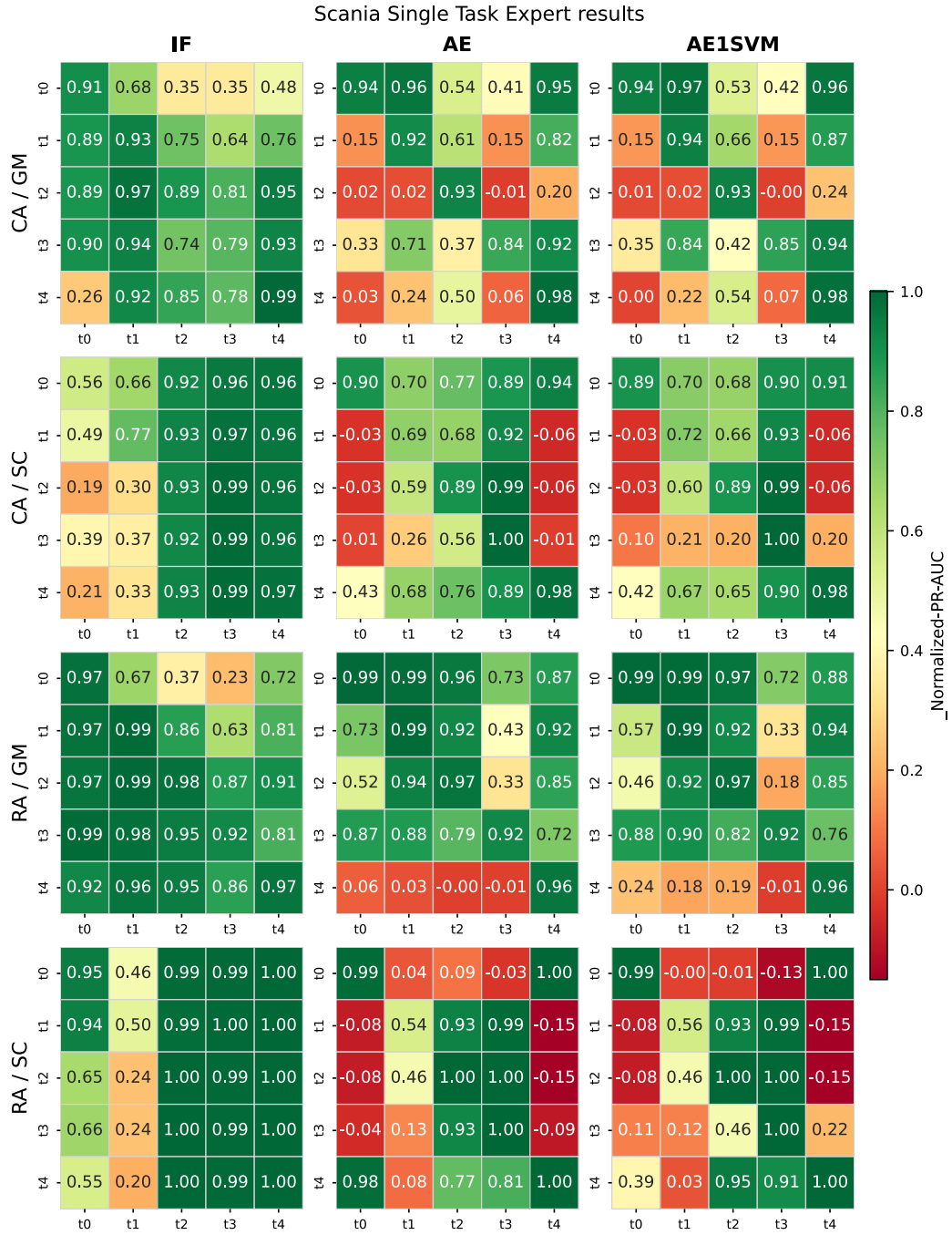


Figure 35: STE evaluation heatmaps for the Scania candidate splits measured with normalized PR-AUC.

K.6 CAD-TCM

Table 49: Scenario CAD-TCM: Task filtering results per task discovery configuration. BC = Both-Classes, CA = Closest Anomalies, RA = Random Anomalies, GM = Gaussian Mixture, KM = K-Means, SC = Spectral Clustering. By File = One task per source file.

Config	IT	FC2	FC3	FC4	FC5	FT	MT	Final tasks.
By File	6	t2	t0, t1	t0, t1, t2, t3	t0, t1, t2, t3	2	✗	—
BC / GM	3	—	—	—	—	3	✓	t0, t1, t2
CA / GM	10	—	—	—	—	10	✓	t0, t1, t2, t3, t4, t5, t6, t7, t8, t9
CA / KM	10	t2, t6	t0	—	t0	7	✓	t1, t3, t4, t5, t7, t8, t9
CA / SC	9	—	t3	—	—	8	✓	t0, t1, t2, t4, t5, t6, t7, t8
RA / GM	10	—	—	t8	—	9	✓	t0, t1, t2, t3, t4, t5, t6, t7, t9
RA / KM	10	—	t8	t5	—	8	✓	t0, t1, t2, t3, t4, t6, t7, t9
RA / SC	9	—	—	—	—	9	✓	t0, t1, t2, t3, t4, t5, t6, t7, t8

TCM is the only non-cybersecurity dataset with a natural task boundary as each source file of the simulator can be treated as a task. This „By File” split is rejected, because four of its six tasks are covered by, dominate, or duplicate other tasks, leaving too few tasks. It means that the files therefore do not correspond to sufficiently distinct regimes. Two clustering configurations, Both Classes with k-means and with spectral clustering, yielded only two initial tasks each and were not evaluated further, so they do not appear in Table 49. The remaining seven clustering-based splits keep enough tasks, and split selection chooses RA / KM, which retains eight of ten tasks. Figures 36 and 37 show the corresponding STE matrices.

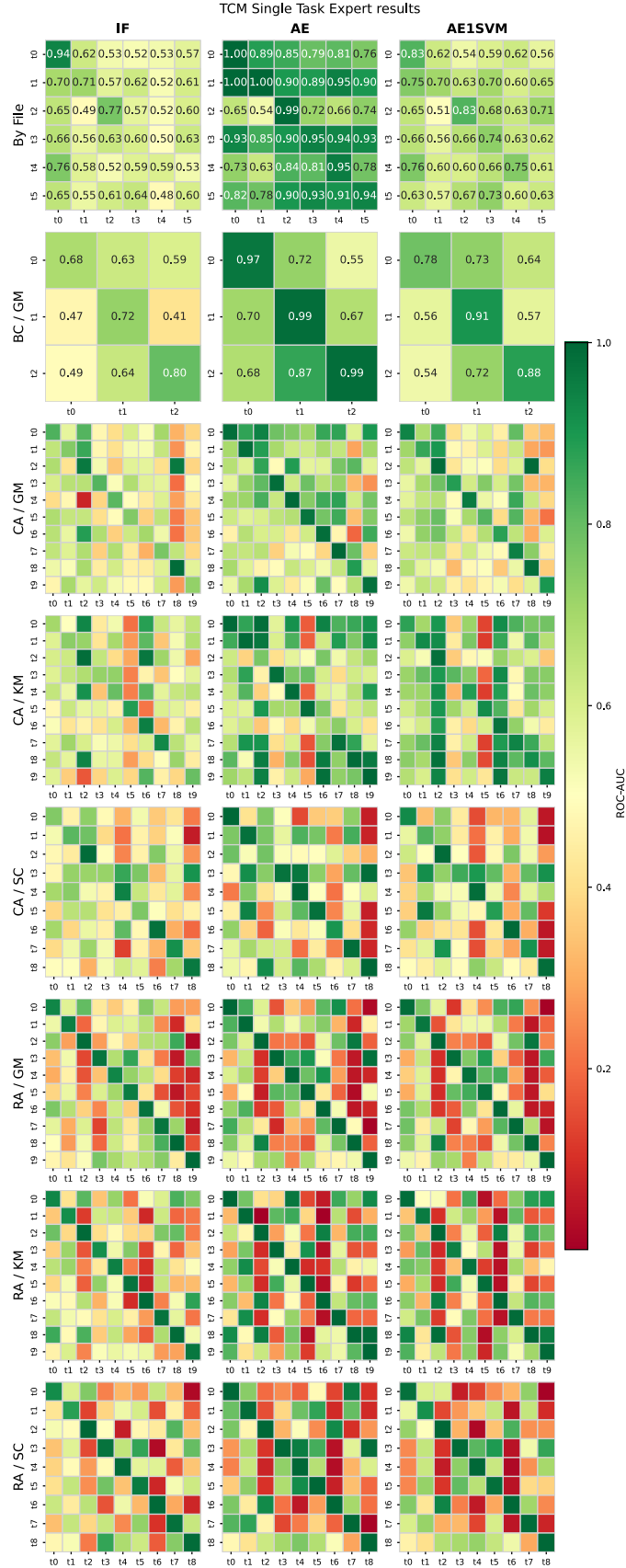


Figure 36: STE evaluation heatmaps for the TCM candidate splits measured with ROC-AUC. In the By File split, the Autoencoder transfers almost perfectly between most files, whereas the clustering-based splits show more varied cross-task structure.

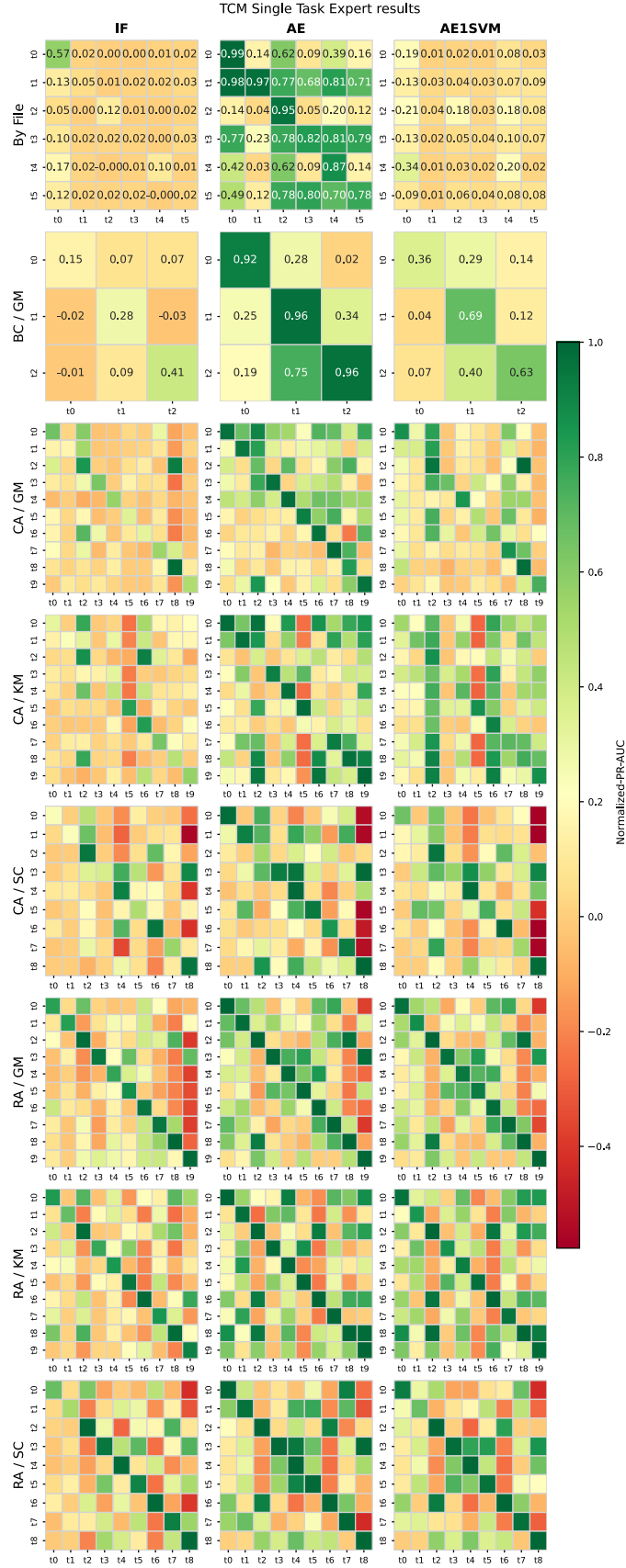


Figure 37: STE evaluation heatmaps for the TCM candidate splits measured with normalized PR-AUC. The qualitative structure is preserved under imbalance-aware evaluation.

K.7 MCAD-CIC-3x1

Table 50: Scenario MCAD-CIC-3x1: Task filtering results

Config	IT.	FT	MT	Final tasks.
Datasets	3	3	✓	t0, t1, t2

The MCAD-CIC-3x1 setting provides a simpler multi-dataset construction report in which each source dataset acts as a single task. Even in this reduced setting, the STE structure is useful for verifying that the three resulting tasks are not trivially interchangeable. Figure 38 summarizes these cross-dataset relations.

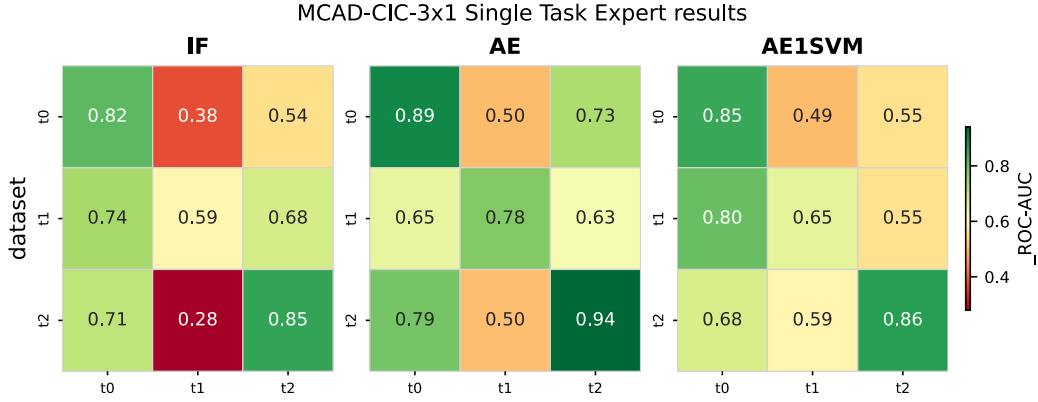


Figure 38: STE evaluation heatmaps for the MCAD-CIC-3x1 scenario measured with ROC-AUC. With one task per dataset, the figure highlights the cross-dataset transfer structure that motivates the multi-dataset continual setting.

K.8 MCAD-CIC-3xN

Table 51: Scenario MCAD-CIC-3xN: Task filtering results

Config	IT.	FC2	FC3	FT	MT	Final tasks.
Tasks from datasets	16	t3, t12	t2	t3	✓	t0, t1, t4, t5, t6, t7, t8, t9, t10, t11, t13, t14, t15

The MCAD-CIC-3xN report illustrates the most complex task-construction case in the paper. Because this scenario mixes multiple datasets and multiple retained concepts per dataset, the STE heatmaps are particularly informative for showing where transfer is preserved and where regime shifts become severe. Figures 39 and 40 confirm that the resulting split is far from trivial and therefore suitable as a harder benchmark extension. The retained indices $t0, t1, t4, \dots, t15$ are remapped in Table 23 to dataset-prefixed identifiers $c17_i, c18_i$, and cu_i for readability.

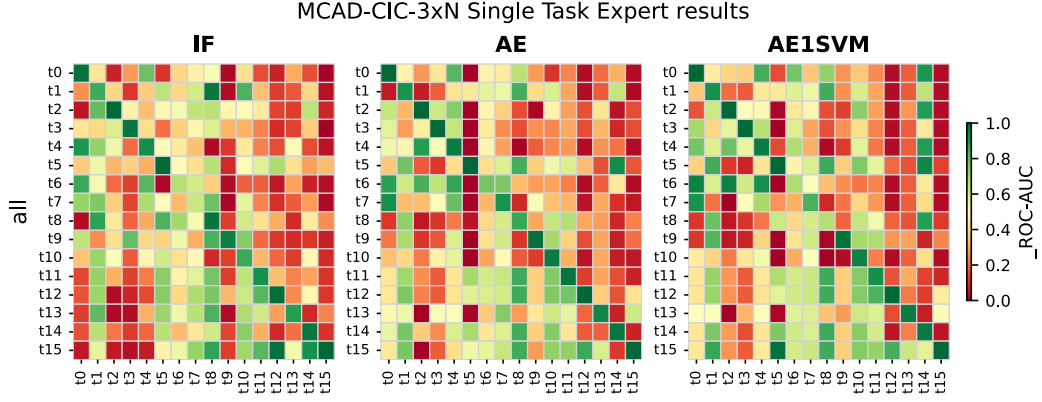


Figure 39: STE evaluation heatmaps for the MCAD-CIC-3xN scenario measured with ROC-AUC. The figure reveals a richer and harsher transfer structure than in the single-dataset scenarios, consistent with the increased difficulty of the multi-dataset setting.

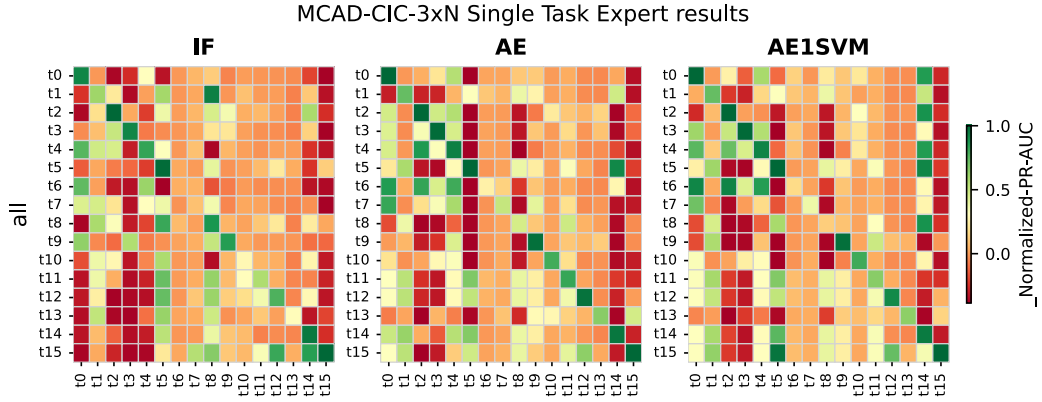


Figure 40: STE evaluation heatmaps for the MCAD-CIC-3xN scenario measured with normalized PR-AUC. The same broad structure persists under imbalance-aware evaluation, further supporting the validity of the retained split.